Physiology

Workbook

Dr. Anupama Chowdhry Devgan

MD, DNB (Physiology)

MEDICAL INSTITUTE FOR SCREENING TEST

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INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
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DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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GENERAL PHYSIOLOGY

HOMEOSTASIS & FEEDBACK MECHANISM

MILIEU INTERIEUR

- Mileu interieur refers to .....
- Homeostasis is .....
- Term coined by Walter Cannon

Disease

- Disease is a state of disrupted homeostasis
- Even in the presence of disease homeostatic functions continue to operate and maintain vital functions

Feedback Systems

NEGATIVE FEEDBACK	POSITIVE FEEDBACK

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NEGATIVE FEEDBACK SYSTEM

Positive Feedback

- Initiating stimulus is enhanced
- Examples of positive feedback: ["CLAPS"]

C:

L:

A:

P:

S:

Parturition:

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LH Surge

Shock

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CONCEPTS OF OSMOLALITY & OSMOTIC PRESSURE

Normal plasma osmolality: .....

Contributors to plasma osmolality

1. Sodium and its associated anions (Cl-, HCO3-) contribute .....
2. Glucose contributes .....
3. Urea contributes – .....
4. Plasma protein contributes .....
5. Remaining ions –.....

   Total = ...

Difference between ECF and ICF

	ECF	ICF
Osmolality		
Most osmotically active particle		
Major cation		
Major Anion		
pH		
Proteins		

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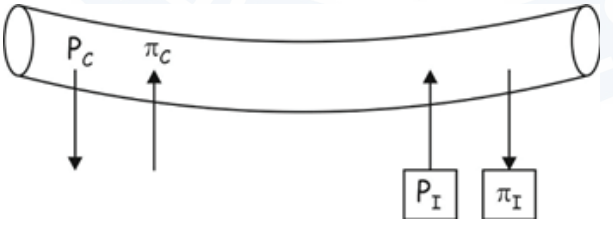
Osmotic pressure

- Definition: .....
  .....
- 1 milli osmole exerts osmotic pressure of 19.3 mmHg
- Osmotic pressure of plasma = ...

Colloid osmotic pressure or oncotic pressure

- Osmotic pressure exerted by protein in plasma
- Normal value = ...
- Protein which contributes max to oncotic pressure is .....
  .....

STARLING FORCES IN TISSUE CAPILLARY



PC – Hydrostatic pressure

π c – Colloid Osmotic pressure

Pi = Interstitial fluid hydrostatic pressure

π I = interstitial colloid osmotic pressure

Net Pressure = _____ –_____ –_____+_____

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Normal Values

PC = ...

π C = ...

Pi = ...

π i = ...

Kf π Ultrafiltration constant

= ..... × .....

Net pressure = 35 – 25 – (-1) + 0 = + 11 mmHg

Rate of tissue fluid formation = Kf (PC – π C – Pi + π i)

Right ventricular failure	LVF	Liver Cirrhosis

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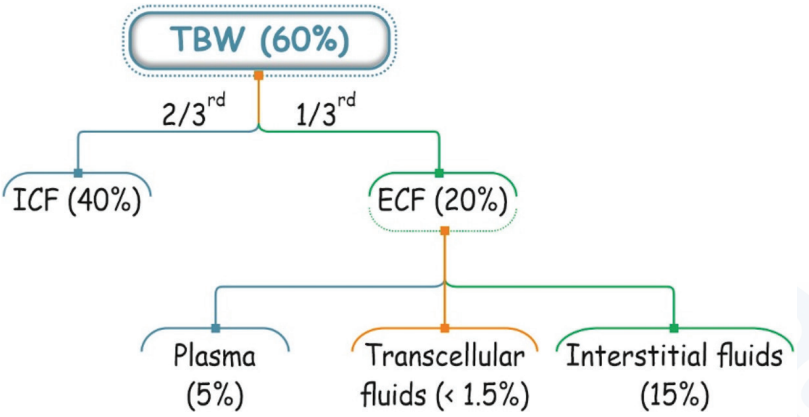
DISTRIBUTION OF TOTAL BODY WATER (TBW)

TBW = 60% of body weight



Example

ECF volume in a 70 kg Man =

60 - 40 - 20 Rule

↓ ↓ ↓

TBW ICF ECF

Transcellular fluids

These are parts of ECF but contained in their own spaces

Examples:-

All transcellular fluids put together are .....

Factors affecting Total Body Water (TBW)

1. Age	
2. Fat	

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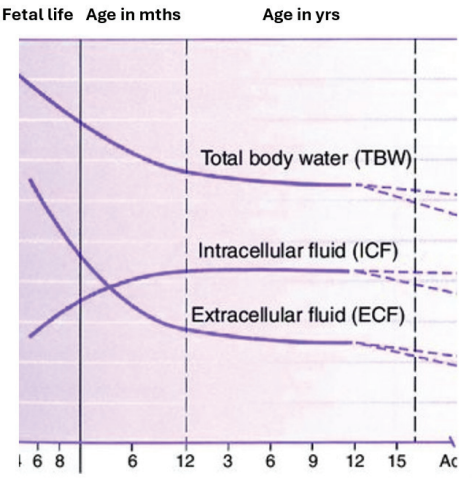

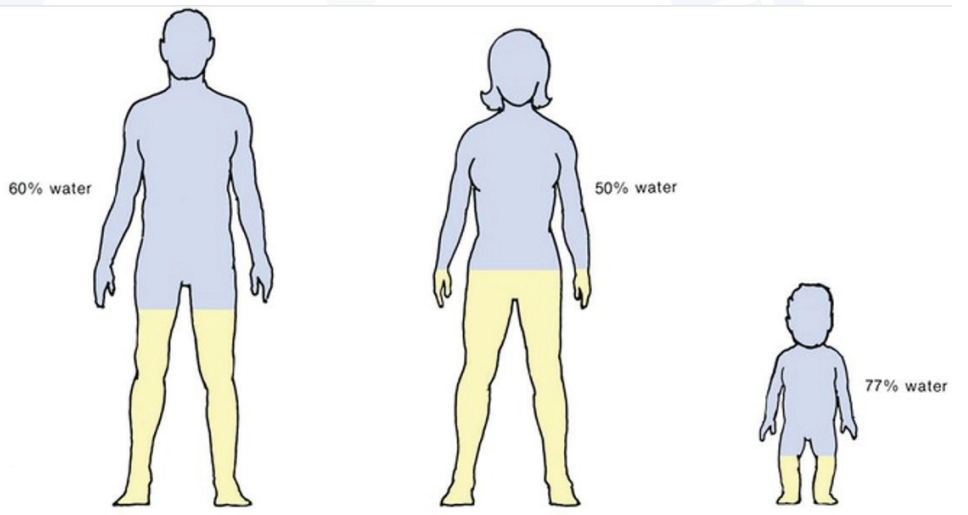

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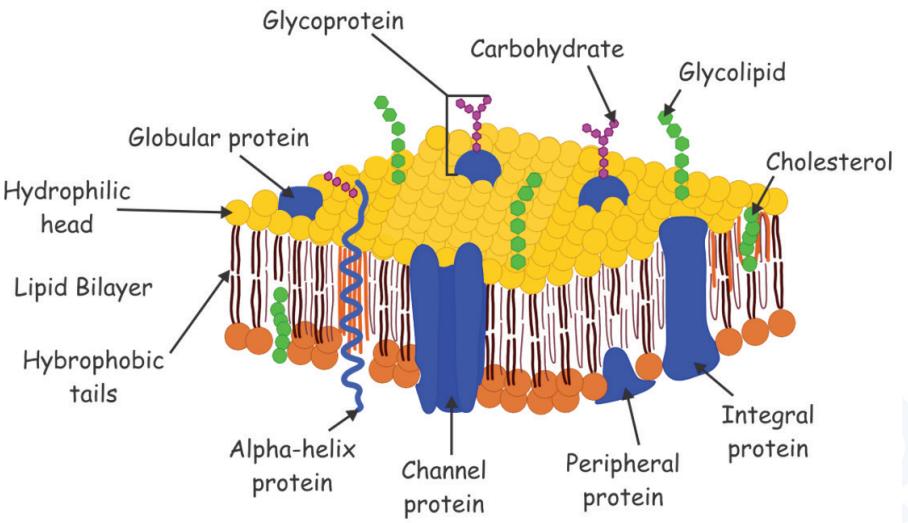
CELL MEMBRANE AND INTERCELLULAR JUNCTION



Cell Membrane: Thickness is .....

- Cell membrane has lipid bilayer with hydrophilic heads and hydrophobic tails

Protein in cell membrane can be	(a) Receptors (b) Antigens (c) Transport proteins (d) Channels (e) Structural proteins (f) Enzymes	
Intercellular communication can be	(a) Autocrine	
	(b) Juxtracrine	(Eg:TGF a)
	(c) Paracrine	
	(d) Endocrine	
CAMS (Cells Adhesion Molecules):	(a) ..... (b) ..... (c) ..... (d)	

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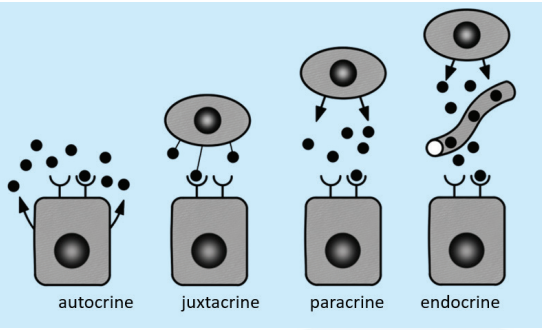
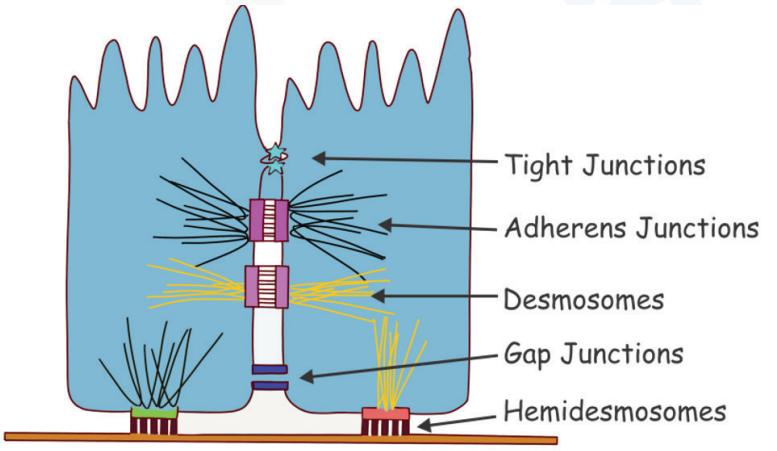


TIGHT JUNCTIONS

✦ Tight junctions are found in- Choroid plexus, between endothelial cells of cerebral capillaries



Proteins of tight junction:

1. .....
2. .....
3. .....

Tight junctions do not permit paracellular movement of substances

Tight junction forms zonula occludens

(Extra Edge: - Vibrio cholera produces 2 types of toxin

(a) Cholera toxin

(b) Zona occludens toxin[Q]

Zona Occludens toxin will disrupt Zona occludens)

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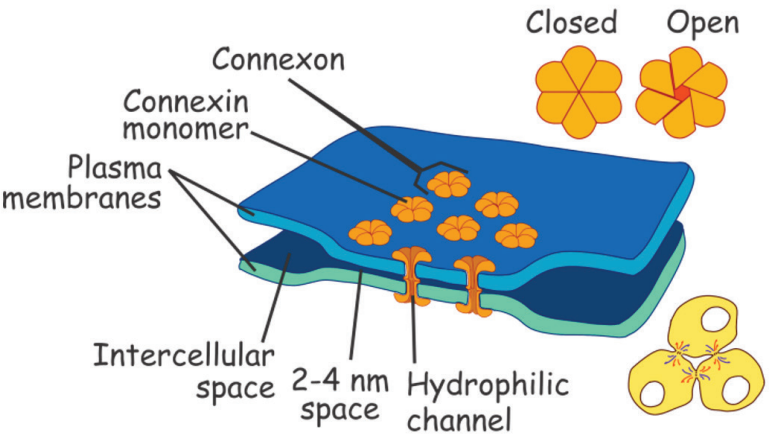
Gap Junctions

Gap junctions are made up of connexon proteins- each connexon has 6 subunits known as connexins which surround a central pore

- ✦ Two connexons together form GAP Junction which allows bidirectional movement of substances of size up to 1000 Daltons
- ✦ GAP Junctions form ..... synapses[Q]
- ✦ GAP Junctions are found in retina, myelin near the nodes of ranvier, cardiac muscles, single unit type of smooth muscles



Connexin mutation can give rise to

Charcot Marie tooth disease, and other inherited disorders

CYTOSKELETON

Made up of

- ✦ Microfilaments
- ✦ Intermediate filaments
- ✦ Microtubules

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TRANSPORT ACROSS THE CELL MEMBRANE

TYPES OF TRANSPORT ACROSS CELL MEMBRANE

Passive Transport	Active Transport
Types: 1. 2. 3. 4.	Types: 1. 2.

Passive Transport

1. Simple diffusion

Substances transported by simple diffusion:

Simple diffusion is governed by Fick's Law of Diffusion

Fick's law of Diffusion:

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J =

D =

A =

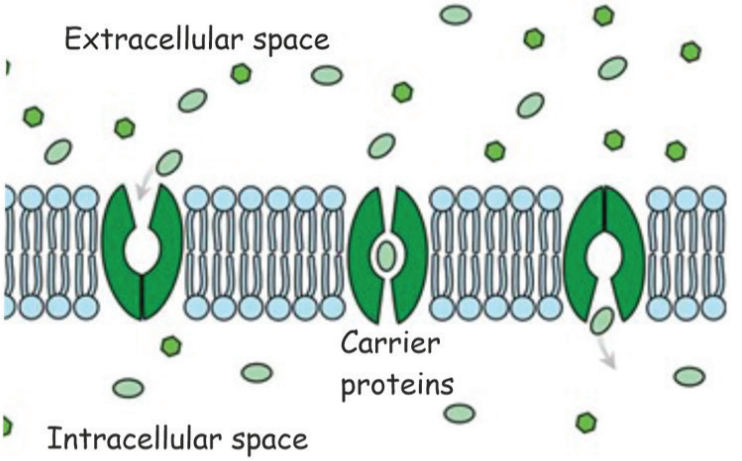
Δc =

Δx =

- Jaa diffusion coefficient, J a surface area, J a concentration - gradient, $J \alpha \frac{1}{\Delta x}$
- Applied:
- In emphysema:
- In pulmonary fibrosis:

2. Facilitated diffusion

- Done with the help of carrier protein



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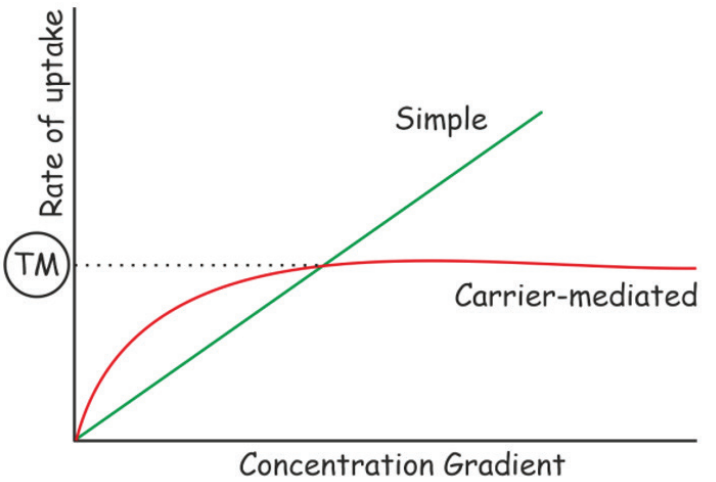
Characteristics of carrier-mediated transport:-

i.

ii.

Graphical representation of simple and facilitated diffusion



Examples of facilitated diffusion

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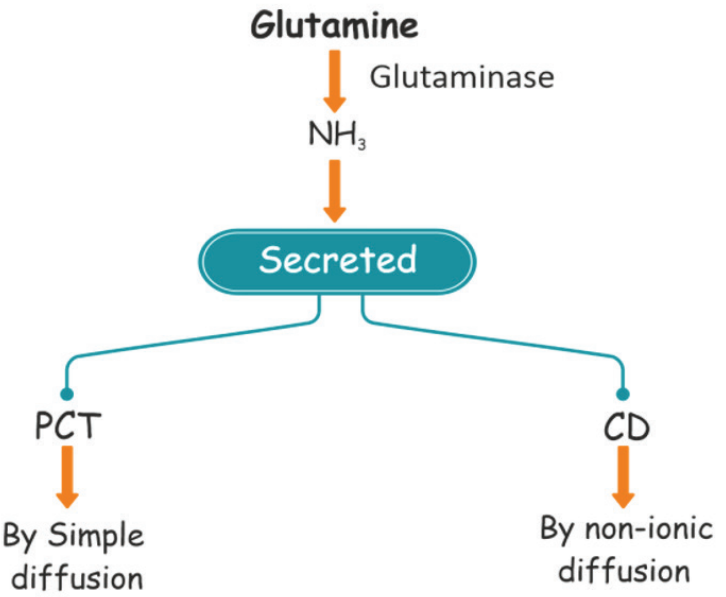
3. Non-ionic diffusion



In the CD,

Other examples of non-Ionic diffusion:

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4. Osmosis

Active transport

2 types

a. Primary active transport

b. Secondary active transport

Examples of primary active transport:

Sodium potassium ATPase pump

- Coupling Ratio is .....
- It is an electrogenic pump-

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- In cells where the RMP is -90mV, the contribution of Na+-K+ ATPase pump is .....
- Associated with Cell volume regulation since it causes a net loss of one osmotically active particle
- Contributes significantly to B.M.R. (Basic Metabolic Rate)

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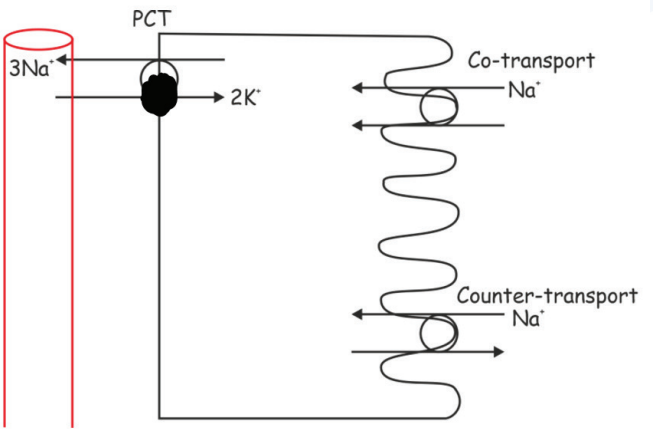
Structure

It is a Trimer with 3 subunits α, β, □

SECONDARY ACTIVE CO – TRANSPORT AND COUNTER – TRANSPORT

- Activity of Na – K ATPase on the basolateral membrane creates a concentration as well as an electrical gradient for Na+
- On the opposite side membrane, sodium will move from outside to inside along its electrochemical gradient created by the Na – K ATPase pump- this movement of sodium provides the energy for the active transport of some other substance
- If other substance moves in the same direction as sodium- it is called as .....
- If the other substance moves in a direction opposite to the direction of sodium, it is k/as .....



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Example of secondary active Co – Transport:

In PCT:

In TAL (Thick Ascending Loop):

In intestinal Epithelium:

Distal Ileum:

Thyroid:

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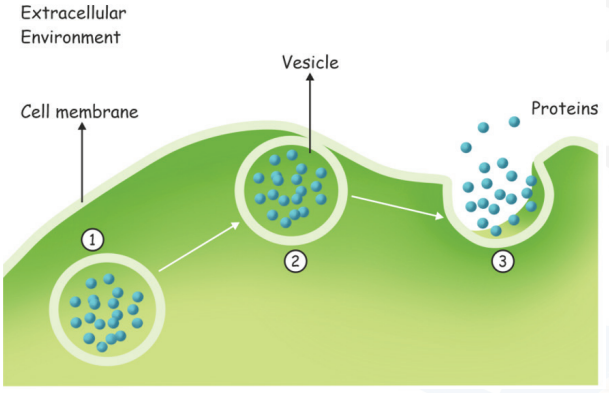
Secondary active counter-transport:

- In PCT →
- In myocardial cell →

Applied:-

EXOCYTOSIS



- Results in ..... in total area of the cell membrane
- For exocytosis cell needs ....., ...
  and .....
- Ex of docking protein

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Example of exocytosis: - Exocytosis of acetylcholine in the neuromuscular junction

- Botulinum toxin causes flaccid paralysis by inactivating docking proteins needed for the exocytosis of Ach

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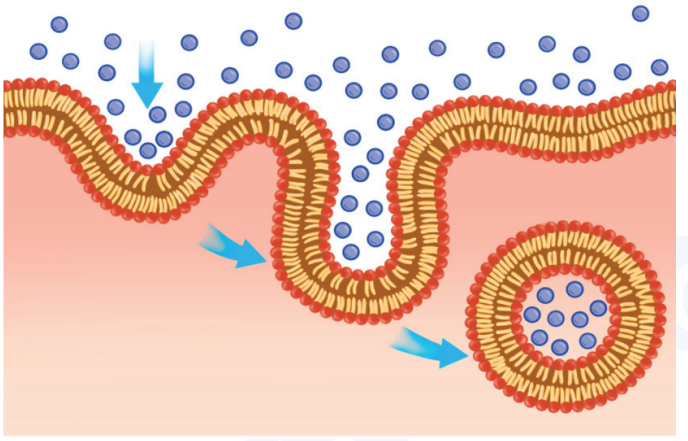
ENDOCYTOSIS



- Endocytosis ..... the total area of the cell membrane
- Can be phagocytosis (cell eating), pinocytosis (cell drinking)

Substance requires for endocytosis:

- Ca2+, ATP, Clathrin, caveolin, dynamin, pinchase
- Clathrin → Required for .....
- Caveolin → Present in .....

ION Channels:

- It is for the movement of charged particles

a. Leak Channels - these are open at rest

b. Gated Channels

Gated Channels further divided into four sub-types:

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Resting Membrane Potential

RMP of Different Cells

Cells	RMP
Most Neurons	
Large motor Neurons	
Skeletal muscle Cell	
Cardiac muscle Cell	
Pacemaker cell	
Smooth muscle cell	
Inner hair cell (Cochlea)	
Rods and cones	
Epithelial cell	
RBC	

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Effect of hyperkalaemia and hypokalaemia on RMP

Hypokalaemia causes a hyperpolarization and decrease in excitability

In hyperkalaemia:

Mild to moderate hyperkalaemia causes depolarization and increased excitability

Severe Hyperkalaemia:

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Effect of Hypocalcaemia on Excitability

- Calcium has membrane-stabilising action

Mechanism of Tetany in case of hyperventilation

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NERVE AND SKELETAL MUSCLE PHYSIOLOGY

LOCAL POTENTIAL AND ACTION POTENTIAL

Stimulus → Causes opening of Na^+ gated channel → Na^+ moves from higher to lower concentration i.e. from outside to inside → Potential decreases (becomes less negative)

- Resting membrane potential of most neurons is .....
- Subthreshold stimuli produce .....
- Threshold stimuli produce .....
- Threshold stimulus changes the potential of the neuron from the resting level (-70mV) to the firing level (-55mV)

Local Potential	Action potential

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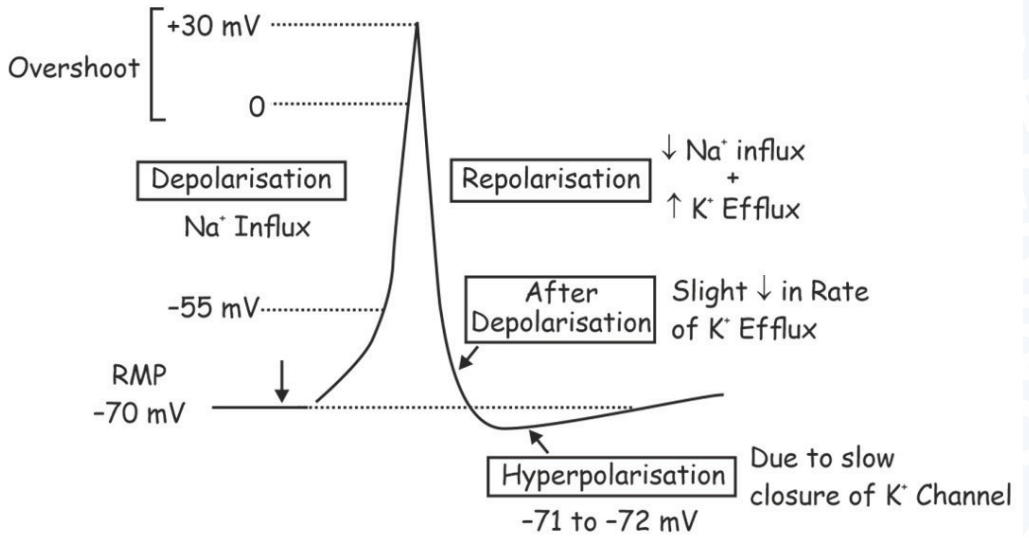
- **Spatial Summation:** The number of subthreshold stimuli given simultaneously resulting in an action potential is the summation over space or spatial summation

- **Temporal Summation:** Increase frequency of subthreshold stimuli resulting in an action potential is the summation over time or temporal summation

PHASES OF ACTION POTENTIAL



PHASE	DUE TO
1. Depolarization	
2. Repolarization	
3. After-depolarization	
4. Hyperpolarization	

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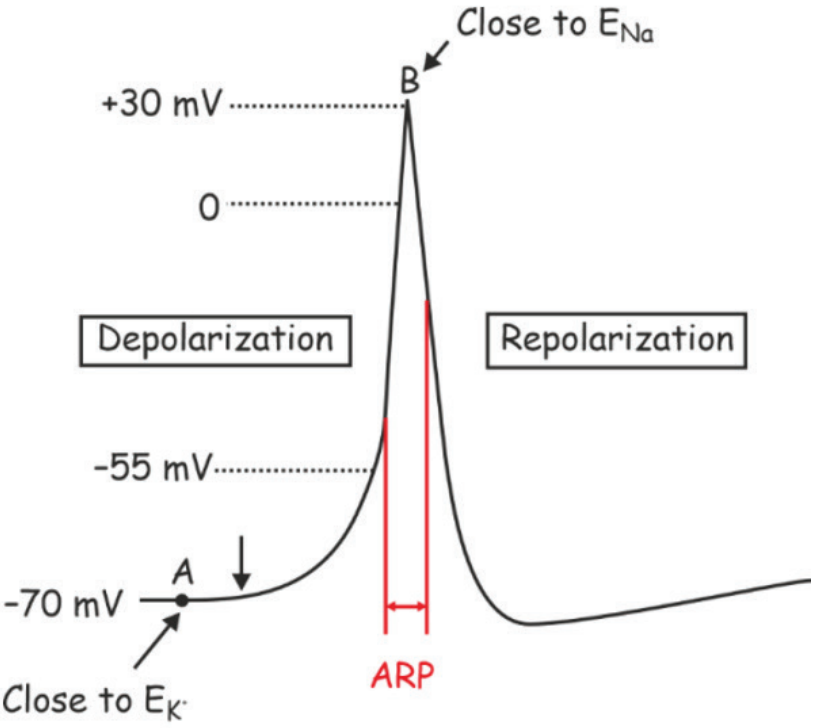
EXOCYTOSIS

- Refractory period is the period during which cell does not respond to a second stimulus

- Refractory period can be absolute or relative refractory period



- Neuron is least excitable during _____ phase of action potential[Q]

FACTORS AFFECTING VELOCITY OF CONDUCTION OF NERVE

Factors Affecting The Velocity Of Conduction Of Nerve Impulse

1. **Diameter:**

   a. Increased diameter, Increased surface area, Increased total number of Na+ channels, Increased velocity

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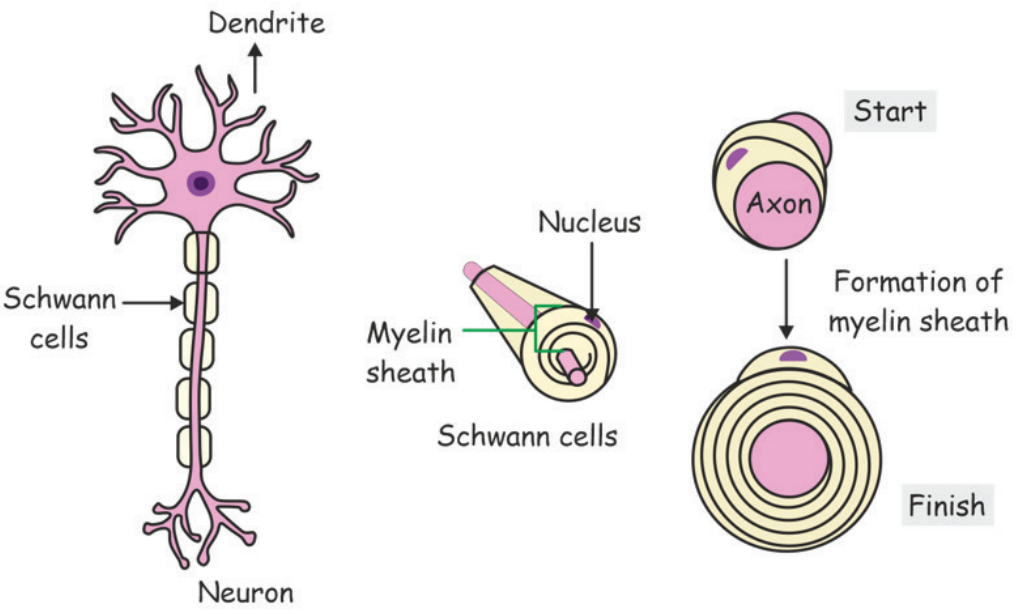
2. Myelin



- Myelin is produced by ..... in the central nervous system and by ..... in the peripheral nervous system
- Schwann cell wraps its membrane around an axon up to 100 times-the layers of the membrane are compacted to form myelin
- Myelin envelops the axon except at the nodes of ranvier
- Myelin is a protein-lipid-rich complex
- Myelin has an insulating function
- Myelin ..... membrane resistance (Rm) of the axon
- Myelinated nerve fibres have high membrane resistance
- Nodes are areas of low resistance; therefore, action potential jumps from node to node (saltatory conduction), increasing the velocity of conduction

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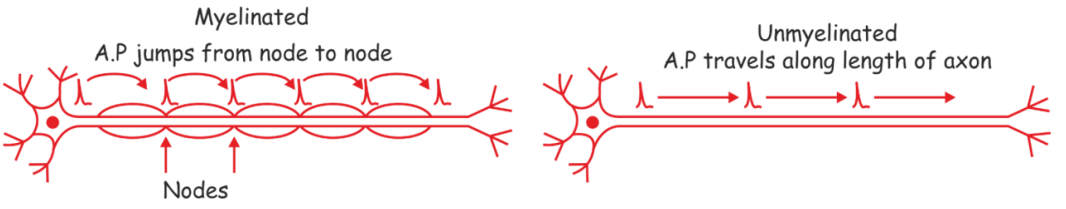

NERVE FIBRE CLASSIFICATION

Erlanger & Gasser Classification	Lloyd & Hunt's Numerical Classification
All types of nerve fibres- sensory, motor and autonomic	Only for sensory
A → A_α, A_β, A_γ, A_δ B C Decreased diameter & velocity from A to C A and B are myelinated & C is unmyelinated	Type I Type II Type III Type IV

Aα		
Aβ		
Aγ		
Aδ		
Type B		
Type C		

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SUSCEPTIBILITY OF NERVE FIBRES

Pressure	A (Aα) > B > C Example: Saturday night palsy/Sunday morning palsy
Hypoxia	B > A > C
Local anaesthetic	C > B > A A(Aγ) > B > C [as per academic research]

AXOPLASMIC TRANSPORT & NEUROTRANSMITTERS

- The neuron is a special type of cell- the cell body and the end of the neuron are at a distance from each other, so there is a need for a transport mechanism
- Axoplasmic transport is the physical transport of substances that occurs through the axoplasm
- There is a set of microtubules & molecular motors which are involved
- The transport can be from the cell body to the end of the axon or in the reverse direction
- Axoplasmic transport does not refer to the transmission of action potential (which is due to the movement of charges)

Microtubules

- Microtubules are hollow tubes made of proteins known as ....
- In axons, microtubules serve as tracks
- Molecular motors move on these tracks and transport substances from one end of the neuron to another

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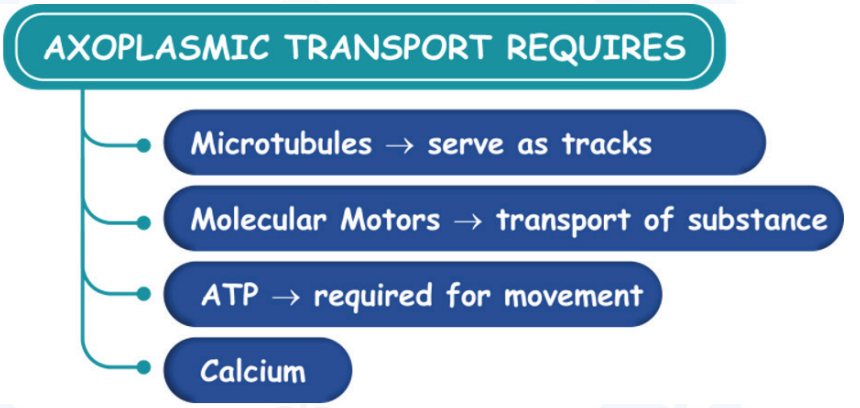
MOLECULAR MOTORS

- Small vehicles for the transport of substances from one end of the axon to another
- They are:
  1. .....
  2. .....
  3. .....
- They also have ATPase enzyme required for ATP production



TYPES OF AXOPLASMIC TRANSPORT

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SYNAPSE

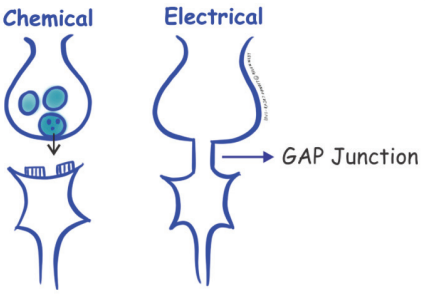
Synapse is a connection between

1. Either 2 neurons
2. Neuron and a muscle
3. Neuron and a gland

Anatomical Classification

1. Axodendritic : Axon of one neuron terminating on the dendrite of the next neuron
2. Axo somatic : Axon of one neuron terminating on the soma of the next neuron
3. Axo axonal : Axon of one neuron terminating on the axon of the next neuron

Classification according to the nature of transmission of impulse

Chemical		
Electrical		

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Neurotransmitters (can be)

Small molecule rapidly acting	Large molecule slowly acting (Neuropeptides/ growth factors)
✦ Class I:	✦ Hypothalamic releasing hormone : TRH, LHRH, GnRH, Somatostatin
✦ Class II:	✦ Pituitary peptides : ACTH, -endorphin, -MSH, Prolactin, LH, TSH, Growth hormone, Vasopressin, Oxytocin
✦ Class III:	✦ Peptides that act on gut and brain: Leucine-enkephalin, methionine- enkephalin, Substance – gastrin, CCK, VIP, Nerve growth factor, Brain derived Neurotropic factor, neurotensin, insulin
✦ Class IV:	✦ From other tissues: Angiotensin-II, bradykinin, sleep peptides, calcitonin

AUTONOMIC NERVOUS SYSTEM

Autonomic Nervous system (ANS) is of two types - sympathetic & parasympathetic

Sympathetic

- Thoracolumbar outflow
- Sympathetic ganglia are paravertebral
- Sympathetic system has.....preganglionic fibre and ..... postganglionic fibre
- Preganglionic fibres secrete ....., postganglionic fibres secrete .....

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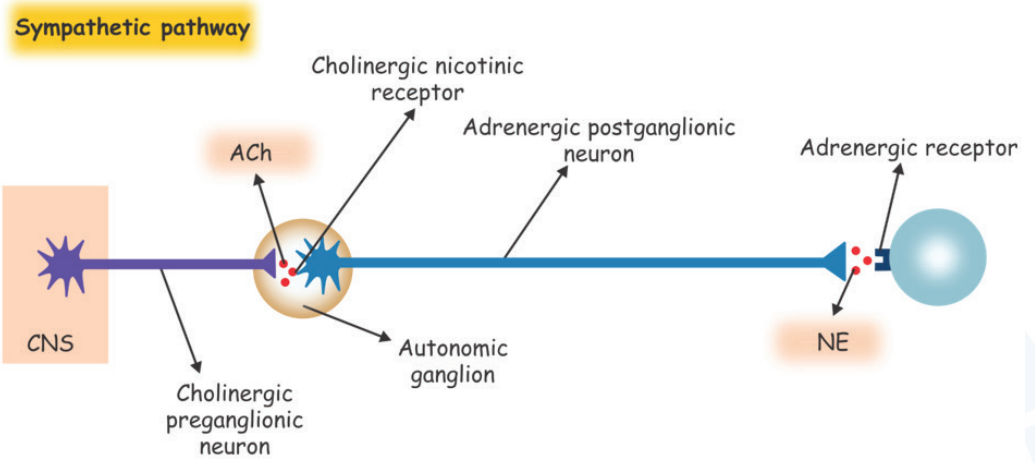
- Exception:

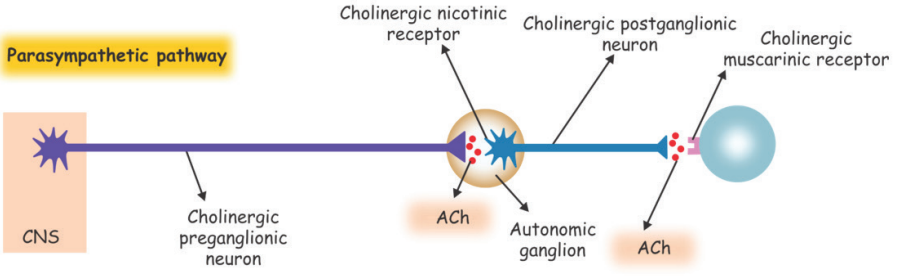
.....

.....



Parasympathetic

- Craniosacral outflow
- ..... preganglionic fibres and ..... postganglionic fibres
- Parasympathetic ganglia are located in or near the organ they innervate
- Pre as well as postganglionic fibres secrete .....



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MUSCLE PROTEINS

SKELETAL MUSCLE

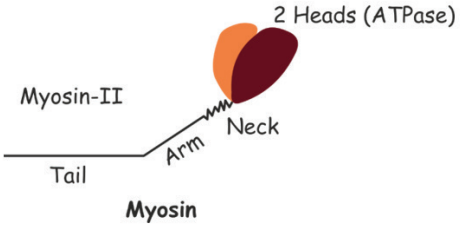
- Striated voluntary muscle
- Proteins

1. Contractile Proteins

a. Actin: 2 strands of actin are arranged in the double helix

b. Myosin II: Has 2 Heads, Neck, Arm, Tail

- Out of 2 heads of myosin, only one attaches with actin at a given point in time
- Heads also contain ATPase enzyme, which catalyses the breakdown of ATP to form ADP + Pi + Energy



Actin

2. Regulating Proteins

a. Tropomyosin : 1 molecule of tropomyosin covers 7 active site on actin

b. Troponin : Has 3 parts troponin I, C & T

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3. Structural Proteins

a. Actinin : Binds .....

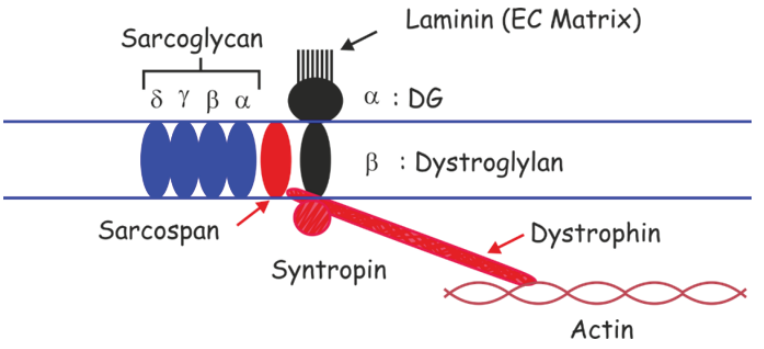
b. Titin:

- Binds.....
- Provides scaffolding (structural support) for sarcomere
- Responsible for muscle elasticity[Q]
- Largest known protein with a molecular mass of nearly 3,000,000 Daltons
- Titin mutation is associated with .....
  and in skeletal muscle with .....

c. Desmin: Binds Z – Line to Sarcolemma

4. Sarcolemmal Proteins

- Also known as Dystroglycan – Sarcoglycan complex



1. Dystrophin- connects ..... to .....,
   which in turn is connected to a-dystroglycan, and these connect to laminin 211 in the extracellular matrix

2. _____

3. _____

4. _____

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5. ..... complex of four glycoproteins-α-, β-, γ- and δ-sarcoglycan Laminin is a protein in the extracellular matrix

Syntropin, sarcospan, and sarcoglycan are all associated with dystroglycan complex

Dystrophin amplifies the force generated in actin & myosin, hence its absence & reduction cause diseases.

- ✦ In the absence of dystrophin, there is severe proximal muscle weakness, and the condition is called ..... . It is X-linked and usually fatal by the age of 30
- ✦ In a milder form of the disease, ..., dystrophin is present but altered or reduced in amount
- ✦ Limb-girdle dystrophies are associated with mutations of the genes coding for .....

SARCOMERE

- Structural or functional unit of muscle
- Present between 2 Z – lines
- M-Line – Dark or A band (.....)
- I-band – .....
- H-zone – .....

Sarcomere = 1/2 'I' Band + A – Band + 1/2 I – Band

Active space

- Z – line → Actin is attached at Z-line(Tropomyosin + troponin move along with actin)
- M – Line → Middle of sarcomere and connects the different myosin molecules
- Dark band or 'A'(Anisotropic to light) – Band → Consists of .....
- H – Zone → Present within A – band; slightly lighter area with .....
- Light Band or I(Isotropic to light) – band → Present on either side of A – band with .....
- Half of I – band lies in one sarcomere and half in the neighbouring sarcomere

Muscle contraction

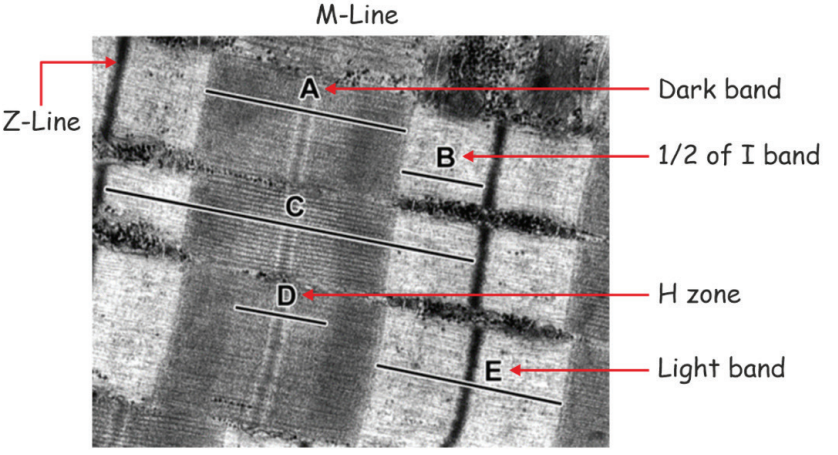
During muscle contraction the I – band slides into the A – band

↓

Width of I – band reduces

H – Zone decreases / disappears

..... remains unchanged[Q]



Active space

- C – Sarcomere (between 2 Z-lines)

TRANSMISSION ACROSS NEUROMUSCULAR JUNCTION

- Neuromuscular junction is formed between the Aα motor neuron & the sarcolemma.

Lambert Eaton syndrome	Myasthenia Gravis

Active space

SARCOTUBULAR SYSTEM AND EXCITATION CONTRACTION

Sarcotubular system

- Sarcolemma dips into muscle fibre membrane, i.e. it invaginates and forms T-tubule or transverse tubule; T- tubule in skeletal muscle is at the junction of A band, and I band
- On either side of T – tubule is the L – tubule (longitudinal tubule), which is the sarcoplasmic reticulum
- Sarcoplasmic reticulum (SR) is very well developed in skeletal muscle
- End of L – tubule dilates to form a cistern which is a storehouse of Ca2+
- DHPR (Dihydropyridine receptor) present on ..... of skeletal muscle is a .....

Ryanodine receptor, present on ....., is a .... with .... as its natural ligand

Active space

EXCITATION CONTRACTION COUPLING

Excitation-contraction coupling refers to the conversion of electrical activity (action potential) into a mechanical activity (contraction)

Action Potential reaches DHPR
↓
Conformational change in DHPR
↓
Mechanical coupling (physical interaction/molecular connection) between DHPR & Ryanodine receptor causes Ryanodine receptor to open
↓
Ca^{2+} released from Sarcoplasmic reticulum into sarcoplasm

Ca^{2+} binds with troponin-C
↓
Conformational change in troponin
↓
Causes tropomyosin to slide
↓
Exposes active sites on act in
↓
Actin-myosin cross bridge formation (muscle contraction starts)

Ca^{2+} binds with more ryanodine recepto
↓
Ca^{2+} induced Ca^{2+} release (Positive feedback)
↓
Muscle contraction

Ca^{2+} is the excitation-contraction coupling agent

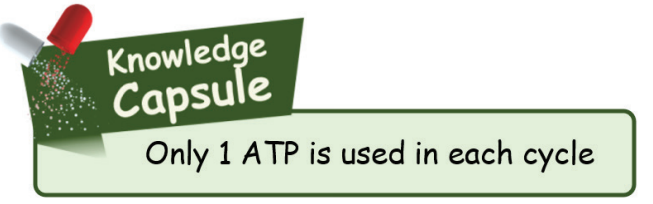
Muscle relaxation is by removal of sarcoplasmic Ca^{2+}

- Removed by .....
- SERCA pumps ..... ions into the sarcoplasmic reticulum for each ATP hydrolysed
- Muscle relaxation is an .....
- Inhibition of SERCA → ..... duration of muscle contraction because calcium is available for a longer duration

Active space

ACTIN-MYOSIN CROSS-BRIDGE CYCLING

When the muscle is at rest, myosin head is in high energy high-affinity state



Sequence of steps[Q]:

1. Power stroke
2. _____
3. _____
4. _____
5. _____

When myosin heads bind with actin → The heads pull actin towards the centre → Repeated actin

myosin cross-bridges → Causes actin to slide over myosin → 2 Z-lines come closer → Length of

sarcomere decreases → Hence, length of muscle decreases (muscle contraction)

Active space

TYPES OF MUSCLE FIBRES

Type - I	Type - II
✦ S → slow, small diameter ✦ O → oxidative ✦ R → Red fibre ✦ R ✦ Y	✦ Fast, large diameter ✦ Anaerobic/glycolytic metabolism ✦ White fibre
✦	✦
✦	✦
✦	✦
✦	✦
✦	✦
✦	✦

- All muscles have both Type I and Type II muscle fibres, depending on the activity of muscle Type I or Type II may predominate

E.g. all posture-maintaining muscles ® contract throughout the day

—

Type I fibres will predominate

Size principle

- During a graded activity, the ..... fibre will be recruited first followed by.....
- Type II are reserve muscle fibres here

Active space

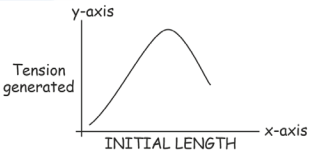
Motor unit

- .....
- Only one type of muscle fibre is present in the motor unit
- Muscles with finer control have ..... motor units
- Muscles which do not need a fine control have ..... motor unit
  - ❑ Extra ocular m/s fibre: Fewer muscle fibres/motor unit (5 – 6 m/s fibres/motor unit)
  - ❑ Muscles of back: 600 muscle fibre/motor unit

Isotonic	Isometric

LENGTH TENSION RELATIONSHIP

Frank Starling's Law



- Applicable to ..... muscle contractio
- It is the relationship between
  - ❑ Initial length of the muscle (length of muscle before it begins to contract) on the X-axis
  - ❑ Tension generated during isometric muscle
  - ❑ Contraction on the y-axis

Active space

Length tension relationship:

- .....
  .....
  .....
  .....
- Applicable to skeletal and cardiac muscle
- In Cardiac muscle:

Optimum length:

- It corresponds to a sarcomere length of .....
- It is also called the .....
- At optimum length, there is maximum overlap between actin and myosin
- At optimum length, there are maximum number of actin and myosin cross-bridge

CARDIAC MUSCLE

- Striated, involuntary, extensively branched
- Has intercalated Discs:
  - ✦
  - ✦

Active space

- Has GAP Junctions:
  - ❑ At .....
  - ❑ Made of proteins called .....
  - ❑ Gap junctions make the heart a .....

Differences between skeletal and cardiac muscle:

Structure	Skeletal muscle	Cardiac muscle
T- tubule		
Sarcoplasmic Reticulum(SR)		
Source of calcium		
Dihydropyridine receptor		
Relaxation by		
Phospholamban		

(Phospholamban → Inhibitory protein which inhibits SERCA)

Types of Cells in the Heart:

1.
2.

Active space

The Conducting System of the heart has 6 components:

1. SA Node
2. Internodal tracts
   - ❑ Anterior
   - ❑ Middle
   - ❑ Posterior
3. AV Node (Smaller diameter fibres with fewer gap junctions)
4. Bundle of His
5. Bundle branches
   - ❑ Right
   - ❑ Left
     - o Anterior Fascicle
     - o Posterior Fascicle
6. Purkinje Fibers

Active space

Workbook Sheet

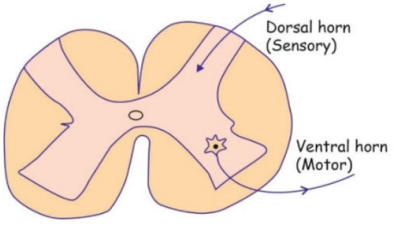
MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

INTRODUCTION TO CENTRAL NERVOUS SYSTEM

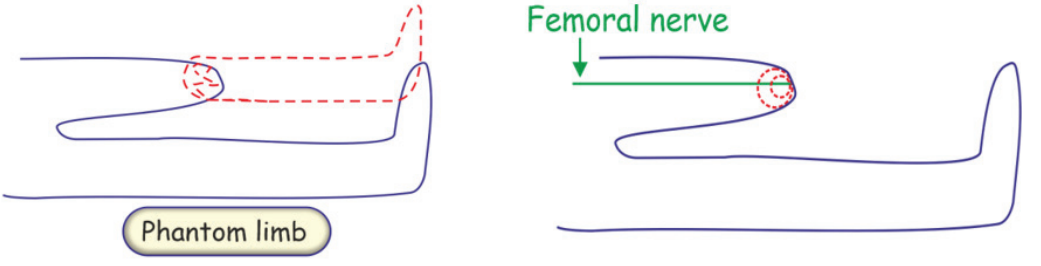
LAWS OF SENSORY PHYSIOLOGY

1. Bell-Magendie Law



2. Law of projection [Phantom Limb]

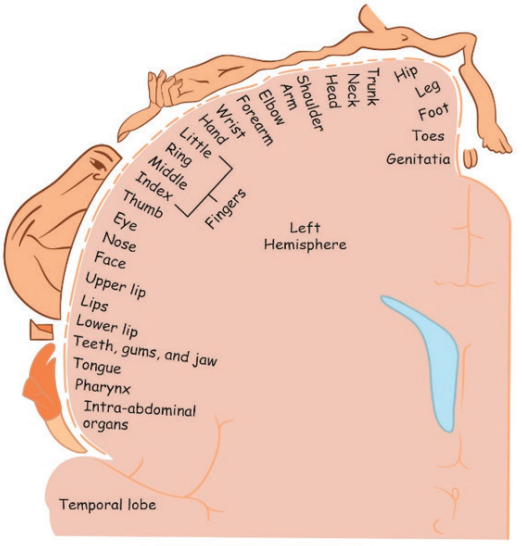
Example: Cut end of the nerve is stimulated by the release of inflammatory chemicals in stump → patient complains of pain from the non-existing foot (phantom limb)



Active space

Recent advances suggest that the phantom limb is due to the phenomenon of cortical plasticity.

Cortical plasticity: Means the cortex and thalamus have the capacity to form new synapses.



Sensory homunculus

E.g., Following a below-knee amputation, neurons of the neighbouring areas (hip, trunk) in the sensory cortex project to form new synapses with leg and foot areas. If a sensory stimulus is given on the hip or trunk areas, the patient perceives it as though the sensation is arising from the non-existent foot.

3. WEBER-FECHNER Law:
   - ✦ Sensation felt _____
   - ✦ Sensation felt = _____

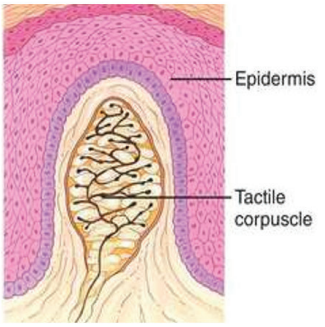
Active space

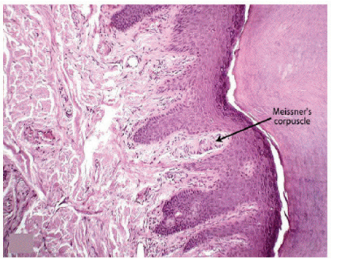
4. STEVEN'S POWER Law:
   - ✦ Sensation felt _____
   - ✦ Sensation felt = _____

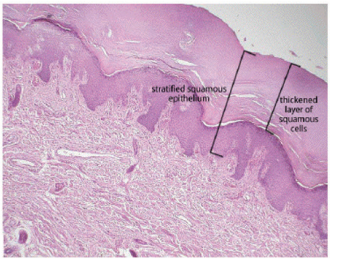
SENSORY SYSTEM- RECEPTORS AND ASCENDING TRACTS

RECEPTOR

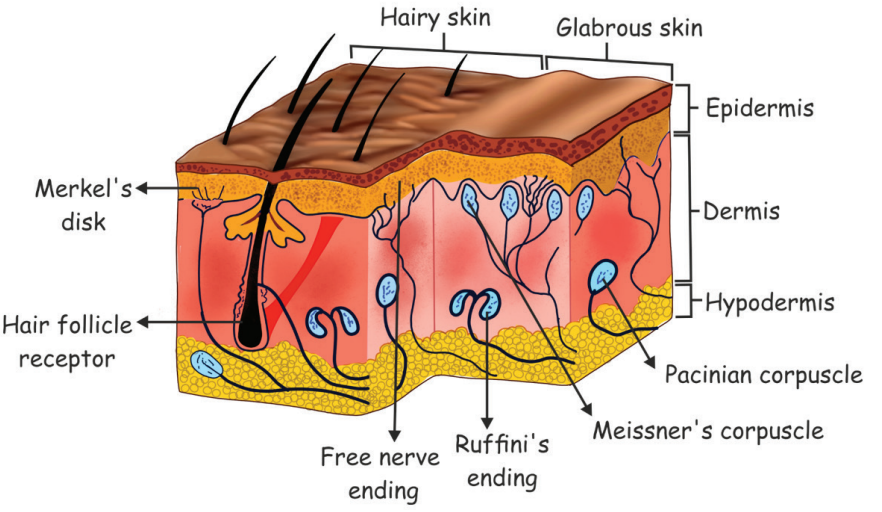
- Receptor = biological transducer
- Subthreshold stimulus: Generates .....
- Threshold stimulus: Generates .....







Distribution of receptors:



Active space

MUSCLE SPINDLE

- Receptor for muscle length and stretch
- Extrafusal (outside muscle spindle) muscle fibres are responsible for muscle contraction/movement; motor supply to extrafusal muscle fibres is Aα
- Intrafusal muscle fibres are present within the muscle spindle
- Intrafusal muscle fibers are of two types:- .....
  .....
- Sensory / afferent fibers from muscle spindle are :- .....
  .....
- Motor/ efferent fibers to ends of intrafusal muscle fibers are .....
  .....
- Only receptor with both sensory and motor fibers

ASCENDING OR SENSORY TRACTS

Divided into 3 types:

1. Dorsal column tract/Posterior column tract/dorsal column medial lemniscal system (DCML)
2. Spinothalamic tracts
3. Spinocerebellar tracts

Active space

DORSAL COLUMN TRACT/DORSAL

COLUMN MEDIAL LEMNISCAL SYSTEM (DCML)

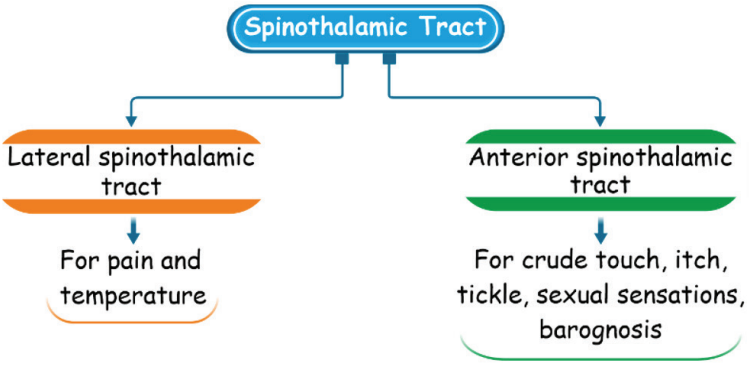
For

-
-
-
-
-
-
-
-
-

Knowledge Capsule

Tract specific for stereognosis[Q] is fasciculus cuneatus)

SPINOTHALAMIC TRACT (STT)



SPINOCEREBELLAR TRACT

- For unconscious proprioception
- From the lower limb, there are 2 tracts –
  - A. Dorsal spinocerebellar tract:
  - B. Ventral spinocerebellar tract:

Active space

From upper limb 2 tracts –

A. Cuneo cerebellar tract

B. Rostral spinocerebellar tract

All spinocerebellar tract terminate in the ipsilateral cerebellar hemisphere.

Dorsal column-medial lemniscal pathway and Spinothalamic tract

Active space

	DCML	Spinothalamic tracts
First order sensory neuron		
Second order sensory neuron		
Third order sensory neuron		
First synapse		
Second synapse		
Level of crossing		
Arrangement of fibres from medial to lateral		

Arrangement of fibres

In the posterior columns, the arrangement of fibres from medial to lateral- sacral, lumbar, thoracic, and cervical (fibres from the lower half of the body are pushed medially).

In the STT, fibres from the lower half of the body are placed laterally, from lateral to medial-sacral, lumbar, thoracic, and cervical.

Active space

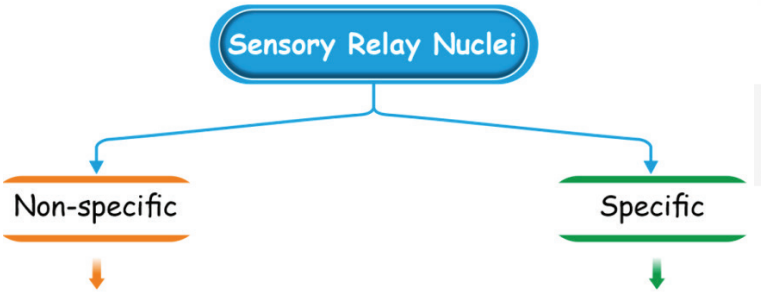
CLINICAL

- Spinal Tumours are broadly of two types- medullary and extramedullary

- Medullary spinal tumour- h/o descending loss of pain and temperature
- Extramedullary spinal tumour- h/o ascending sensory loss

THALAMUS

Thalamus is k/a ' ..... '- all sensations relay in the thalamus



MOTOR NUCLEI

- _____
- _____

Anterior thalamic nuclei form a part of the Papez circuit which plays an very important role in emotions, learning & memory.

Active space

Functions of Papez Circuit:

- Conversion of short-term memory into long-term memory (Hippocampus)
- Learning, especially skills meant for survival
- Emotions

REFLEXES

INTRODUCTION & REFLEX ARC

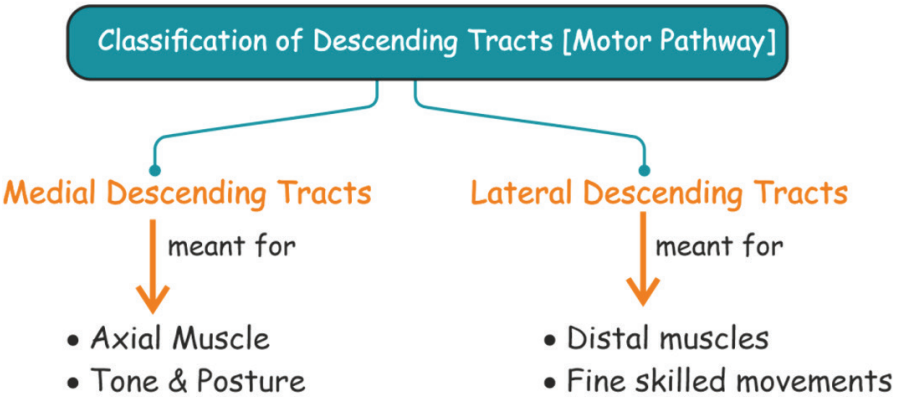
- Reflex is defined as an automatic and reproducible response to a stimulus

	Stretch Reflex	Inverse Stretch Reflex
Also Known as		
Receptor		
Afferent/Sensory		
Centre		
Efferent/Motor		
Response		
No. of synapses		

Active space

MOTOR SYSTEM- DESCENDING TRACTS

CLASSIFICATION OF DESCENDING TRACTS (MOTOR PATHWAYS)

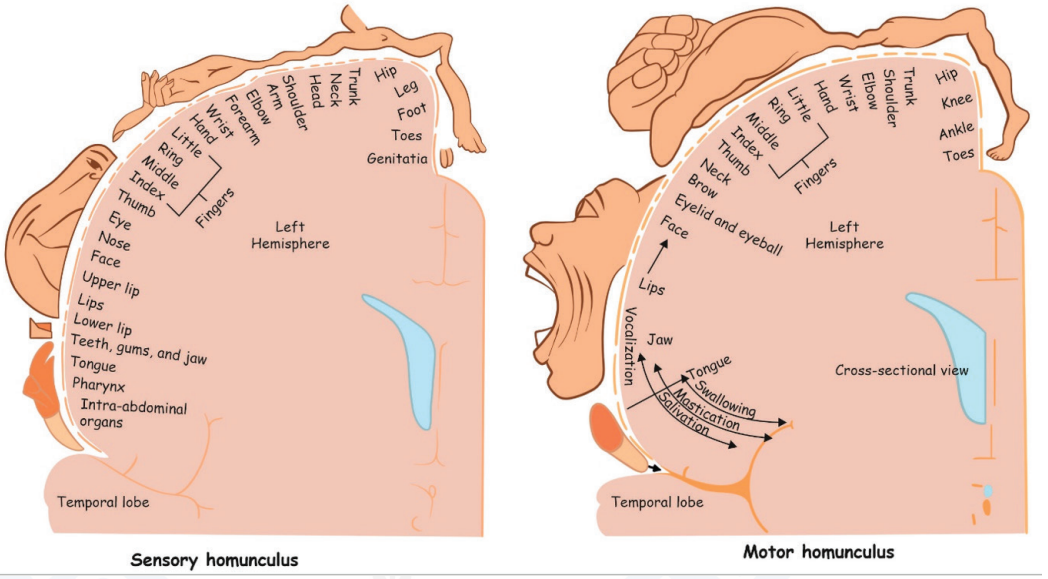


CORTICOSPINAL TRACT

Origin:-

- Primary motor cortex
- Primary motor cortex contributes to ..... of fibres
- Primary motor cortex is located in the "..... gyrus"
- Also known as Area .....
- Motor Homunculus
- More complicated movements have larger cortical representations, such as:
  - Muscles of _____
  - Muscles of _____
  - Muscles of _____

Active space



2. Pre-motor and supplemental motor area

   This is "Area-_____"

   Contributes _____of fibres

3. Primary Sensory Cortex and sensory association areas

   Areas 3, 1, 2

   Contribute_____ of fibres

   Sensory Association Areas

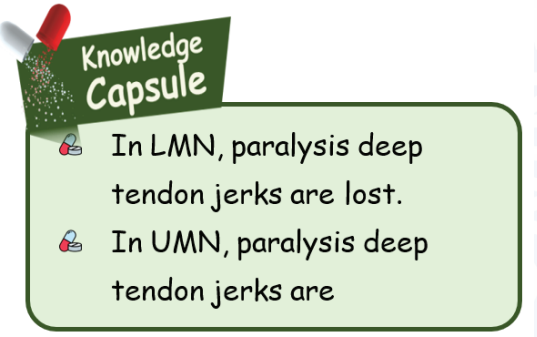
   - Located in the parietal lobe

Active space

MIST

DESCENDING TRACTS

- ..... corticospinal tracts are "Crossed tracts", and ..... corticospinal tracts are uncrossed tracts



- Commonest lesion in the CNS involves the Middle Cerebral Artery (MCA) [there could be a thrombus, embolus or haemorrhage of MCA]

- The lesion in the MCA will affect the supply of internal capsules, so the patients present with:
  - ✦ Dense hemiplegia of the opposite side
  - ✦ Loss of fine skilled movement of the opposite side

Active space

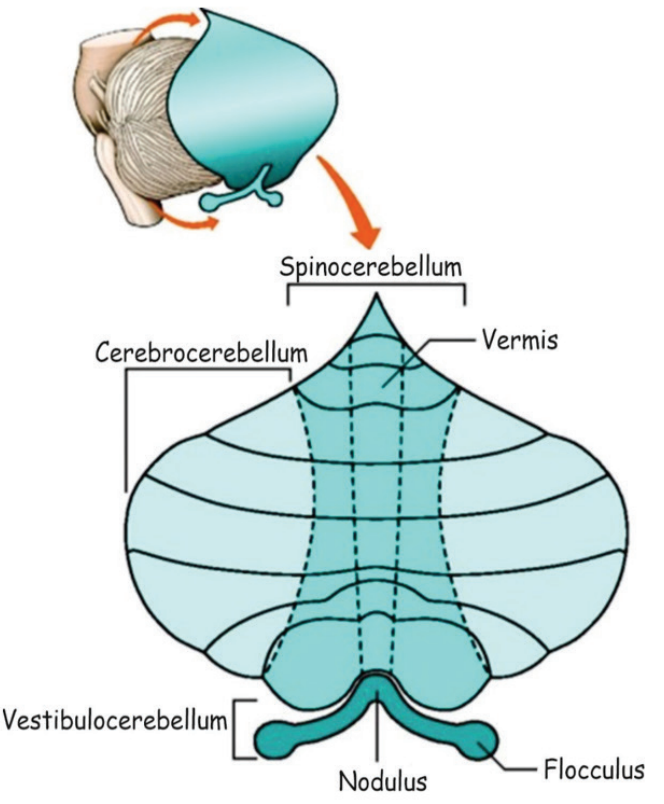
BROWN SEQUARD SYNDROME

- Hemisection of the spinal cord:

1. Mnemonic: BS TOP – Brow Sequard Temperature, Pain of Opposite side is lost (below level of lesion)
2. ..... is the least affected
3. All other sensations are lost on ..... (below the level of the lesion) (Posterior Column Sensations)
4. Motor paralysis on the ..... side, i.e., ... corticospinal signs

CEREBELLUM, BASAL GANGLIA

CEREBELLUM



Active space

- Has 3 parts
  - ✦ Vestibulocerebellum:
    - ➨ Known as flocculonodular lobe
    - ➨ For .....
- Spinocerebellum:
  - ✦ For .....
  - ✦ Medial part (k/a medial lobe) → Control axial muscles; the deep nucleus associated with the medial lobe is .....
  - ✦ Lateral part (k/a intermediate lobe) → Control distal muscles; deep nuclei associated with intermediate lobe- ..... and ..... (together called as "Nucleus Interpositus")
- Cerebrocerebellar:
  - ✦ K/a lateral lobe
  - ✦ For .....
  - ✦ Nucleus associated is .....
  - ✦ Dentate Nucleus is the largest nucleus in the cerebellum
- 4 pairs of deep Nuclei

  _____, _____,

  _____, _____

Knowledge Capsule

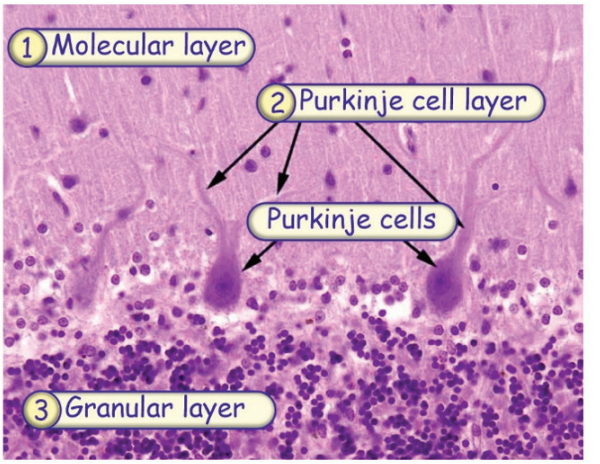
Mnemonic: for arrangement of cerebellar nuclei from lateral to medial - DOCTORS EAT GOOD FOOD
Dentate, Emboliformis, Globose, Fastigeal

- 5 Cells

Active space

Layers of cerebellum

1. Molecular layer
   - Stellate cells
   - Basket cells
2. Purkinje cell layer
   - Purkinje cells
3. Granular layer
   - Granule cells
   - Golgi cells



BASAL GANGLIA

Functions of basal Ganglia:

- Planning and programming of movement
- Involved in the process which converts an abstract thought into a voluntary movement

- Control and coordinate complex patterns of movements, automatic, skilled movements and associated movements

  E.g. swinging of arms while walking, facial expressions while talking

Active space

- In association with the cerebral cortex, basal ganglia coordinate timing (how rapidly a movement has to be performed) and scaling (how large a movement will be) of movements.

Knowledge Capsule

Cerebellum controls and coordinates deliberate, voluntary movements. It controls the rate, range, force and direction of movement.

Parts of Basal Ganglia

Part	Neurotransmitter	Lesion causes
1. Caudate nucleus		
2. Putamen		
3. Globus Pallidus ✦ External segment ✦ Internal segment		
4. Substantia nigra ✦ SN pars compacta ✦ SN pars reticularis		
5. Subthalamic nucleus of Lewy's		

HYPOTHALAMUS

- Hypothalamus is:
  - ✦ Vegetative brain
  - ✦ Controls ANS

Active space

- Functions of hypothalamus:

T - T .....

E - E .....

N - N .....

H - H .....

A - A .....

R - R .....

R .....

T - T .....

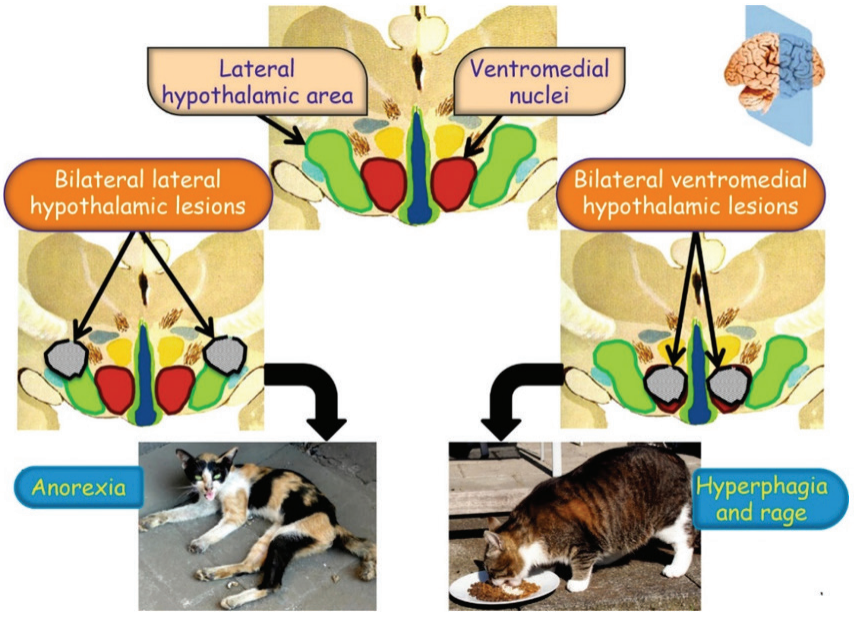
S - S .....

S .....

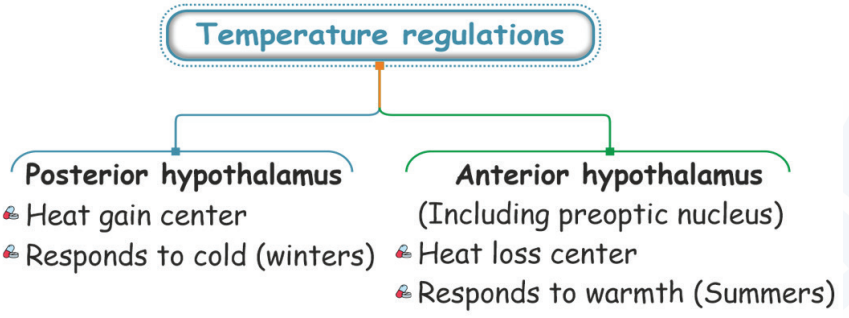
Circadian rhythm → .....

Function	Hypothalamic nucleus
Hunger, thirst, rage, aggression	
Satiety, reward	
Head gain centre	
Heat loss centre	
Circadian rhythm	
Sexual activity	
Neurosecretion	

Active space



Temperature regulations

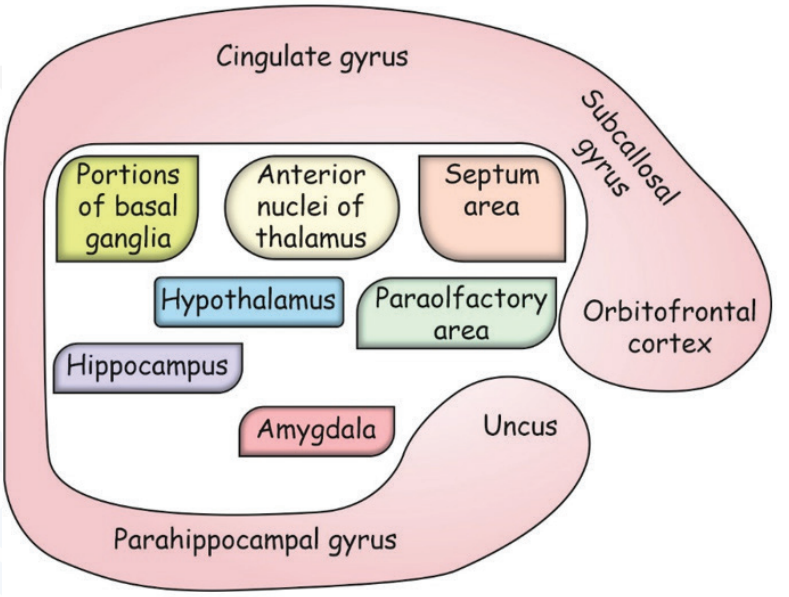


Posterior Hypothalamus (heat gain centre):

Active space

Anterior Hypothalamus

LIMBIC SYSTEM



- The word 'limbus' means 'border'
- The term limbic system includes the entire neuronal circuit that controls emotional behaviour and motivational drives
- Middle of all the structures of the limbic system is the hypothalamus, which from a physiological point of view, is one of the central elements of the limbic system

Active space

Kluver Bucy Syndrome

1. Is not afraid of anything
2. Has extreme curiosity about everything
3. Forgets rapidly
4. Increased oral tendency
5. Hypersexuality
6. Decreased emotional reactions

HIGHER FUNCTIONS- DOMINANT AND NON- DOMINANT HEMISPHERES, MEMORY, SPEECH, EEG

DOMINANT HEMISPHERE

- Also called as ..... Hemisphere. This is responsible for sequential-analytical processes.
- In about 95% of all people, the left hemisphere is dominant
- Wernicke's & Broca's areas are more developed on the left side
- Language, verbal symbolism, ability to read, ability to perform mathematical calculations, and ability to think through logical problems

Active space

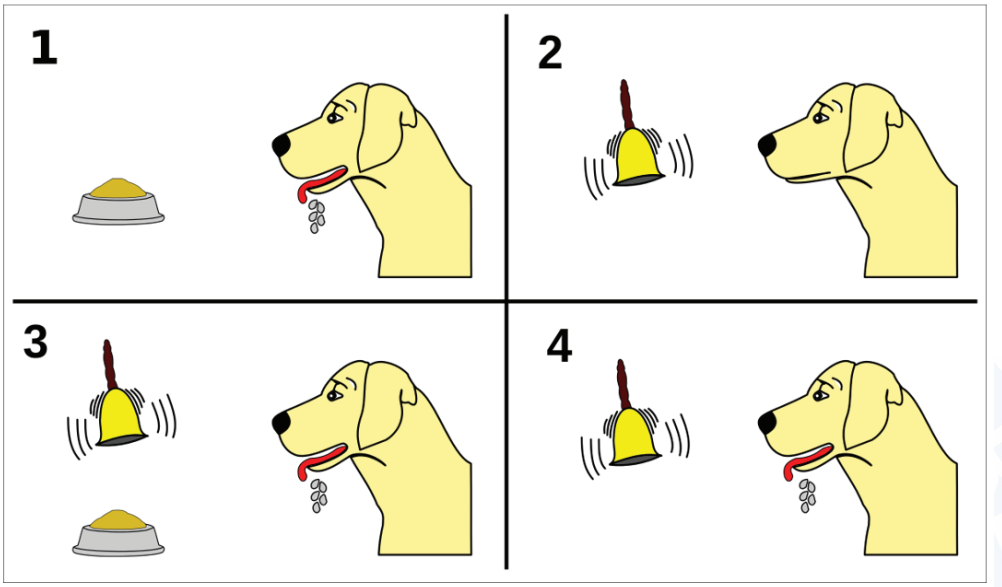
NON-DOMINANT HEMISPHERE

- Also called as ..... / representational hemisphere
- This hemisphere is important for understanding and interpreting music, non-verbal visual experiences (especially visual patterns), identifications of objects by their form, facial recognition, spatial relations between the person & surroundings, and somatic experiences related to the use of limbs and hands (dance)
- Right inferior temporal lobe – recognition of faces (lesion of right inferior temporal lobe causes inability to recognize faces k/a ....)

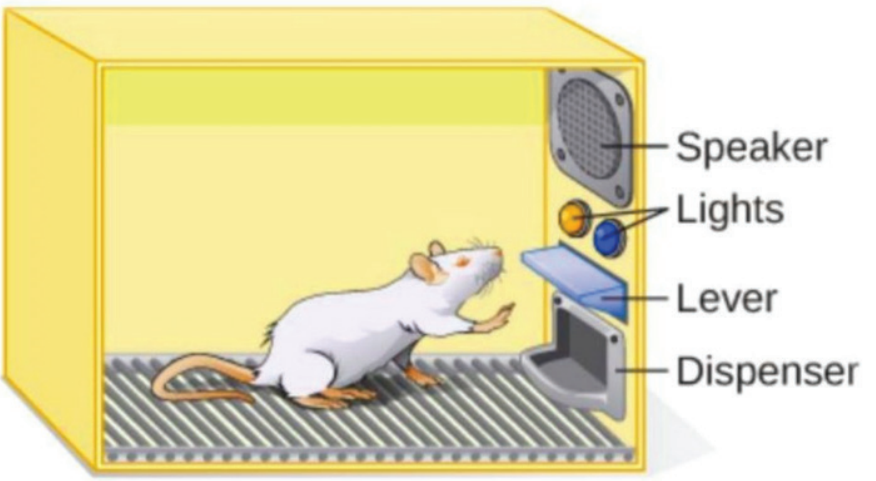
MEMORY

Active space

Classical Conditioning



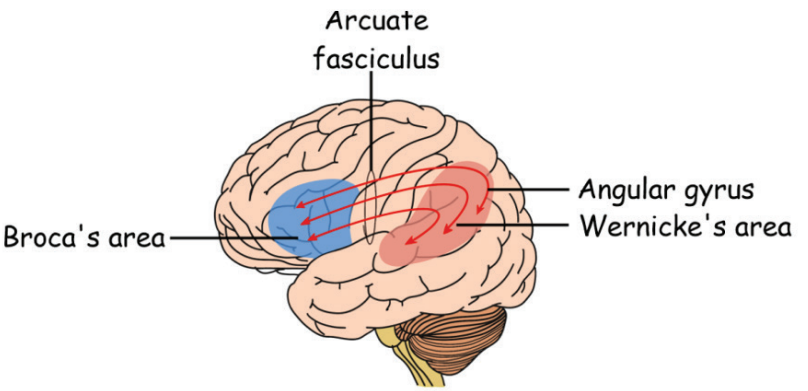
Operant Conditioning



Active space

MEMORY

Broca's Area	Wernicke's Area
Lesion:	Lesion: .....



- Angular gyrus: Processing of visual information and written word
  - ✦ Lesions: .....

ELECTROENCEPHALOGRAPHY (EEG)

Delta Waves

D	T	A	B
1-3 Hz	4-7 Hz	8-12 Hz	13-30 Hz

→

↑ Frequency ; ↓ Amplitude

Active space

Theta waves

Alpha waves

Beta waves

PHYSIOLOGY OF SLEEP

Ascending Reticular Activating System:

- Reticular formation is a loose network of neurons which is present in the brainstem
- Reticular formation includes the following:
  - ✦ Raphe Magnus Nucleus (secretes serotonin)
  - ✦ Locus Ceruleus (secretes norepinephrine)
- Fibres from reticular formation project to the cortex, especially from Raphe Magnus Nucleus and Locus Ceruleus
- Activity of reticular formation is responsible for the β-wave of EEG

Active space

Types of sleep

Nrem sleep	Rem sleep

Active space

Sleep Cycle & Eeg In Sleep

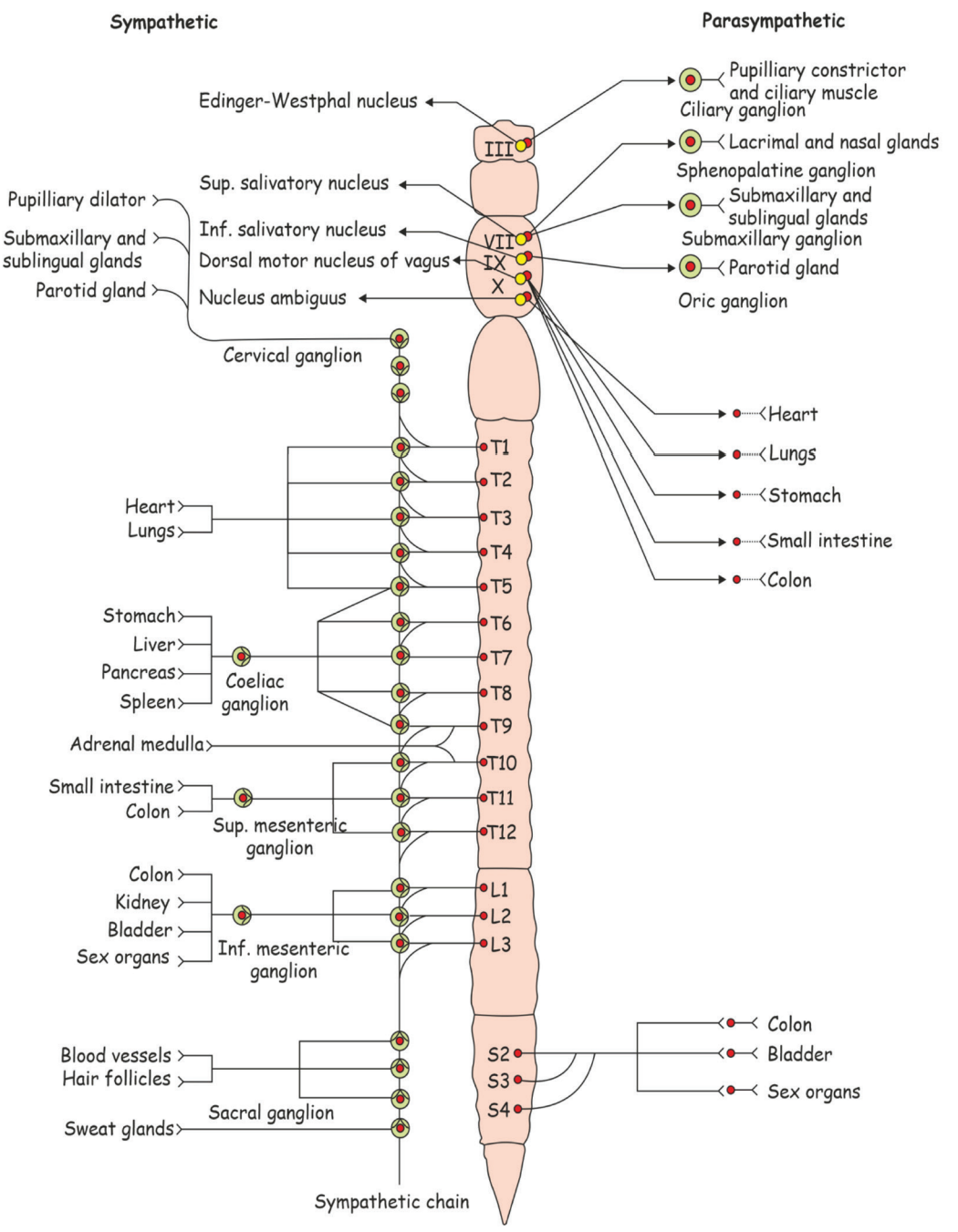
- At rest, awake, eyes closed, the mind is wandering means we decided to go to sleep – a wave (8-12 Hz).

AUTONOMIC NERVOUS SYSTEM

CONTROL OF ANS BY HYPOTHALAMUS

- Parasympathetic is Cranio-sacral outflow
- The cranial nerves which carry parasympathetic are **III, VII, IX, and X (3, 7, 9, 10);** sacral segments are involved **S2, S3, S4**

Active space



- Sympathetic is thoracolumbar outflow (T1–L3)
- Sympathetic is excitatory to CVS
- Parasympathetic is excitatory to GIT

MIST

Active space

Sympathetic vs Parasympathetic Nervous System Master table

Effector Organs	Parasympathetic Nervous System	Sympathetic Nervous System Receptor type	Sympathetic Nervous System Response
Eyes Radial muscle of iris Sphincter muscle of iris Ciliary muscle	- Contraction (miosis) Contraction of near vision	α_1	Contraction(mydriasis) - -
Heart SA node Atria & ventricle AV node & Purkinje	Decreased heart rate Decreased arterial contractility Decreased conduction velocity	β_1 β_1 β_2 β_1	Increased heart rate Increased contractility Increased conduction velocity
Arterioles Skin, splanchnic vessels Skeletal muscle	- -	α_1 α_1/β_2 M	Constriction Constriction/Dilation
Systemic veins	-	α_1, α_2/β_2	Constriction/Dilation
Bronchial smooth muscle	Contraction	β_2	Relaxation
Stomach & Intestine Motility and tone Sphincters Secretion	Increased Relaxation Stimulation	α_1, α_2, β_2 α_1 -	Decreased Contraction
Gall bladder	Contraction	β_2	Relaxation
Urinary bladder Detrusor Sphincter	Contraction Relaxation	β_1 α_1	Relaxation Contraction
Uterus (Pregnant)	-	α_1/β_2	Contraction/Relaxation
Male sex organs	Erection	α_1	Ejaculation
Skin Pilomotor muscles Sweat glands	- -	α_1 M	Contraction Secretion
Liver	-	α_1, β_2	Glycogenolysis
Pancreas Acini Islet cells	Increased secretion -	α α/β_2	Decreased secretion Decreased/Increased secretion
Salivary glands	Profuse, watery secretion	α_1/β	Thick, viscous secretion/Amylase secretion
Lacrimal glands	Secretion		-
Adipose tissue	-	β_2	Lipolysis

Active space

RENAL

GLOMERULAR FUNCTION & GFR

KIDNEY

3 main functions that occur in the kidney are:

- ✦ Secretion
- ✦ Filtration
- ✦ Reabsorption

- Excretion = _____ - _____ + _____

GLOMERULAR FILTRATION MEMBRANE

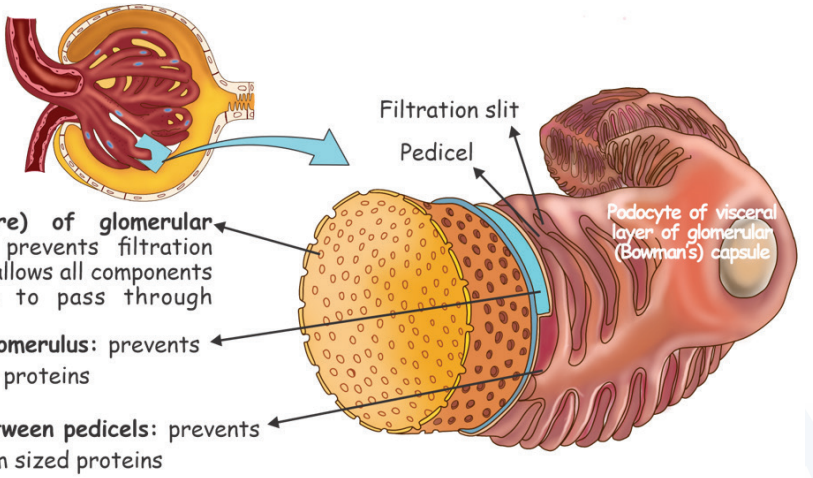
Has 3 layers

1. Glomerular Capillary Endothelium (Fenestrated)

2. Basement membrane (BM):

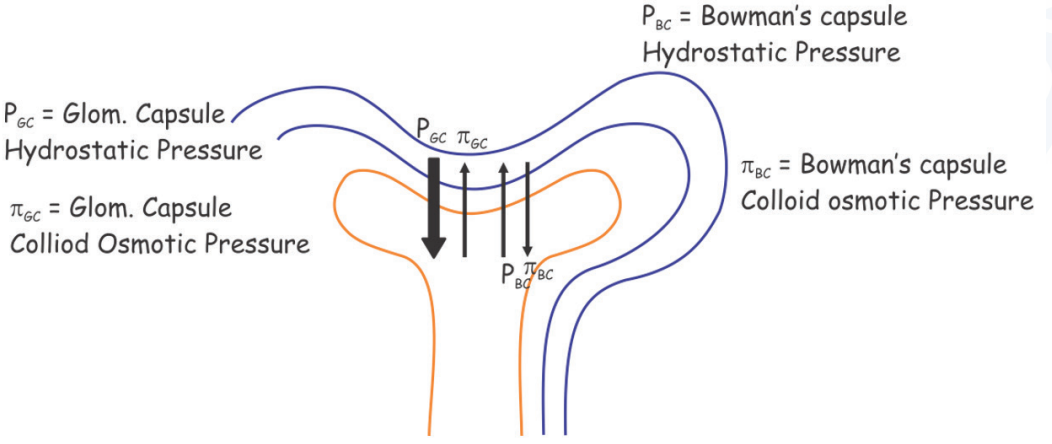
Active space

3. Visceral layer of Bowman's capsule (modified to form podocytes)



- **Fenestration (pore) of glomerular endothelial cell:** prevents filtration of blood cells but allows all components of blood plasma to pass through
- **Basal lamina of glomerulus:** prevents filtration of larger proteins
- **Slit membrane between pedicels:** prevents filtration of medium sized proteins

GLOMERULAR FILTRATION MEMBRANE



Active space

Pressures favouring filtration	Pressures opposing filtration

Net filtration pressure: Net filtration pressure =

=

FACTORS AFFECTING GFR

1.
2.
3.
4.
5.
6.

Clinical Situation

- Ureteric stone →

Active space

- Renal artery stenosis →

- Increased Sympathetic discharge →

(i)

(ii)

(iii)

Filtration Fraction

- Renal Blood flow (RBF) =
- Renal plasma flow (RPF) =
- Glomerular filtrate = ultrafiltrate of plasma
- Filtration fraction =

Effect of Angiotensin-II on GFR

In low concentration	In high concentration

Active space

Clinical correlation

- In patients with bilateral renal artery stenosis

,

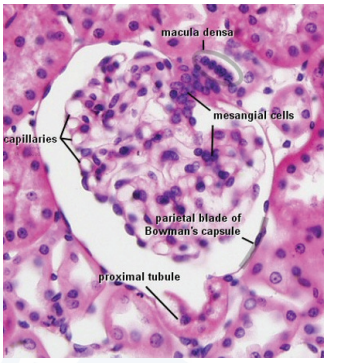
Autoregulation of GFR

- Between MAP (Mean Arterial Pressure) of 90 and 200 mmHg, GFR is constant

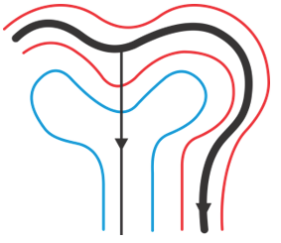
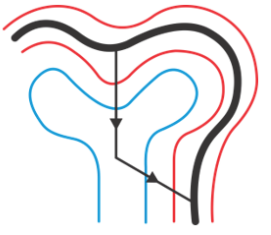
GFR is autoregulated by 3 mechanisms:		
Myogenic mechanism of autoregulator	TGF – Tubuloglomerular feedback	GTB – Glomerular tubular balance

Active space

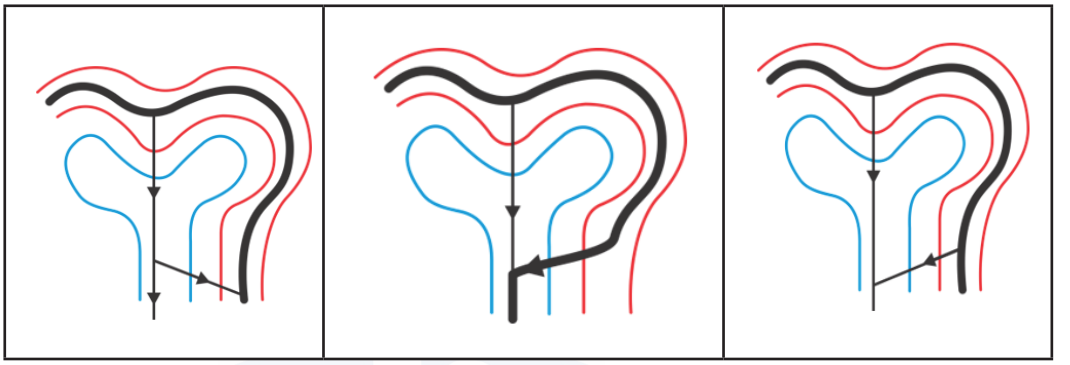
JG Apparatus



RENAL HANDLING OF DIFFERENT SUBSTANCES

1st Category	2nd Category	3rd Category
Not filtered at all	Freely filtered but not reabsorbed and not secreted	Freely filtered, completely reabsorbed
		
1st Category	**5th Category**	**6th category**
Freely filtered, partly reabsorbed	Freely filtered, completely secreted	Freely filtered, partly secreted

Active space

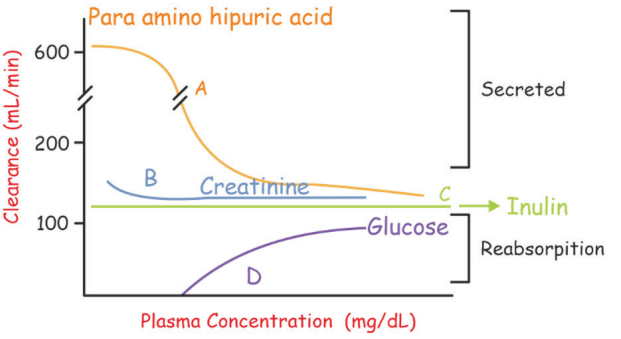


CLEARANCE

Clearance

- It is the volume of plasma cleared of that substance or free of that substance in unit time
- Unit is .....
- Clx =

- Clearance of inulin =
- Clearance of Para amino hippuric acid (low concentration) =
- Clearance of substances which are reabsorbed is .... clearance of Inulin
- Clearance of substances which are secreted is ..... clearance of Inulin



Active space

Creatinine Clearance

- Also used for the estimation of GFR
- Creatinine is an endogenous substance

$$CL_{CR} = \frac{U_{CR} \times V}{PL_{CR}}$$

- Gold standard for measuring GFR is _____

TUBULAR HANDLING OF DIFFERENT SUBSTANCES

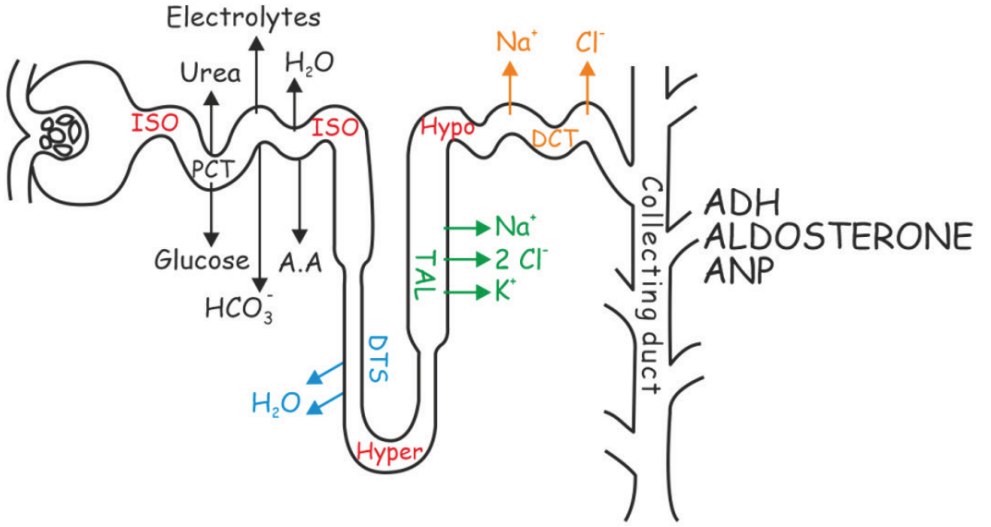
GLOMERULAR FILTRATE

- Glomerular filtrate is isotonic with plasma

Site	Reabsorption of	Fluid
PCT (Proximal Convoluted Tubule)		✦ Isotonic
Descending Limb of Henle		Hypertonic

Active space

Thick Ascending Limb		✦ Hypotonic (Also known as a diluting segment of nephron)
DCT (Distal Convoluted Tubule)		✦ Hypotonic
Collecting duct		✦ Hormone regulated segment

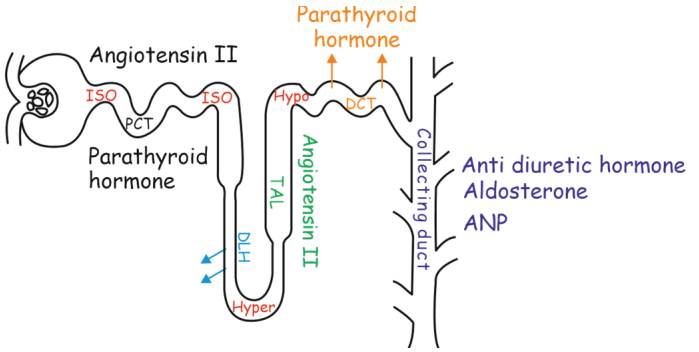


DTS - Descending thin segment

Active space

HORMONES AND THEIR ACTION ON DIFFERENT PARTS OF NEPHRONS

Site	Hormone	Action
PCT	✦ Angiotensin II ✦ PTH (Parathyroid hormone)	
TAL (Thick Ascending Limb)	✦ Angiotensin II	
DCT	✦ PTH	
Collecting Duct (CD)	✦ ADH (Anti-diuretic Hormone) ✦ Aldosterone ✦ ANP	



RENAL HANDLING OF SODIUM

✦ Sodium is reabsorbed in all parts of nephrons except .....
.....

✦ Filtered load of sodium = GFR × Plasma concentration of sodium =
.....

Active space

1. Sodium reabsorption in PCT (60% of filtered sodium is reabsorbed in PCT):

- ✦ Secondary active co-transport:

- ✦ Secondary active counter-transport:

2. Sodium reabsorption at TAL (30% of filtered sodium is reabsorbed in TAL):

- ✦ Secondary active transport:

- ✦ Barterr syndrome is defective Na+, K+, Cl- reabsorption in TAL. Symptoms similar to Lasix overuse.

3. Sodium reabsorption at DCT

- ✦ Secondary active co-transport

Active space

✦ Gitelmann syndrome due to loss-of-function mutation of the gene for ..... symptoms are similar to ....

4. Sodium reabsorption at CD

P cell (principal)	I Cell (Intercalated)

✦ Diuretic, which inhibits aldosterone → .....

✦ Diuretic, which inhibits epithelial sodium channel → ....

Active space

RENAL HANDLING OF POTASSIUM

- Filtered load of potassium = ...
- K+ (Potassium): Both reabsorbed and secreted

Reabsorption of K+ at		Secretion of K+ at
PCT	TAL	CD

TUBULAR HANDLING OF GLUCOSE

- Site – .....
- By .....
- Amount reabsorbed .....

90% of glucose reabsorption occurs in _____by _____and _____

Active space

Renal threshold

-
- Normal value- _____

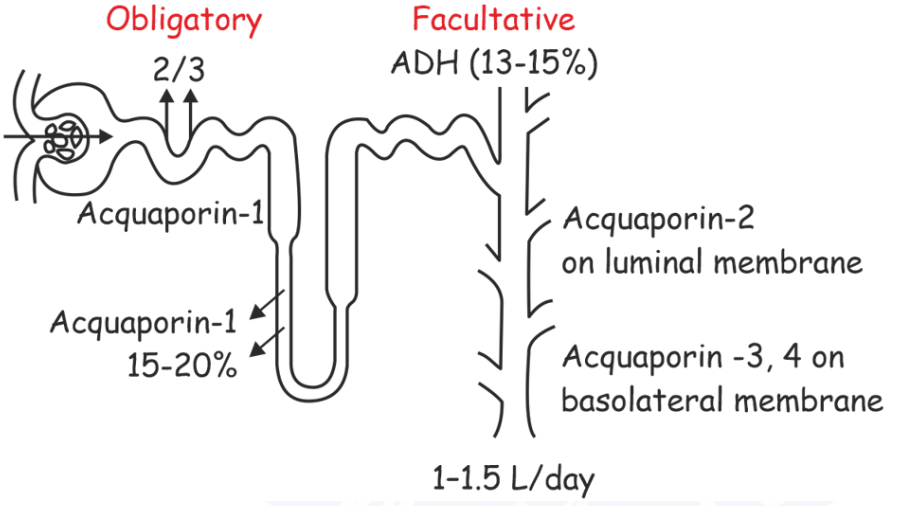
TmG

-
- In males → _____

  In females → _____

TUBULAR HANDLING OF WATER

- GFR = 180 L/day
- Normal urine production → 1-1.5 l/day
- 2/3rd of water is reabsorbed in PCT, known as .....
- Water reabsorption in the CD is controlled by ADH and is k/a .....



TUBULAR HANDLING OF PAHA (PARA AMINOHIPPURIC ACID)

- Secretion of PAHA occurs in .....
- At low plasma concentration (PAHA < 20mg/dL), PAHA is .... secreted but at high concentration (PAHA > 20mg/dL) PAHA is ..... secreted.
- Transport maxima (TM) of PAHA → 80 mg/min
- Cl PAHA (Low conc.) = ...

Active space

- PAHA is secreted by facilitated diffusion, which is carrier mediated
- Increase in plasma concentration of PAHA will cause saturation of carrier protein resulting in incomplete secretion of PAHA, and therefore, decreased clearance of PAHA

TUBULAR HANDLING OF H+

- Since there is no free H+ in plasma, filtration of H+ ions does not happen
- Urine is acidic due to the secretion of H+
- Maximum secretion of H+ occurs in .....
- Urine acidification occurs in .....
- **H+ secretion in PCT**
- For every H+ secretion at PCT, one HCO3 is reabsorbed

Reabsorption of HCO3-:

- HCO3- is freely filtered
- If plasma HCO3- <24 mmol/L, all the filtered HCO3- is reabsorbed
- 80% happens in PCT
- 20% at TAL, DCT by the same mechanism as in PCT

H+ secretion in CD

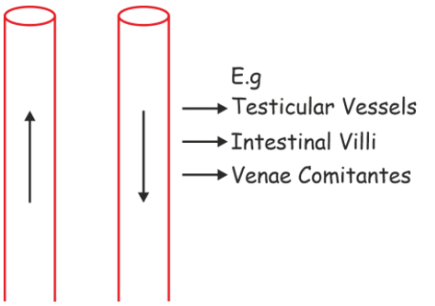
Active space

	PCT	CD
H+ secretion/day		
Carbonic anhydrase		
Mechanism of H+ secretion		
Buffers		
Acidification of urine		

Active space

COUNTERCURRENT MECHANISMS IN KIDNEY, MICTURITION REFLEX

Counter current mechanism in Kidney

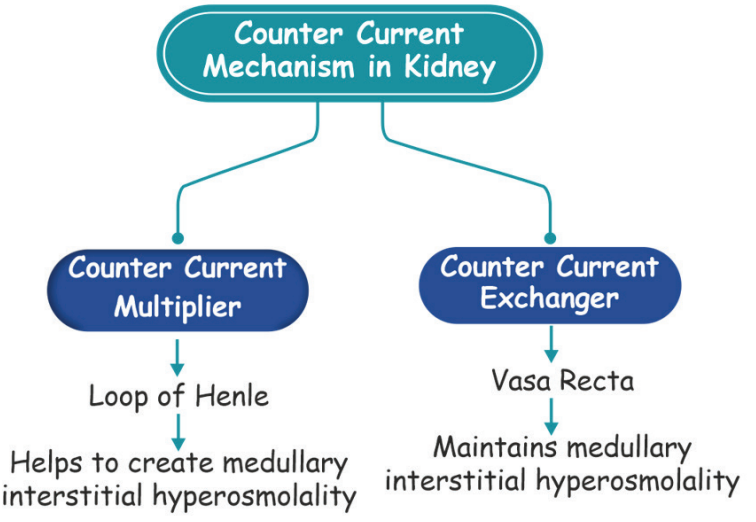


Counter current:

Flow in opposite directions in two closely located vessels

E.g.:

- Testicular vessels
- Intestinal villi
- Alveoli
- Venae comitantes (associated with certain arteries)



Counter current multiplier

- The two limbs of loop of Henle have **differential permeability**
- From descending thin segment, there is the absorption of only water. From the thick ascending limb, there is absorption of only solutes **(Na+K+2Cl-), but** the absorption of solutes from thick ascending limb is more than the absorption of water from descending thin segment

Active space

- Fluid that enters the loop of Henle is **isotonic**, but as it moves down after the absorption of water from descending limb, it becomes **hypertonic**, and as it goes above the ascending limb and absorption of solutes causes the tubular fluid becomes **hypotonic**

- Interstitial fluid around the loop of Henle becomes concentrated as more solutes than water enter the medullary interstitium

- Collecting duct passes through the cortex, outer medulla, and inner medulla

- The collecting duct is surrounded by cortical interstitial fluid in the cortex and medullary interstitial fluid in the medulla

- Cortical interstitium is isotonic, but the medullary interstitium is hypertonic; deep in the medulla, the osmolality is nearly 1200mOsm/L

- To produce concentrated urine, 2 things are required- ADH and concentrated/hypertonic medullary interstitium

- ADH increases the insertion of water channels (Aquaporin 2) in the luminal membrane

- Water will only move if the interstitial fluid surrounding the collecting duct is concentrated (therefore, the need for a concentrated medullary interstitium)

- Max urine osmolality which can be achieved is .....

- Normal range of urine osmolality is .....

- Possible range of urine osmolality is .....

Active space

MIST

Solutes responsible for hyperosmolal medullary interstitium

- Na+, K+, Cl- → From thick ascending limb
- Urea from the medullary collecting duct

Urea cycling

- Urea is a freely filtered substance
- In PCT, 50% of urea will be reabsorbed (by facilitated diffusion)
- **UTA1, UTA3** → Urea Transporters make the medullary collecting duct permeable to urea

- From the medullary CD, urea enters into the medullary interstitium increasing the medullary interstitial osmolality. Urea re-enters the tubule from the descending thin segment. This is k/a urea cycling.
- Advantage of Urea cycling is it increases medullary interstitial hyperosmolality

Active space

Permeable to urea	Impermeable to Urea

Counter current exchanger

✦ Counter current exchanger is vasa recta

Vasa recta

- Only supply of blood and oxygen to the medulla
- 2 limbs of vasa recta are counter current with each other and with vasa recta
- Both limbs of vasa recta are freely permeable to solutes and water
- Vasa recta receive less than 5% of renal Blood flow; therefore, the rate of vasa recta blood flow is very slow

Active space

Cystometrogram and micturition

- 1st urge to pass urine is at a bladder volume of .....
- Uncontrollable urge comes at a bladder volume of .....
- Initially, as the bladder gets filled with urine, the intra-vesical pressure rises, and gradually the detrusor muscle relaxes, so even though the bladder is getting filled up, there is not much change in pressure
- When the urge becomes uncontrollable at 400 ml, the detrusor will contract, and micturition starts

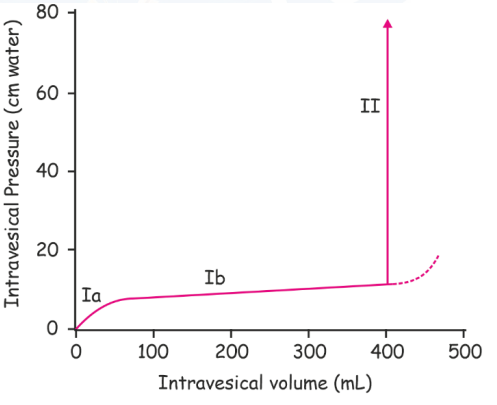
LA Place's Law

For sphere $T = \frac{p.r}{2w} = P = \frac{2TW}{r}$ (P = Intravesical Pressure)

T = Wall tension, P → Distending pressure, r – Radius, w → wall Thickness

- With ↑se of volume: both wall tension and radius increase (bladder distends), resulting in hardly any increase in intravesical pressure (segment Ib)

La place's law is applicable to segment Ib in the diagram

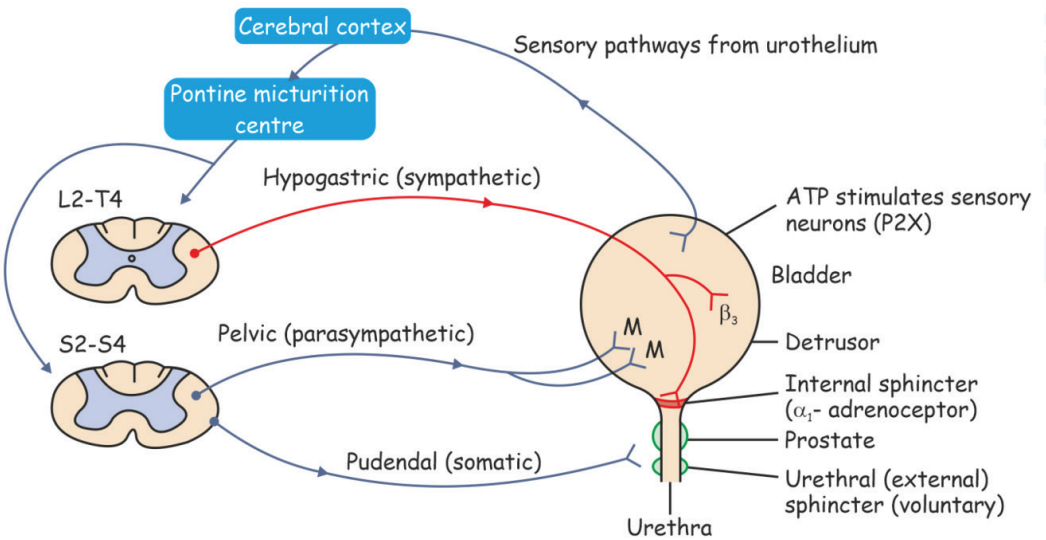


Active space

Ia	Bladder fills up Pressure increases
Ib	Bladder fills up Hardly any change in Pressure
II	Sharp increase in intravesical pressure leading to the evacuation of bladder

Micturition Reflex

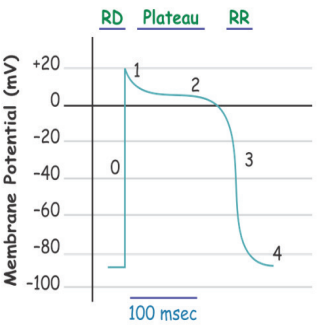
- Is a spinal reflex



Active space

ORIGIN AND CONDUCTION OF CARDIAC IMPULSE

PLATEAU PHASE

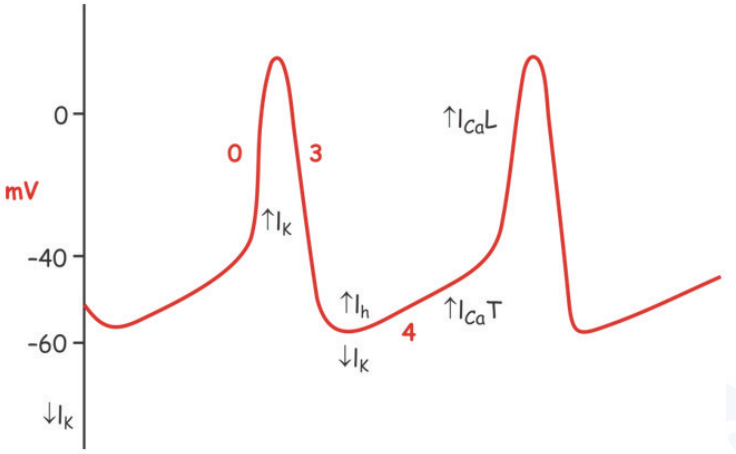
Phase 0			
Phase 1			
Phase 2			
Phase 3			
auses 4			

- Plateau phase causes prolongation of the absolute refractory period
- Due to prolonged absolute refractory period cardiac muscle cannot be tetanised

Active space

PACEMAKER POTENTIAL

Can be recorded from SA node and AV node



Phase 0		
Phase 3		
Phase 4		

EFFECT OF SYMPATHETIC STIMULATION ON THE PACEMAKER

↑Sympathetic Stimulus

Via β_1 receptor

Opening of additional Ca^{2+} channel including phosphorylation channel

↑↑ Ca^{2+} influx

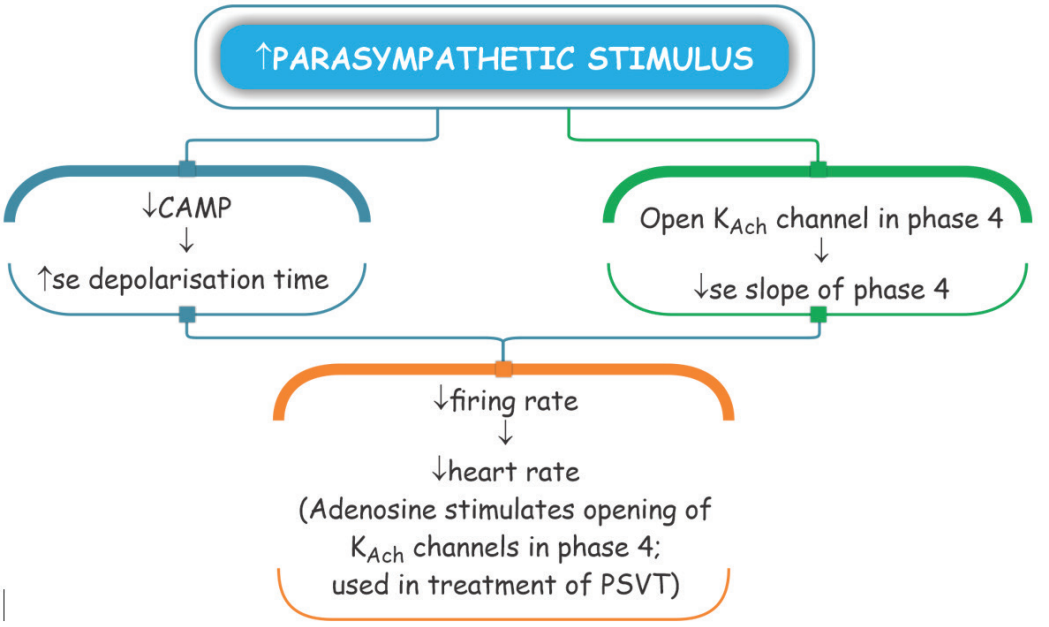
↓se depolarisation time

↑se firing rate

↑heartrate (positive chronotropic action)

Active space

EFFECT OF PARASYMPATHETIC STIMULATION ON THE PACEMAKER



CONDUCTING SYSTEM OF HEART

(Conducting cells are modified contractile cells)

1. SAN
2. Internodal tracts
   - Anterior
   - Middle
   - Posterior

3. AV Node (Slowest 0.05 m/sec)
   - Small diameter fibres
   - Very few GAP junctions
   - Duration of nodal delay is .....
   - Advantage of nodal delay .....

Active space

4. Bundle of His
5. Bundle branches
   - Right
   - Left
     - Anterior
     - Posterior fascicle
6. Purkinje fibre (Fastest 4.0 m/sec)

	Heart Rate	Force of contraction
Sympathetic stimulation	-	-
Parasympathetic stimulation	-	-

- Para sympathetic supply is present at all places in the heart except the ventricle myocardium

(Remember- 'No vagus to ventricles')

VENTRICULAR DEPOLARISATION AND REPOLARISATION

Depolarisation

- 1st .....
- Septal depolarisation .....
- Impulse travels from .....
- Last part of the ventricle to depolarise .....

.....

- Sequence of depolarisation[Q]: .....

Active space

Repolarisation

- Starts from .....to .....
- 1st part of the ventricle to repolarise – .....

Last part of the ventricle to repolarise – .....

BLOOD VESSELS & HAEMODYNAMICS

BLOOD VESSELS

1. Windkessel Vessels = aorta and large arteries

- Windkessel vessels have ..... tissue >..... muscle
- Aorta and large arteries have the property of elastic recoil
- Diastolic BP is due to:

1.

2.

2. Resistance Vessels = arterioles

Arterioles have plenty of smooth muscles, less elastic tissue

Arterioles also have a very good sympathetic supply

Active space

Mechanism

↑ Sympathetic discharge

↓

Vascular smooth contraction

↓

Radius of vessel reduces

↑↑↑ Resistance

In the arterioles, small changes in radius produce large changes in resistance

3. Exchange Vessels = capillaries

- Single cell in its wall
- All exchange between blood and tissues (respiratory gases, nutrients, waste products) happens at the level of capillaries
- Do not have any smooth muscle or sympathetic nerve supply
- Local metabolites generated in tissues can increase or decrease the blood flow in capillaries

↓ PO_2
↑ PCO_2
↑ H^+
↓ pH
↑ K^+
↑ Temperature
↑ Adenosine

→ Produce relaxation of pre-capillary sphincter and terminal arterioles

↓

Capillary blood flow increases

Active space

Skeletal muscle blood flow – At rest blood supply of muscle tissue ...., but during exercise, it increases to .... primarily due to the action of ....

4. Capacitance Vessels = venules, veins, vena cava

- Thin walled with very little smooth muscle
- Capacity to expand
- 2/3rd of blood volume is present on venous side

Effect of sympathetic discharge on veins:

↑ Sympathetic discharge
↓
Smooth muscle contraction
↓
Venoconstriction
↓
Decrease in venous compliance
(Distensibility reduces)
↓
↑ venous return

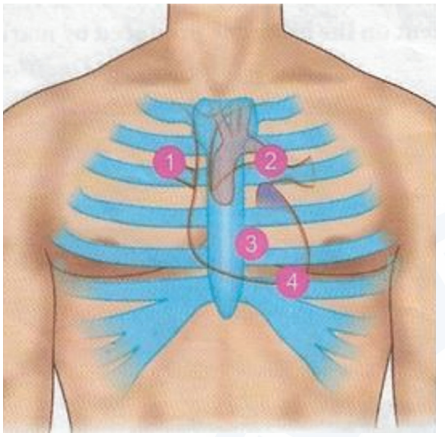
5. Shunt Vessels = arteriovenous anastomoses

- Present at a few places like ....., etc.
- Supplied by sympathetic nervous system
- Function → .....

Active space

Auscultatory areas

Auscultatory areas refer to the specific locations on the chest where heart sounds are best heard during auscultation. These are:-



1. Aortic area
2. Pulmonic area
3. Tricuspid area
4. Mitral area

Differences between the first and second heart sound

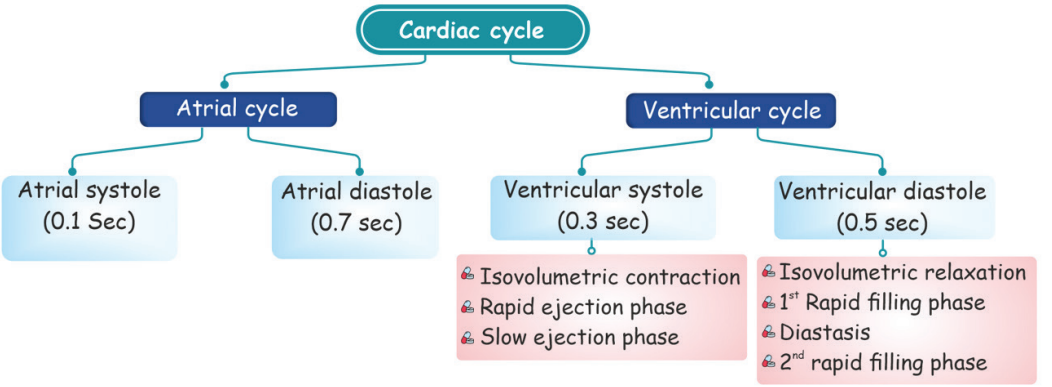
	First heart sound	Second heart sound
Cause		
Characterstics	Long and soft Low pitched "LUBB"	Short and sharp High pitched "DUBB"
Position in cardiac cycle		
Duration	0.10 – 0.17s	0.10 – 0.14s
Relation with ECG		
Site of examination		

Active space

CARDIAC CYCLE

Cardiac cycle

- Definition: .....
- .....
- Normal cardiac cycle time .....
- Cardiac cycle time =
-



- Atria function as PRIMER PUMPS; the atrial contraction is responsible for 20%to 30% of ventricular filling in the second rapid filling phase

Atrial pressure

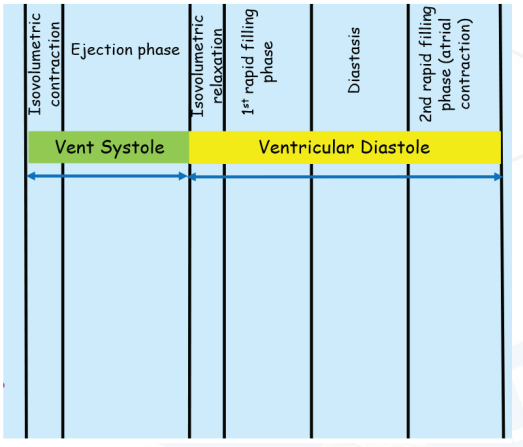
- Atria are low-pressure zones
- Diastolic pressure in atria → 0 - 3 mmHg
- During systole
  - ✦ RAP (Right atrial pressure) 4 – 6 mmHg
  - ✦ LAP (Left atrial pressure) 6 – 8 mmHg
- Systolic pressure (SP) is the maximum pressure during systole
- Diastolic pressure (DP) is the minimum pressure during diastole
- Mean arterial pressure = Diastolic blood pressure + 1/3rd pulse pressure

Active space

MIST

	Systolic pressure	Diastolic pressure	MAP
Systemic circulation	120 mmHg	80 mmHg	93 mmHg – 100 mmHg
Left ventricle	120 mmHg	0 (0-5mmHg)	-
Pulmonary circulation	25 mmHg	9 mmHg	15 mmHg
Right ventricle	25 mmHg	0 (0-5mmHg)	-

Heart sounds with Cardiac cycle



S1:

- At the onset of .....
- Due to the ..... of atrioventricular value

S2:

- At the onset of .....
- Due to the closure of .....
- Second heart sound is normally split

Active space

- A2 > P2 (A2 – Aortic Value, P2 – Pulmonary value); aortic valve closes first, followed by the closure of the pulmonary valve resulting in the splitting of the second heart sound
- Splitting of S2 widens with .....

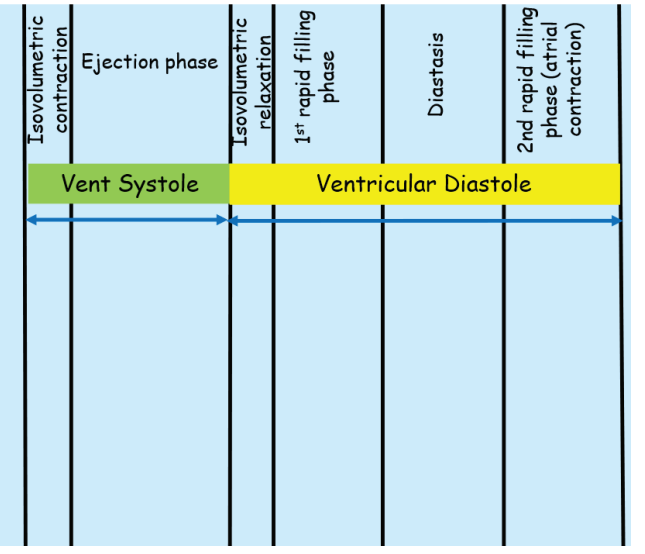
S3:

- Heard in the .....
- Commonly heard in young adult males with a thin chest wall
- Due to .....
- Became prominent in all conditions of volume overload

S4:

- Normally S4 is not heard
- If present, S4 coincides with . .....
- Heard in conditions associated with .....
- E.g.-: ventricular hypertrophy, cardiac failure

ECG with Cardiac Cycle



Active space

- P wave - .....
- QRS complex – .....
- T wave – .....
- Atrial repolarisation is not represented in the ECG because atrial repolarisation merges with the QRS complex
- Peak of the R wave is followed by the onset of isovolumetric contraction
- S1 is heard just after the peak of R wave
- S2 is heard at the end of T wave
- S3 is heard between T and next P wave
- S4 is heard in the PR Interval

Left ventricle (LV) Pressure and LV volume changes with Cardiac cycle

At point A:

- LV EDV (End Diastolic Volume) = 120 ml
- LV EDP (End Diastolic Pressure) = 5mmHg
- LV pressure > LA pressure
- MV closes producing S1

A → B is isovolumetric contraction phase:

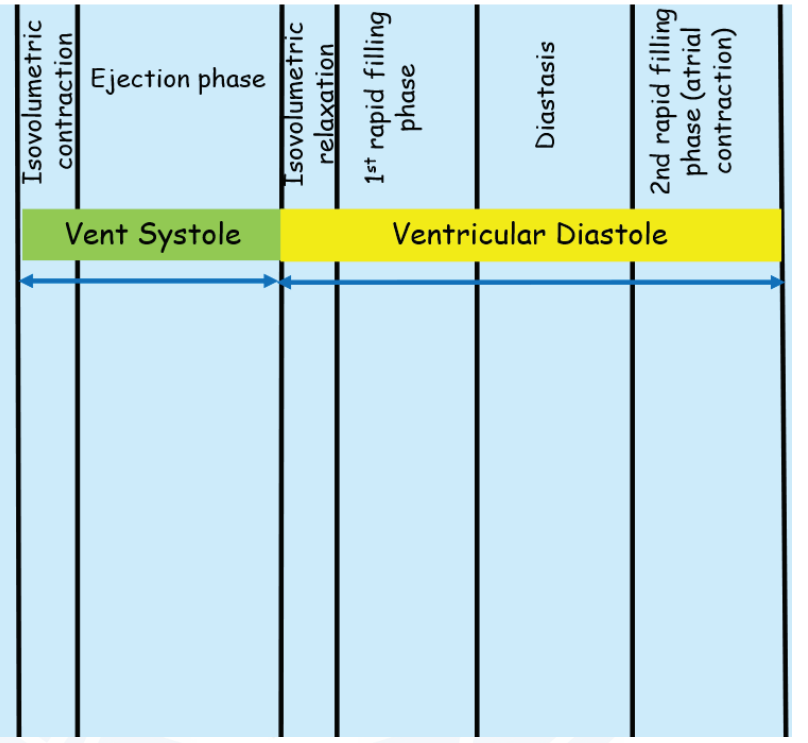
- Sharp increase in left ventricle pressure from 5 to 80 mmHg
- No change in left ventricle volume

At Point B:

- LV pressure became > Diastolic Blood pressure → Aortic value opens → ejection phase
- Ejection phase
  - ✦ Rapid E. Phase
  - ✦ Slow E. Phase

Active space

In Rapid Ejection Phase	In slow Ejection Phase
↑se in LV pressure to 120 mm	↓ se in LV Pressure
Aortic pressure closely follows	↓ se in Aortic Pressure
Rapid ↓ se in LV volume	Slower ↓ se in LV Volume



At Point C:

- LV pressure is slightly lower than Aortic Pressure Closure of aortic valve and S2

C → D is Isovolumetric Relaxation phase:

- Sharp decrease in LV pressure to O mm with no change in LV volume

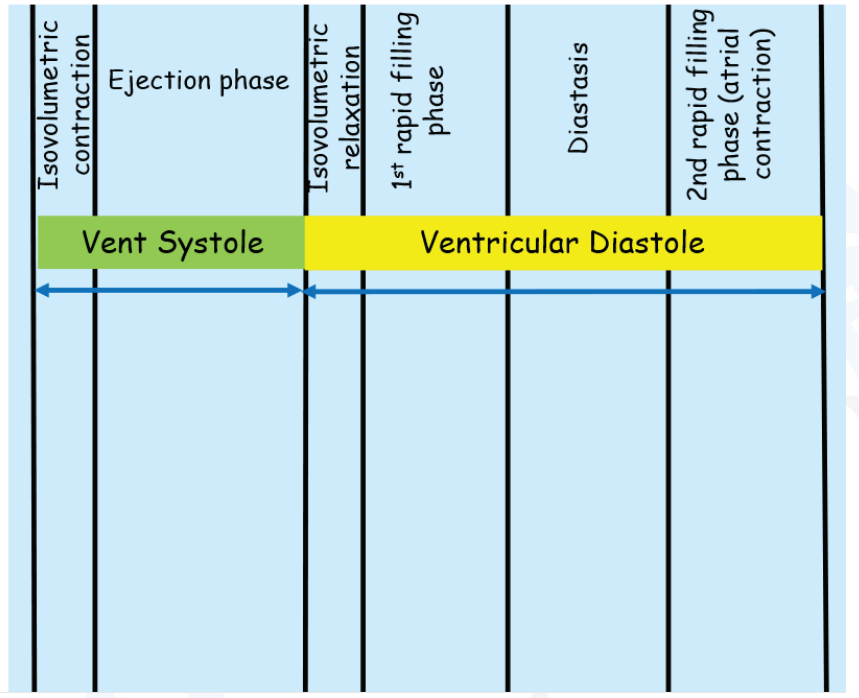
MIST

Active space

D → A is Filling phase

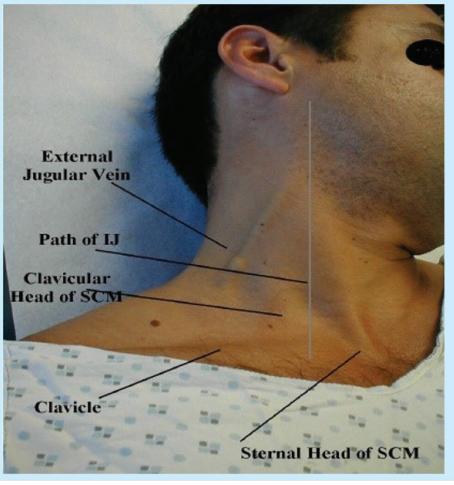
Aortic Pressure changes with Cardiac Cycle

Incisura or Dicrotic Notch:



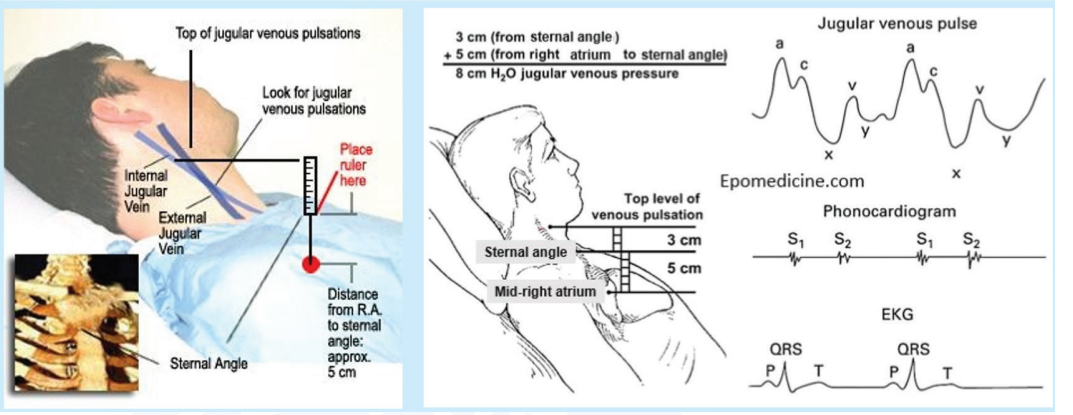
JVP with Cardiac Cycle

Examination of JVP



Active space

MIST



JVP is seen on the right side because .....

.....

JVP is seen along with pressure changes on the right side of the heart

"POLICE"

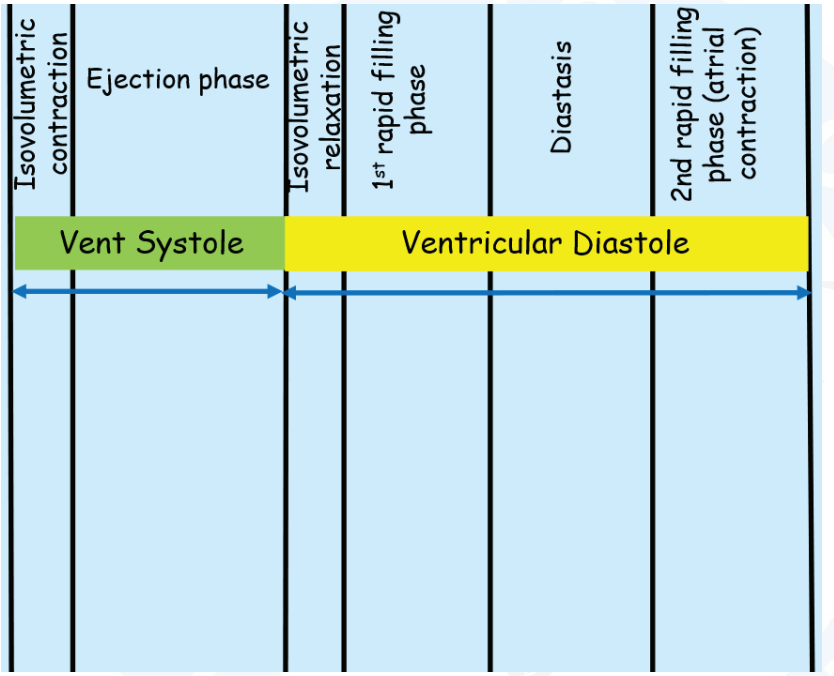
- P Palpation-
- O Occlusion-
- L Location-
- I Inspiration-
- C Contour-
- E Erect position-

- Positive waves
  - ✦ a, c, v
  - ✦ Directed outwardly

- Negative waves
  - ✦ x descent, y descent
  - ✦ Inwardly directed

Active space

- a = ...
- c = ...
- x descent = ...
- v = ...
- Y descent= .....



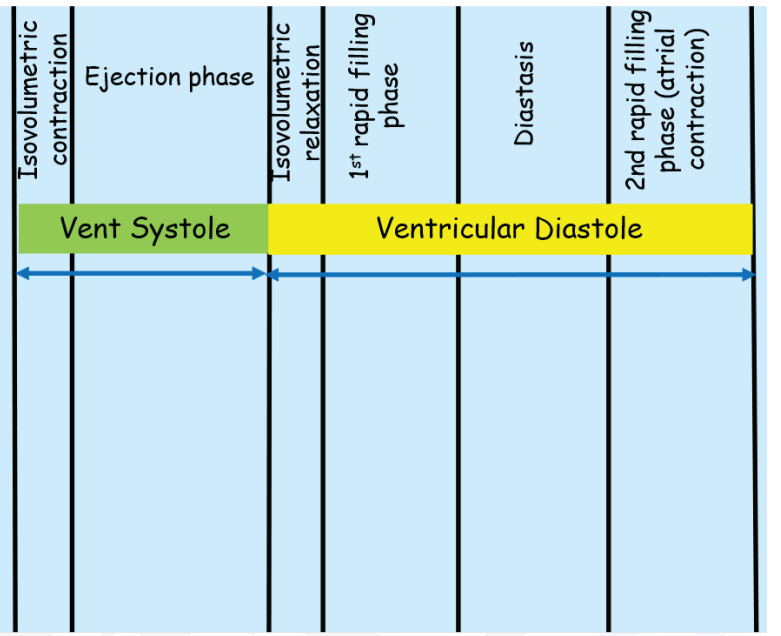
Abnormal JVP

Prominent a wave	
Cannon a wave	
Absent a wave	
Large v wave (accentuated v wave)	
Blunted y descent	
Sharp x descent	
Sharp y descent	

Active space

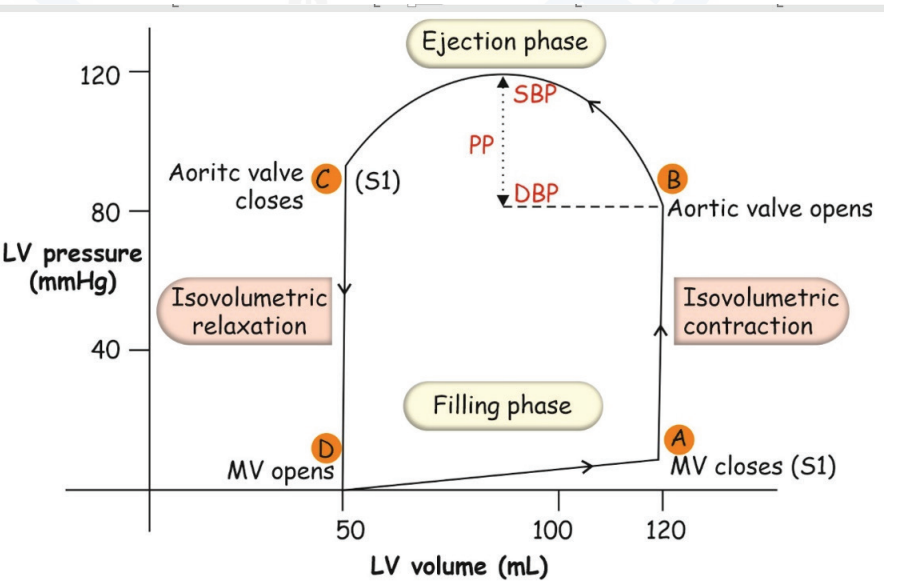
MIST

Cardiac Cycle Summary



- If heart sounds, ECG, aortic pressure changes and JVP are being recorded simultaneously, the sequence of events are best described by the acronym

LV Pressure-Volume Loop



Active space

- Pulse pressure = SBP – DBP
- Width of loop = Stroke volume
- Ejection Fraction = $\frac{\text{Stroke Volume}}{\text{End Diastolic Volume}}$ = 0.55 – 0.65 = 55 – 65%
- Mnemonic to remember which valve opens at which point "*MAAM COCO*"

At point	A	B	C	D
Valve (Mitral/ Aortic))	M	A	A	M
Close/Open	C	O	C	O
Sound	S1	Ejection click (A.S.)	S2	Opening SNAP (M.S.)

CARDIAC OUTPUT & BLOOD PRESSURE

Cardiac Output

Definition: .....

CO = LV output = RV output

CO = Systemic blood flow rate = pulmonary blood flow rate

Cardiac Index = ...

(Normal) CI = _____

Cardiac Output = Heart rate × Stroke volume

$$CO = \frac{MAP}{TPR} \text{ or } MAP = CO \times TPR$$

Active space

Conditions which decrease cardiac output

Conditions which increase cardiac output

CO = HR × SV

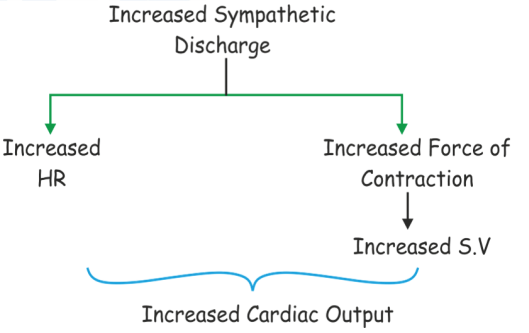
Regulation of Cardiac Output:

i. Factors affecting heart rate

ii. Factors affecting stroke

Volume

Factors affecting HR



Active space

Increased parasympathetic discharge

↓ Rate of firing of SA Node

↓

Decreases HR

Factors affecting SV

1. Myocardial contractility

2. Venous return

Factors affecting venous return

i. Blood volume

ii. Posture:

iii. Sympathetic discharge

iv. Competent Venous Valves, deep fascia

v. Calf muscle contraction during exercise increases venous return (known as ..... pump)

vi. Deep inspiration (k/a ..... pump):

Active space

Blood Pressure

SBP → .....

DBP = ...

PP = ...

MAP =

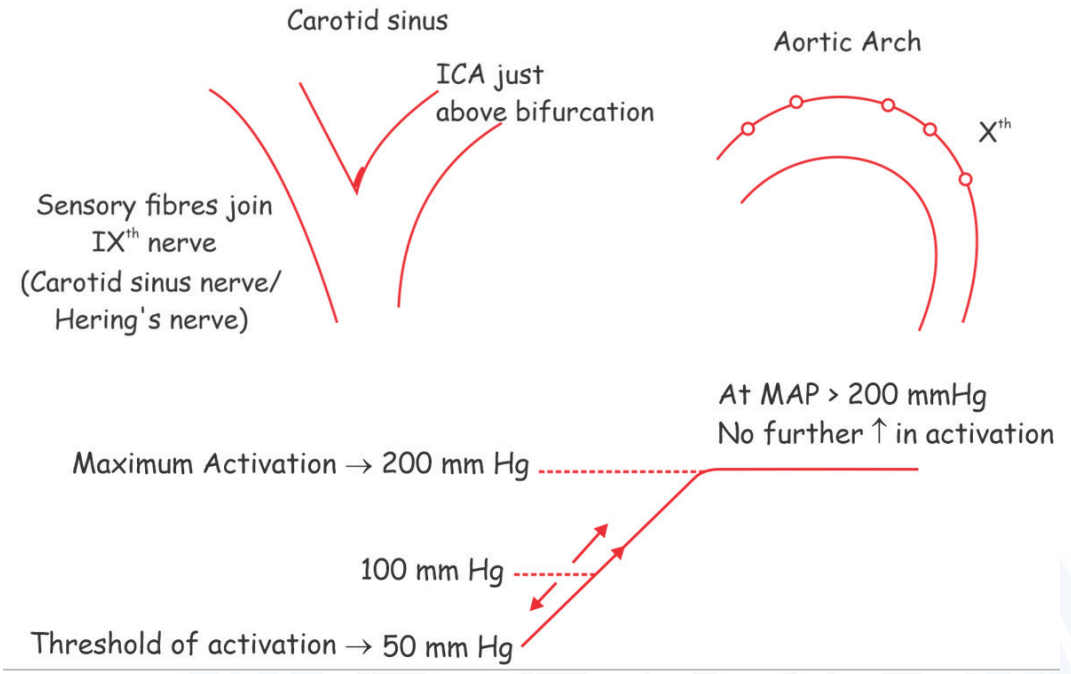
Regulation of BP

1. Via Baroreceptor:

Mechanism - Increase MAP - Increase Stretch of vessel wall - firing rate

- Baroreceptors act via a ..... mechanism
- Very prompt response
- Adaptation present
- Important in ..... -term control of blood pressure
- Important in minute-to-minute control
- Also known as ..... nerves[Q]
- **Carotid sinus baroreceptors-** located in the adventitia of internal carotid artery just above the bifurcation
- Aortic arch baroreceptor present along the arch of the aorta
- Sensory fibres from carotid sinus baroreceptors (carotid sinus nerve or Hering's nerve) join the ..... nerve, sensory fibres from aortic arch baroreceptors join the ..... nerve

Active space



- Carotid sinus baroreceptors are about 5 times more sensitive than aortic arch baroreceptors
- Threshold for activation of baroreceptor is MAP ....
- Max activation is seen at MAP of ..... mm Hg; at MAP > 200 mm Hg there is no further increase in stimulation of baroreceptors

Baroreceptor mechanism for regulation of blood pressure

Active space

Atrial stretch receptor mechanism for regulation of blood pressure

- ..... receptors
- Present in low-pressure zones- ..... and ....

Atrial stretch receptor response

Active space

Other influences on vasomotor (RVLM) centre

- Between MAP of ..... mmHg, cerebral blood flow is constant; CNS ischaemia (at MAP < 65mm Hg) can directly stimulate RVLM (VMC)
- Limbic system, pain receptors, joint proprioceptors via hypothalamus can also stimulate RVLM (VMC), increasing the sympathetic outflow to the cardiovascular system
- Hypothalamus can directly stimulate RVLM(VMC)

Renin angiotensin aldosterone system (RAAS) on Regulation of BP

Angiotensin II peripheral vasoconstriction → TPR MAP

- GFR
- Na+ Reabsorption in PCT, TAL
- Thirst

Active space

Stimuli which increase renin secretion

- .....
- .....
- .....
- .....
- .....
- .....
- .....
- .....

Mary's law –: .....

Ex: Haemorrhage is associated with Hypotension and Tachycardia

Brain bridge reflex

- .....

Bezold Jarisch Reflex/Coronary Chemoreflex

- Injection of the following substances into the coronary arteries:
  - _____
  - _____
  - _____
  - _____

Active space

Causes ..... and .....

Example: Some patients with MI preset with Hypotension AND Bradycardia; probable explanation- accumulation of toxic metabolites triggers the coronary chemoreflex

CIRCULATION

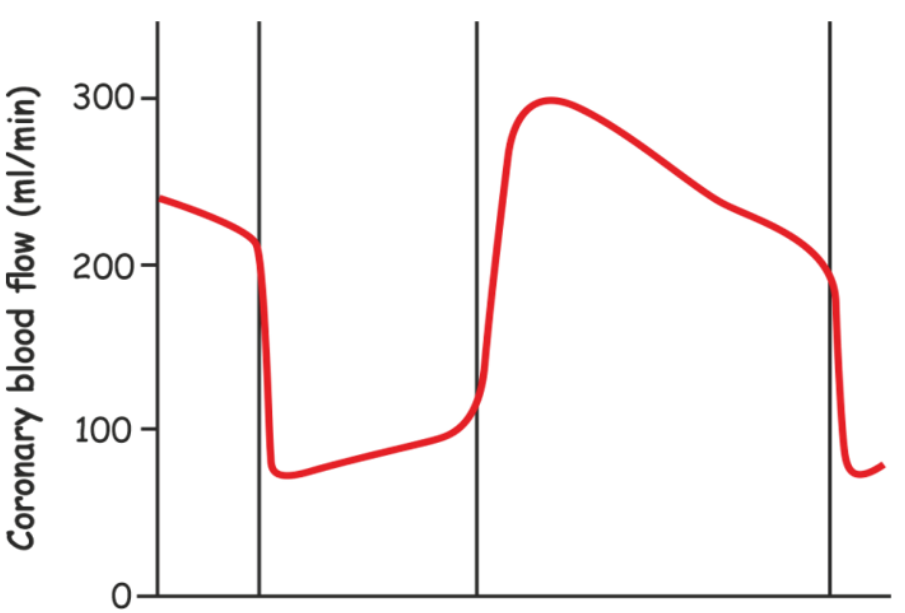
Coronary Circulation

- ..... ml/min
- ..... % of cardiac output
- Heart has a very high oxygen demand because

i cardiac muscle cells have mainly ..... metabolism (1% of the total energy required by the heart is generated by glycolytic metabolism)

ii major fuel for the heart are the ..... which require more oxygen for metabolism

iii heart is a very "hard-working" organ



Blood supply, especially to the left ventricle, occurs mainly in diastole

Active space

Effect of sympathetic stimulation on coronary blood flow:

- Direct effect is vaso.....
- Indirect effect - increase in work done by the heart increases ATP Hydrolysis, and ↑sed Adenosine will cause vasodilation

Factor affecting myocardial oxygen demand:

- Basal myocardial oxygen demand is – ..... of tissue. Basal oxygen demand of skeleton muscle at rest – ....
- Oxygen demand in beating heart at rest – ....tissue

Factors increasing myocardial oxygen demand:

1. _____
2. _____
3. _____
4. _____

Work done = _____ x _____

SV Volume of work, E.g. Aortic regurgitation

MAP Pressure work, E.g. Aortic stenosis

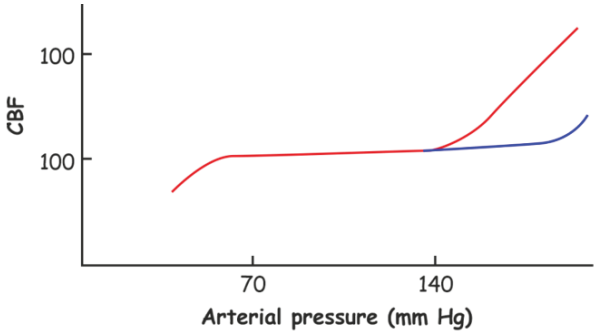
Increase in oxygen demand with an increase in pressure work is more than with se in volume work

Active space

Cerebral Circulation

- ..... ml/min (12 – 13% of cardiac output)
- ml/min/100g

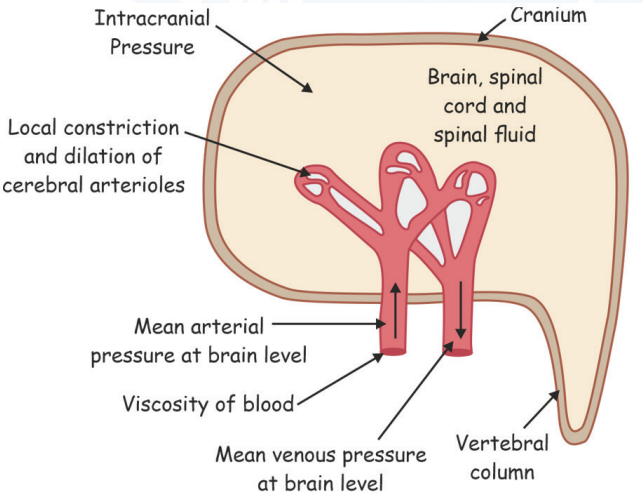
Between MAP of 65 and 140 mmHg, cerebral blood flow is constant because of autoregulation



Monroe – Kellie Doctrine:

- Because brain tissue and spinal fluid are essentially incompressible, the volume of blood, spinal fluid and brain at any time must be relatively constant
- se in intracranial pressure will cause compression of cerebral blood vessels
- se in venous pressure se intracranial pressure
- ↑se in venous pressure ↓se cerebral blood flow
- CPP = MAP – ICP
- (CPP – Cerebral perfusion pressure, ICP = Intra cranial pressure)

Factors Affecting Cerebral Blood Flow



Active space

Following cause an increase in local blood flow

1. ..... PO_2
2. ..... PCO_2
3. ..... H^+
4. ..... K^+
5. ..... Metabolic activity

Active space

RESPIRATORY SYSTEM

TYPES OF CELLS IN THE LUNGS

AGENTS CAUSING BRONCHODILATION & BRONCHOCONSTRICTION

Bronchodilators	Bronchoconstrictor

Important formulae

- Pulmonary ventilation/minute ventilation = ...
  X .....
- Alveolar ventilation = ...
  X .....

Active space

DEAD SPACE

- Definition:- .....
- Normal dead space is ..... ml or .... /kg body weight

Weibel's classification of the tracheobronchial tree:

- It is for anatomical dead space
- Tracheobronchial tree is divided into 23 generations
- Generation 0 → _____
- Generation 1 – 16 → _____
- Generation 17 – 23 → _____
- Generation 0, Generation 1-16 constitute anatomic dead space

Physiological dead space/total dead space:

- Physiological dead space = ... dead space + ... Dead Space.
- In normal individuals, wasted ventilation = ...
- So physiological dead space = Anatomical dead space

Anatomical dead space

- Measured by ..... technique, also known as single-breath nitrogen analysis. (N2 analysis is done in expired air after a single deep breath of 100% O2)

Active space

Physiological dead space

- Measured by _____

LUNG VOLUME & CAPACITIES

Tidal Volume (TV)

Defn:- .....

TV = 500 ml (in both males and females)

Inspiratory Reserve Volume (IRV)

Defn:- .....

In males = 3300 ml, in females = 1900 ml

Expiratory Reserve Volume (ERV)

Defn:- .....

In males = 1000ml, in females = 700ml

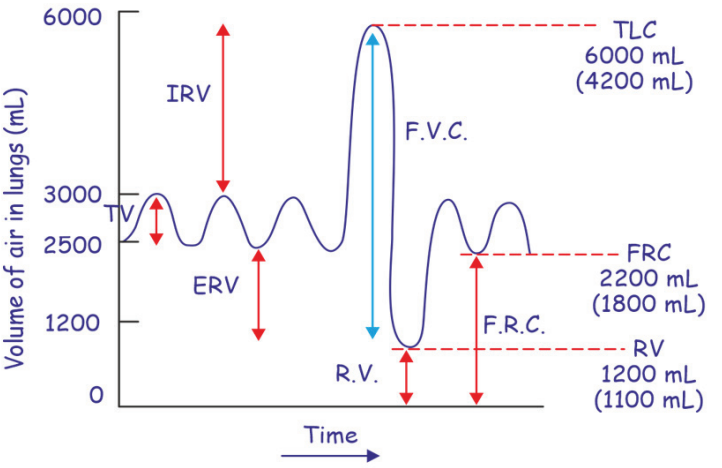
Residual Volume (RV)

Defn:- .....

In males = 1200ml, in female = 1100ml.

- Inspiratory capacity (IC) = ... + ...
- Expiratory Capacity (EC) = ... + ...
- Functional residual capacity (FRC) = ... + ...
  - ✦ FRC is .....
    Forced vital capacity (FVC) = ... + ...
    ..... + ...
  - ✦ FVC or VC is .....

Active space



Lung Volume and capacities which cannot be measured by routine spirometry

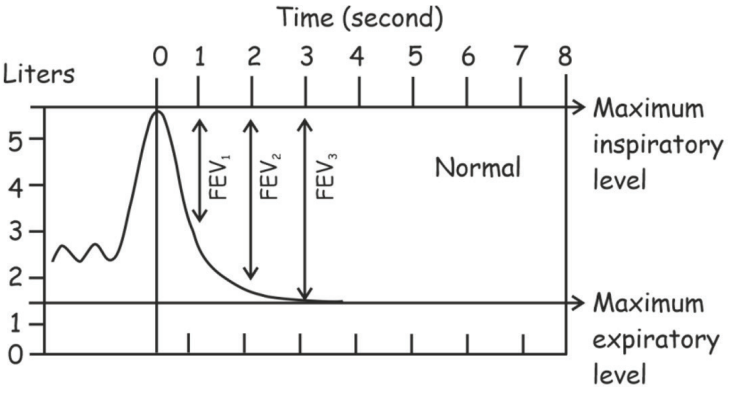
1.
2.
3.

FRC (FRC = ERV + RV) can be measured by:

1.
2.
3.

TIMED VITAL CAPACITY

- Measured as FEV_1, FEV_2, FEV_3



Active space

$\frac{FEV_1}{FVC} = 0.7 - 0.8$ (that is, at the end of 1 second of forceful expiration, a normal individual should be able to breathe out 70 – 80% of his vital capacity)

$\frac{FEV_2}{FVC} = 0.9$ or 90%

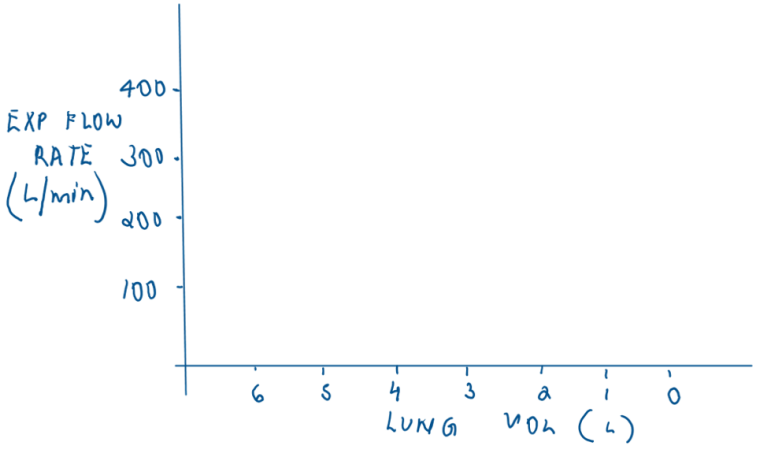
$\frac{FEV_3}{FVC} = 0.97 - 0.98$ or 97 – 98% (that is, at the end of 3 seconds a normal individual should be able to breathe out almost 100% of his vital capacity).

CLINICAL IMPLICATION OF TIMED VITAL CAPACITY

Obstructive Lung Disease	Restrictive Lung Disease
$\frac{FEV_1}{FVC}$ Ratio is	$\frac{FEV_1}{FVC}$ Ratio is .....
Rates are affected more than volume	Volumes are affected more than rates

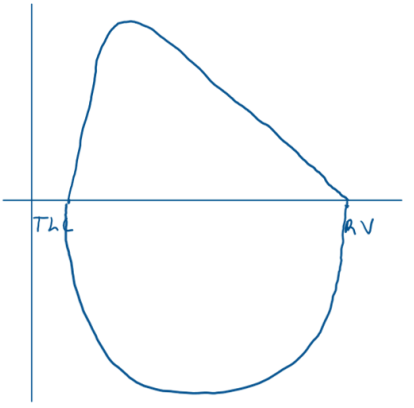
FLOW VOLUME LOOPSY

Expiratory flow volume curve

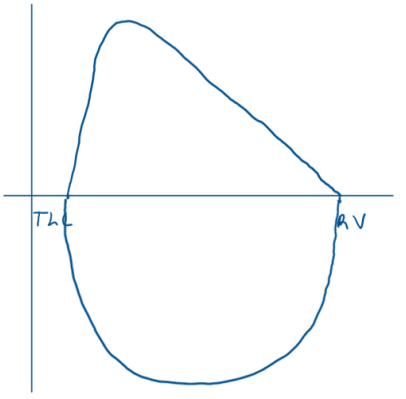


Active space

Flow volume loop in obstructive lung disease

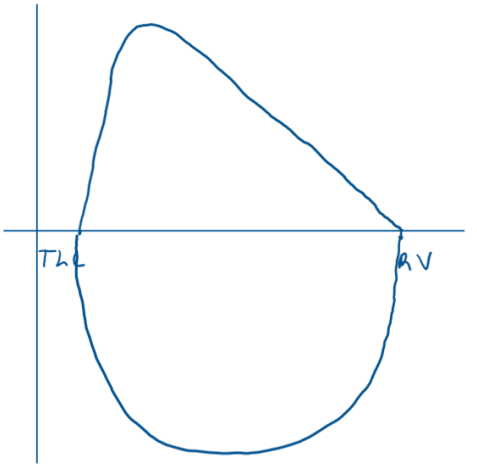


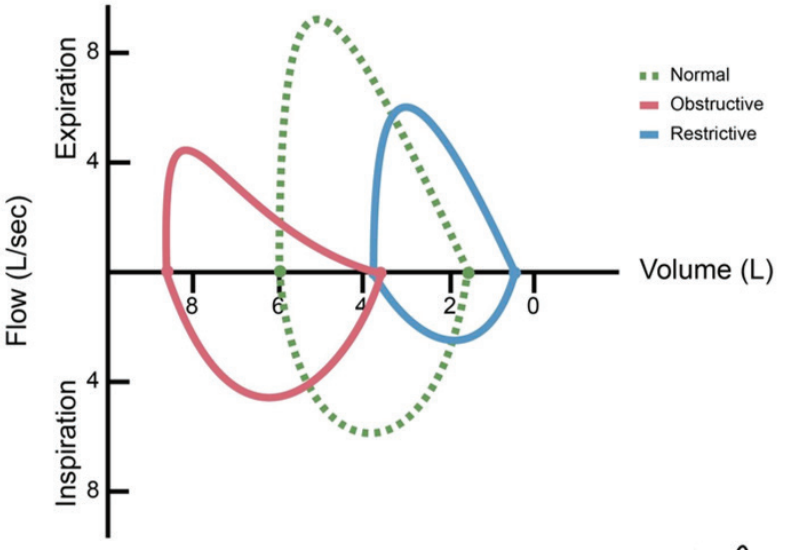
Flow volume loop in restrictive lung disease (intrinsi



Active space

Flow volume loop in restrictive lung disease (extrinsic neuromuscular causes)





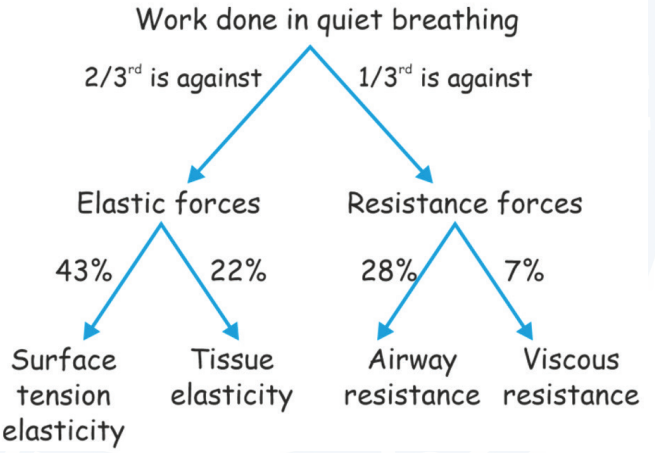
Major processes involved in respiration

1. Ventilation
2. Respiratory Excahnge
3. Transport of gases

Active space

WORK DONE IN QUIET BREATHING

- Work done in quiet breathing = 0.3 – 0.8 kg/m/min



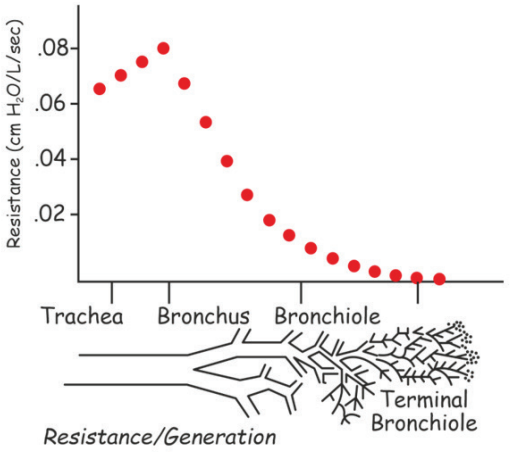
Maximum work done in quiet breathing is against .....

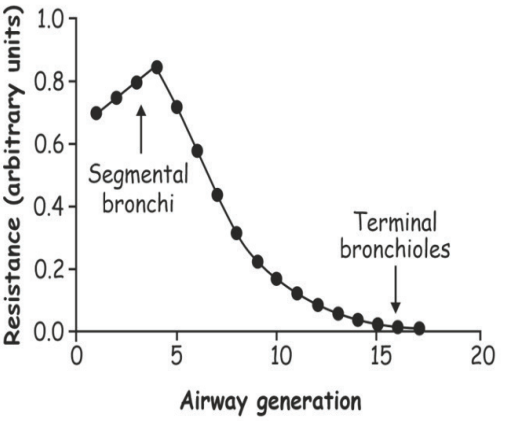
Airways with maximum resistance to airflow:

- Minimum airway resistance is seen in ..... airways
- Max airway resistance is seen in ..... airways
- Gen 0, 1, 2 is large size airway
- Gen 3-5th is medium size airway
- Gen 6-23rd is small size airway

Active space

	Medium size Airways	Small size Airways
Total cross surface area		
Velocity of Airflow		
Reynolds Number		
Tendency for Turbulence		
Resistance		





SURFACE TENSION & SURFACTANTS

Whenever there is an air-liquid interphase, molecules on the topmost layer are attracted to one another creating a surface tension. Surfactant decreases these attractive forces reducing the surface tension

Active space

Surfactants

- Surfactant is secreted by type ..... pneumocytes
- Chemical composition of surfactants – .....
- Surfactant decreases surface tension and, thereby, prevents alveolar collapse
- During expiration → decrease alveolar size → increase concentration of surfactant per unit area → decrease surface tension → prevent alveolar collapse (maintains alveolar stability)

Hormones Affecting Surfactant

Thyroid hormone

-

Insulin

-

Corticosteroids

VENTILATION-PERFUSION GRADIENT IN THE ERECT POSTURE

The following decrease from base to apex

1.

2.

But V/Q ratio .....

from base to

Knowledge Capsule

TB is common at the apex because of higher $\frac{V}{Q}$ ratio at the apex and, therefore, higher PO_2 at the apex.

Active space

apex.

- V/Q ratio at apex = ...→ ↑ PO2, ↑ PCO2
- V/Q ratio at base = ... → ↓ PO2, ↓ PCO2
- V/Q ratio at 3rd rib (hilum) = 1.0
-

PARTIAL PRESSURE

% of O_2 in inspired air is ..... , in expired air is ..... (% of O_2 received by patient in mouth-to-mouth breathing is 16%)

At sea level:

Atmospheric pressure (PB) (760 mmHg) = PO_2 + PN_2

Partial pressure of any Gas = Fractional concentration of gas × PB

Partial pressure of O2 (PO2) = 21/100 ×760=160 mmHg

At high altitudes:

- Atmospheric pressure decreases
- % of O_2 → Unchanged
- PO2 decreases

NORMAL VALUES

	Arterial Blood	Venous Blood
PO_2 (mmHg)		
SO_2 (%)		
Total O_2 (mL/dL)		
PCO_2 (mmHg)		
Total CO_2 (mL/dL)		

Active space

PO_2 refers to dissolved O2 in plasma, and SO2 is saturation of Hb with oxygen

Respiratory Quotient

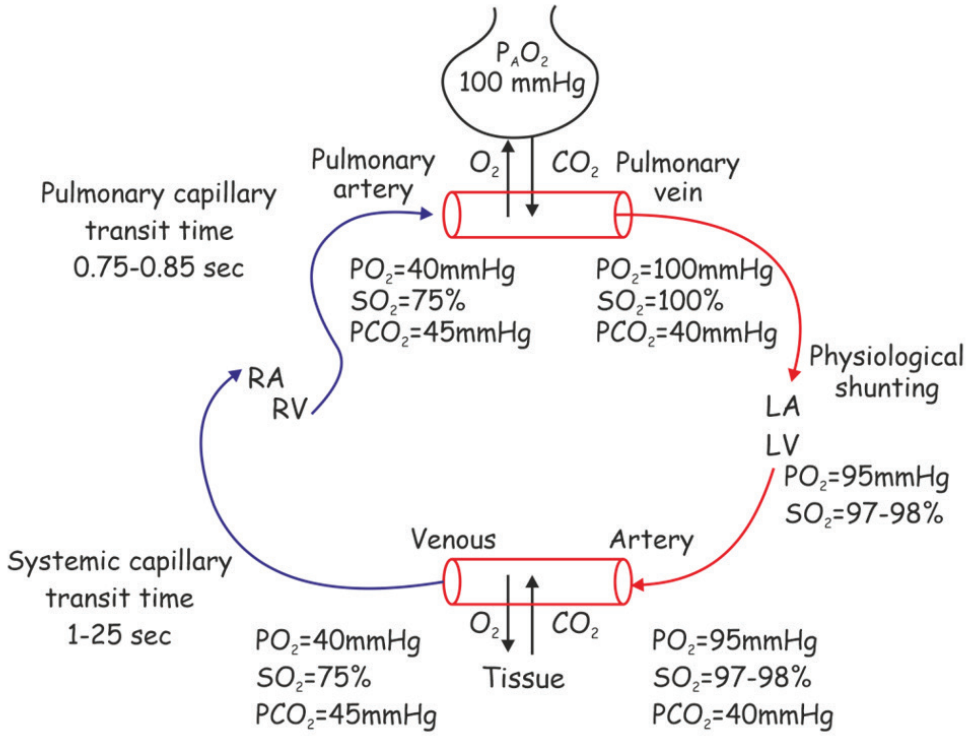
$$\text{Respiratory Quotient}\left(\text{RQ}\right) = \frac{\text{Volume of } CO_2 \text{ produced per min}}{\text{Volume of } O_2 \text{ consumed per min}} = \frac{200 \text{ ml / min}}{250 \text{ ml / min}} = 0.8$$

RQ (for mixed Diet) = 0.8

RQ (for carbohydrate Diet) = 1.0

RESPIRATORY EXCHANGE

Active space



Pulmonary capillary transit time:

- Time for which blood stays in the pulmonary capillary
- Duration:-.....
- Time available for respiratory exchange

Systemic capillary transit time

- 1 to 2 sec

TRANSPORT OF OXYGEN & CARBON DIOXIDE

TRANSPORT OF OXYGEN

Occurs in two ways

1. With Hb
2. As dissolved oxygen

Active space

	Arterial Blood	Venous Blood
Total O2	19mL/dL	14 mL/dL
Dissolved oxygen	0.29 mL/dL	0.12 mL/dL
Oxygen bound with Hb	18.71 mL/dL	13.88 mL/dL

OXYGEN CONTENT

Oxygen Content = ... + ...
.....

= ..... + ...
.....

- Solubility of O2 – 0.003 ml/dL/mmHg
- In hemolytic or iron deficiency or megalobalstic anemia- decreased Hb (but normal oxygen saturation) results in a decreased oxygen carrying capacity of the blood.
- In CO poisoning, decreased saturation (but normal Hb) results in a decreased oxygen carrying capacity of the blood.

OHDC

- Relationship between PO2 and saturation is determined by the oxygen – Hb – Dissociation curve (OHDC)

	PO_2	SO_2 (Saturation)
Venous Blood		
Arterial Blood		

Active space

- When PO2 is 60 mmHg, SO2 is 89%
- Graph of PO2 and SO2 is ..... shaped
- Sigmoid shape of the graph is due to ....

Positive Cooperativity

.....

.....

..... heme has the least affinity for oxygen

P50

.....

Normal - .....

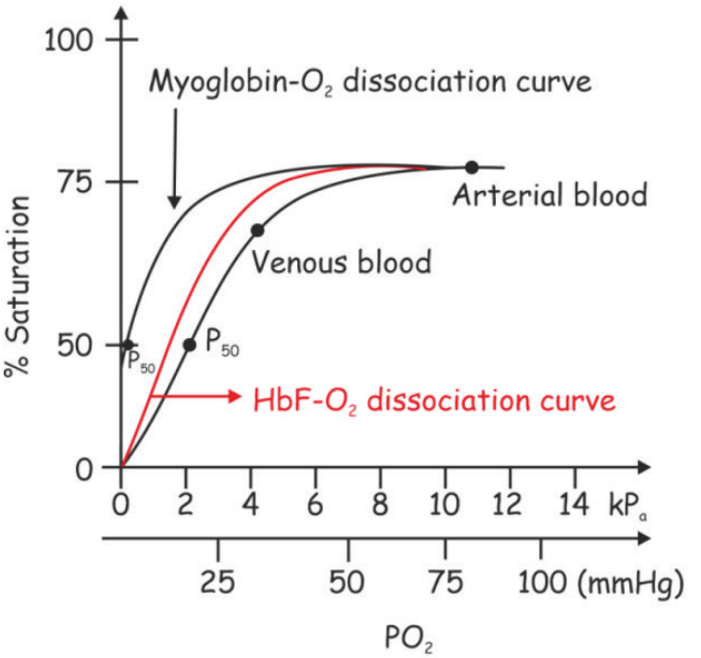
Left shift	Right shift
Increased affinity Decreased P_{50} • • • • • • •	Decreased affinity Increase p_{50} • • • • • •

Active space

2, 3 BPG:

- 2,3 BPG and O2 bind on the same place on the ..... chain of Hb
- ..... mole of Hb binds with ..... mole of 2, 3 BPG
- ..... mole of Hb binds with ..... moles of oxygen

Increase in 2,3-BPG is seen in:



- HbF-O_2 dissociation curve is to the left of the normal OHDC because .....

- Myoglobin – O_2 Dissociation curve is a rectangular hyperbola; myoglobin has a ..... affinity for O_2, P50 for myoglobin is .....

Active space

BOHR'S EFFECT

- Shift of Hb – O_2 dissociation curve due to changes in PCO_2 and H+ Levels
- As the blood reaches the tissues, diffusion of CO_2 from the tissues into the blood increases the PCO_2 and H+ levels in the blood resulting in a shift of the OHDC to the right
- It is responsible for an additional 1 – 2% O_2 delivery to tissues
- As the blood reaches the pulmonary capillaries, diffusion of CO_2 from the blood into the alveoli decreases the PCO_2 and H+ Levels in the blood resulting in a shift of the OHDC to the left- this increases the affinity of the Hb for O_2

CARBON DIOXIDE TRANSPORT

1. Maximum CO_2 transport occurs in the form of bicarbonates
2. With Hb → As carbamino compounds
3. As dissolved CO_2

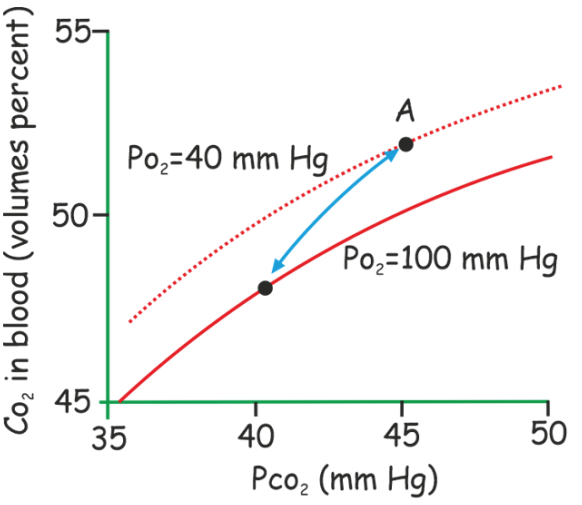
   Dissolved CO_2 = PCO_2 (mm Hg) × 0.067mL/dL/mmHg (Solubility of CO_2 is 20 times the solubility of O_2)

HALDANE'S EFFECT

- Deoxyhaemoglobin (in tissue capillaries) binds more CO_2

Active space

- Oxyhaemoglobin (in pulmonary capillaries) binds less CO_2



TREGULATION OF RESPIRATION

Neural Control of Respiration:

- By medulla
- Pontine Centres
- Higher centres, which include:
  - ✦ Limbic system
  - ✦ Cerebral cortex: involved in voluntary control of respiration

1. Medulla

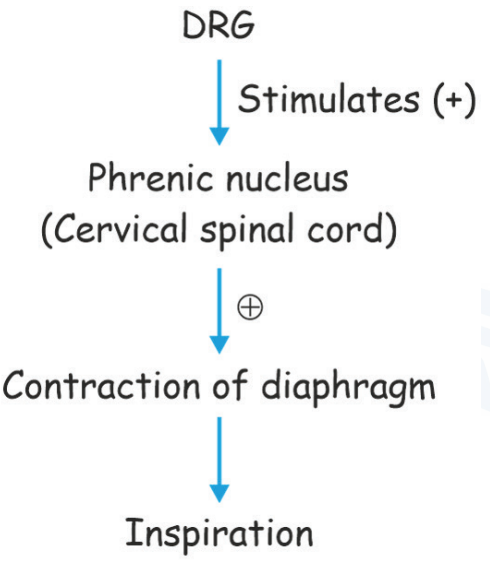
Pacemaker of respiration, the pre-Botzinger complex is located in the medulla

Has Dorsal respiratory group of neurons (DRG) & Ventral respiratory group of neurons (VRG)

Active space

DRG	VRG

DRG → Inspiratory RAMP signal



- Receptors for tidal respiration are muscle spindles in the diaphragm and intercostal muscles- they give a 'switch-off' signal to the DRG, limiting the depth of inspiration

2. For expiration, there are no signals from DRG. Diaphragm relaxes and moves upwards, resulting in lung deflation

3. Pontine Centres
   - Pons has two centres which can regulate the depth of inspiration
     - Pneumotaxic Centre
     - Apneustic Centre

   - Pneumotaxic Centre (and vagal input) ..... depth of inspiration

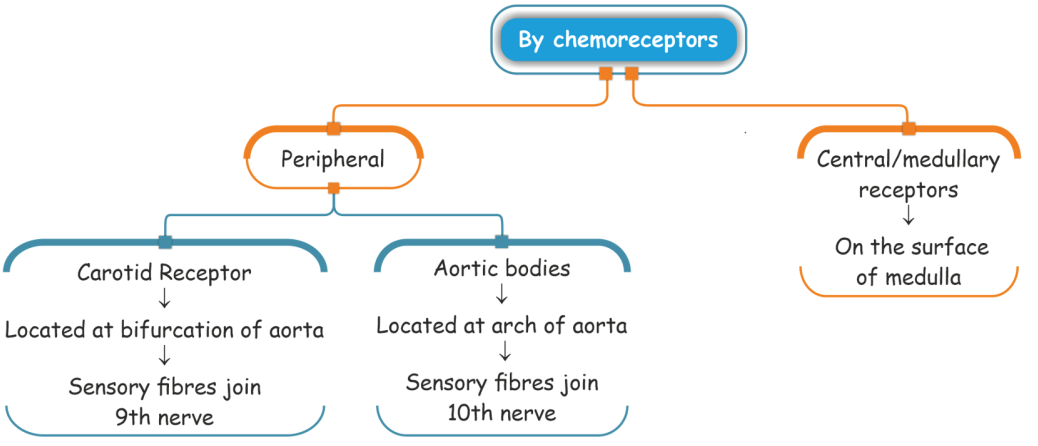
Active space

- Apneustic centre _____ depth of inspiration
- Vagus carries impulses from pulmonary stretch receptors, which are located in the .....
- Stimulation of pulmonary stretch receptors causes the Hering-Breuer Inflation reflex
- Pulmonary stretch receptors are stimulated when lung inflation is more than .....
- Sensory fibres from pulmonary stretch receptors join the IXth nerve, which in turn decreases the depth of inspiration and increases the expiration time

Ondine's Curse/Central Hypoventilation Syndrome:

- Voluntary control of respiration is ..... but involuntary control is .....
- Lesion is probably in the pre-Botzinger complex

Chemical Control of Respiration



Active space

Chemical control of respiration by chemoreceptors

Chemoreceptor stimulation causes:

- Increase in _____
- Increase in _____
- Increase in _____
- Increase in _____

1. Stimulus for peripheral chemoreceptor

- ↓ se PO2
- ↑ se PCO2
- ↑ se H+
- ↓ se PH
- ↑ Lactic Acid
- ↑ K+
- Cyanide

Peripheral chemoreceptors respond to both volatile acid and fixed acid.

- Volatile acid – CO2, fixed acids → lactic acid, keto acid, salicylic acid

Kussmaul Breathing is seen in:

K –

U –

S –

S –

M –

A –

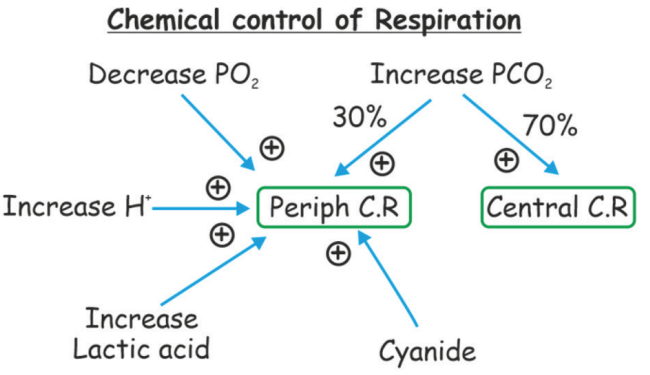
U –

L –

Active space

2. Central Chemoreceptors (CCRs)

- Most potent direct stimulus for central chemoreceptors is H+, but H+ cannot cross the blood-brain barrier
- Most potent stimulus for central chemoreceptors in the blood is ↑ PCO_2
- CCRs respond to only volatile acid (CO2)



Respiratory Reflexes

1. Hering-Breuer Inflation Reflex:

2. Hering-Breuer Deflation Reflex:

   Marked deflation of lungs → ↑se Depth and ↓se expiration time

3. Head's Paradoxical Reflex:

Active space

4. J Receptor Reflex:

Cyanosis

- When deoxyhemoglobin ≥ .....
- When methemoglobin ≥ .....
- When sulfhemoglobin ≥ .....
- Patients with severe anaemia do not present with cyanosis, but patients with polycythaemia can present with cyanosis

Hypoxia

1. Hypoxic Hypoxia:

- Seen in .....
- ↓se PO2, ↓se SO2
- Peripheral chemoreceptor stimulated due to ↓se PO2

2. Anaemic Hypoxia

- Decrease in Hb with normal saturation - Haemolytic Anaemia
- Hb is normal, but saturation is decreased – CO poisoning
- Dissolve O_2 is .....
- No peripheral chemoreceptor stimulation

Active space

3. Stagnant Hypoxia

- Seen in .....
- ..... in Arterio-Venous oxygen difference
- Peripheral chemoreceptors stimulated
- In cardiac failure, ↓se cardiac output and decreased blood flow ↓se total volume of blood and, therefore, O2 delivery per unit time to peripheral chemoreceptors, stimulating them

4. Histotoxic Hypoxia

- Seen in .....
- In histotoxicity, tissues are unable to utilise the oxygen resulting in a ..... Arterio-Venous O2 Difference
- Agents causing histotoxicity, such as cyanide, nicotine cause direct stimulation of the peripheral chemoreceptors resulting in an increase in rate and depth of ventilation

Active space

GIT

SMOOTH MUSCLES

INTRODUCTION

- It is an ..... , ... muscle
- It has actin, myosin, and tropomyosin but lacks .....
- Calcium binding protein present in smooth muscle is ..... Q

TYPES OF SMOOTH MUSCLE

Single Unit Type	Multi-Unit Type
GAP Junctions ..... Present in the wall of a hollow organ Example – GI Tract	GAP Junctions ..... Present in dilator pupillae, constrictor pupillae, ciliaris muscles

- ..... has both types of smooth muscle

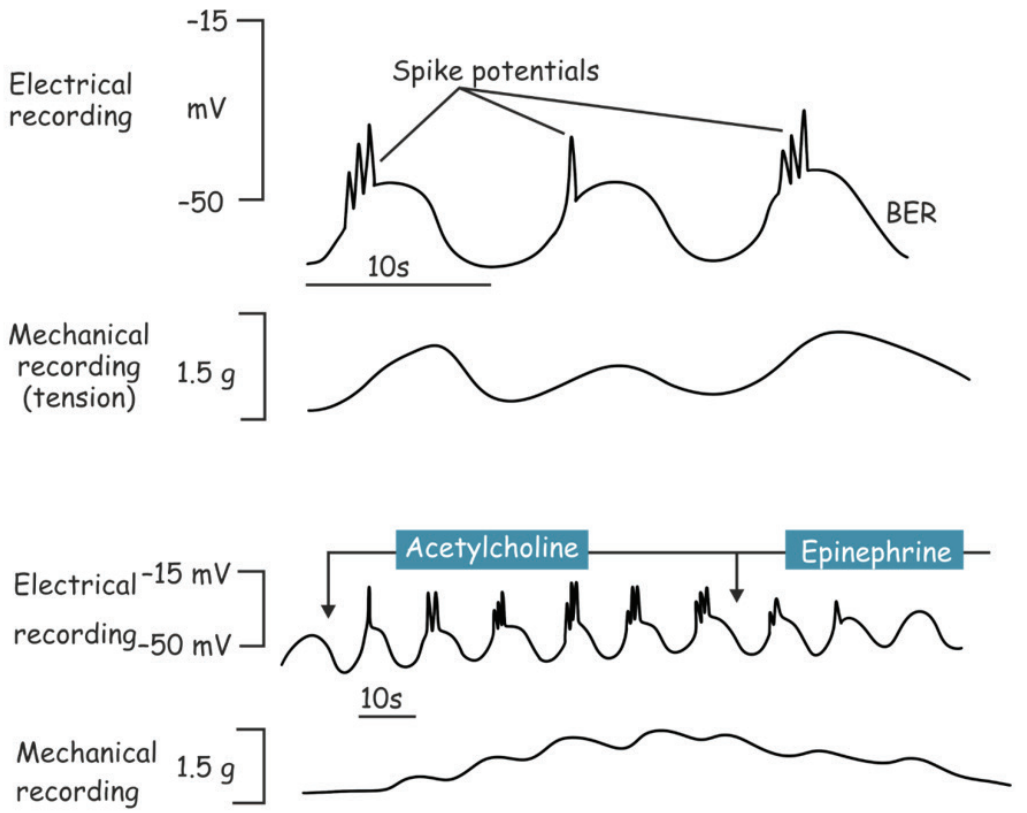
ELECTRICAL ACTIVITY IN SMOOTH MUSCLE

- RMP of smooth muscle is – 40 to – 60 mv
- Oscillatory type of membrane potential is present in smooth muscle

Active space

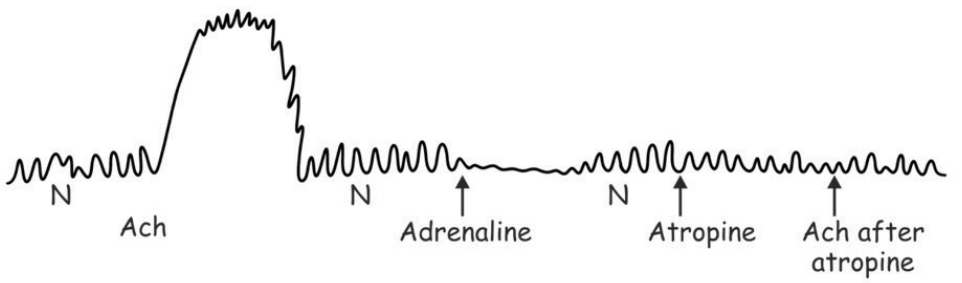
- Oscillation in membrane potential is due to .....
  .....
- This variation in membrane potential is known as BER (Basic electrical rhythm)
- Pacemaker cells of the GI tract are .....
  .....
- ..... are responsible for BER
- Magnitude of BER is .....
- Frequency is variable in different parts of the GI tract
- Max frequency of BER occurs in the .....
  .....
- Min frequency of BER occurs in the .....
  .....
- ..... is responsible for GI smooth muscle contraction
- Depolarisation in the smooth muscle of GIT is due to ..... influx
- Repolarisation is due to ..... efflux.

Active space



Mechanical recording or rabbit intestine

Administration of acetylcholine increases the tone and contractility of the smooth muscle of the small intestine, whereas adrenaline decreases the tone and contractility of smooth muscle



Active space

Mechanism of Smooth Muscle Contraction

Increase in sarcoplasmic calcium (mainly comes from ECF)

↓

Calcium binds to calmodulin

↓ forms

..... complex

↓

Activates .....

↓

..... of myosin light chains in myosin head

↓

Activates .....

↓

Actin-myosin cross-bridge formation

↓

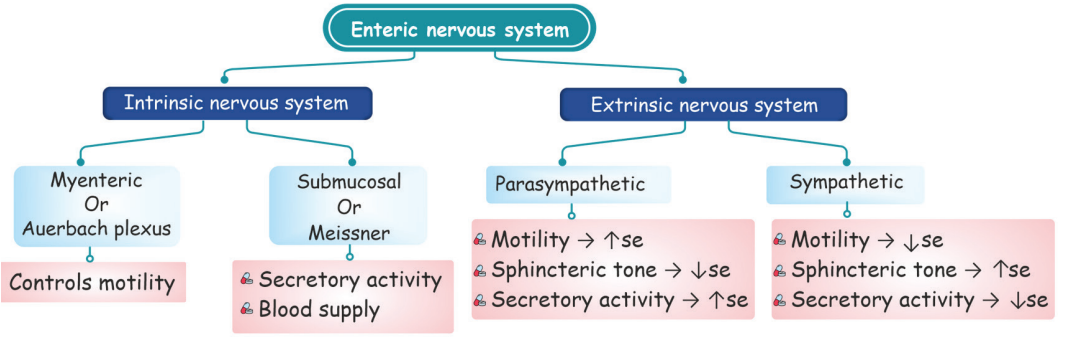
Smooth muscle contraction

- Relaxation of smooth muscle in GIT is by ..... which causes dephosphorylation of the myosin light chain in the myosin head
- Latch mechanism – Slower rate of actin-myosin cross-bridge cycling is known as the ....., seen only in smooth muscle

Active space

GIT

Neural control of GIT



Neural control of GIT

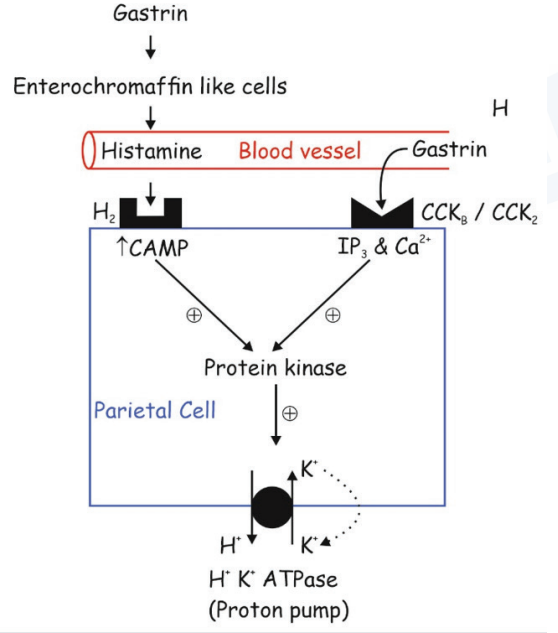
1. Hormones secreted by stomach:

   a. Gastrin:

   Secreted by ..... in the antrum

   Stimulus → ....., ..., ACh

   Function → Increases .....



Active space

b. Ghrelin:

- Secreted by ..... cells in the fundus
- ..... (Increases appetite)
- Increases fat deposition
- Sleep deprivation → Increases ghrelin

2. Small intestinal hormone:

a. CCK:

- Secreted by ..... in the upper small intestine
- Increases ..... contraction
- Increases .....
- Potentiates action of secretin
- Produces satiety
- CCK increases Somatostatin secretion

Stimulus:

- Distension
- Products of proteins and fats digestion

b. Secretin:

- Secreted by .....
- Increases ..... content in bile and pancreatic juice

Active space

c. GIP (.....)

- Secreted by .....
- GIP → Increases .....

d. GLP-1 (Glucagon Like Peptide-1)

- Secreted by .....
- Increases .....
- GLP-1 is known as ..... hormone of GIT.

e. Somatostatin

- Secreted by .....
- Decreases .....
- Decreases .....
- Decreases Gastrin secretion

f. Motilin

- Secreted by .....
- Responsible for MMCS (Migrating Motor Complexes)
- Increases gut motility
- Erythromycin can bind with motilin receptors

Active space

DIGESTION & ABSORPTION

SECRETIONS OF GI TRACT

Substances	Daily Secretion	pH
Saliva		
Gastric		
Pancreatic		
Brunner's gland		
Intestinal		
Colonic		

Note: Colonic secretion is very rich in .....

SALIVA

Three Pairs of major salivary glands are:

- Parotid
- Submandibular (maximum contribution by submandibular gland)
- Sublingual

Three Phases:

1.

Active space

2.

3.

Three Enzymes:

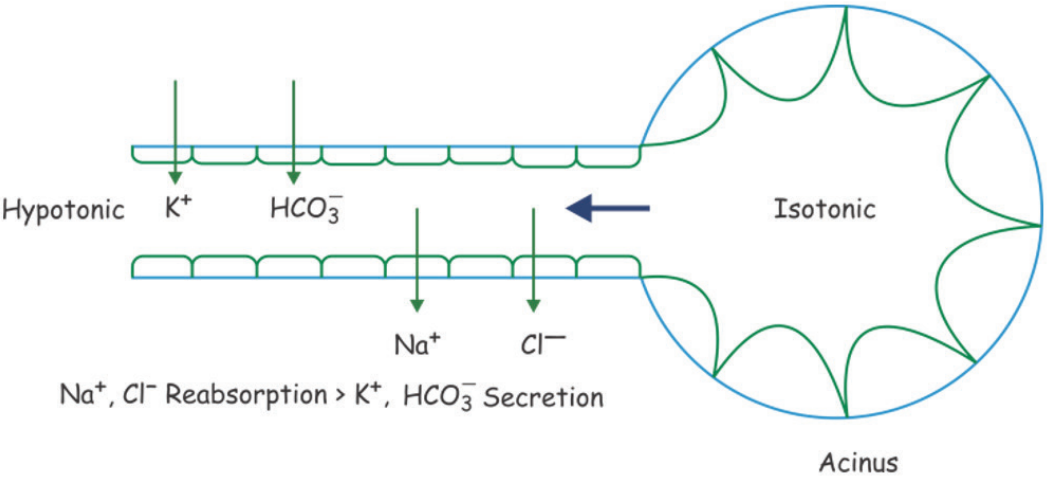
- ..... (Cl– is required for activation) for digestion of carbohydrates
- ..... (bactericidal)
- ..... (by Ebner's glands on the dorsum of the tongue; for digestion of fats)

Mechanism of salivary secretion

- Primary secretion in the acini is .....
- As saliva flows through the ducts,
  - ..... and ..... are reabsorbed
  - ..... and ..... are secreted (ducts are impermeable to water)
  - But sodium and chloride reabsorption is more than potassium and bicarbonate secretion, thus saliva becomes hypotonic.

Active space

- Aldosterone receptors are present in salivary glands
- Aldosterone ..... sodium reabsorption and potassium secretion in saliva



Three Pairs of major salivary glands are:

- Parotid
- Submandibular (maximum contribution by submandibular gland)
- Sublingual

GASTRIC SECRETIONS (HCl)

Three Phases

1.

Active space

2.

3.

2 Cells:

1. Parietal cell/Oxyntic cell

- Secrete ....., ...
- Patients of Pernicious Anaemia have antibodies against the parietal cell mass; they also suffer from hypochlorhydria / achlorhydria

2. Chief Cell/Peptic Cell

- Secrete ..... and .....

Functions of HCL

- .....
- .....
- .....

Once solubilized ferric iron is converted into ferrous iron with the help of vitamin C (iron can only be reabsorbed in the ferrous state)

Active space

PANCREATIC SECRETION

Three Phases:

- Cephalic
- Gastric
- Intestinal → Most important phase; under neural and hormonal control (CCK, Secretin)

Enzymes present in pancreatic juice:

Amylase, Trypsin, Chymotrypsin, Elastase, Carboxypeptidase, Lipases, nucleotidase

INTESTINAL SECRETION

Enzymes in intestinal secretion are:

- Enterokinase/Enteropeptidase

↓

Trypsinogen

↓

Trypsin

↓

Converts other inactive pancreatic enzymes into active form

- Disaccharidases
- Di - and Tripeptidases

Active space

DIGESTION & ABSORPTION OF CARBOHYDRATES

- Carbohydrates undergo
  - ✦ Luminal digestion
  - ✦ Brush border digestion
- Carbohydrates are broken into monosaccharides- glucose, galactose and fructose
- Glucose and galactose are mainly absorbed by .... on the luminal membrane and ..... on the basolateral membrane. Fructose is absorbed by ..... on both luminal and basolateral membranes.

DIGESTION & ABSORPTION OF PROTEINS

- Luminal
- Brush border
- Intra cytoplasmic

- Amino acids are absorbed by .....
- Amino acids leave the cell by .....

Active space

DIGESTION & ABSORPTION OF FATS

Fats are ingested mainly in the form of triglycerides, cholesterol esters and esters of fat-soluble vitamins

Triglycerides $\xrightarrow{\text{Lipases}}$ 2 Monoglycerides + Free fatty acids

Cholesterol esters → free cholesterol

Monoglycerides, free fatty acids, cholesterol, and fat-soluble vitamins are transported into the enterocyte with the help of special transport proteins

Inside the cell, they undergo re-esterification to form chylomicrons

Chylomicrons enter the lymphatics, which drain via the thoracic duct into the systemic circulation to be taken to the liver for further metabolism

Active space

ABSORPTION IN DIFFERENT PARTS OF THE GI TRACT

Short-Chain Fatty Acids

Water absorption

Active space

MIST

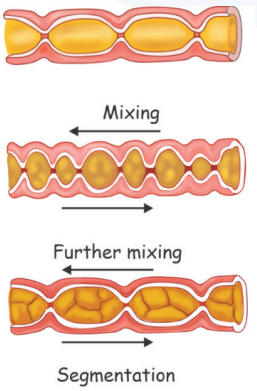
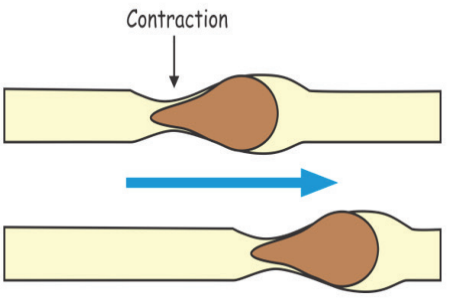
MOVEMENTS OF GI TRACT

Types of movements seen in the GI tract

1. Segmentation contractions
2. Peristalsis
3. Migrating motor complexes

SEGMENTATION CONTRACTION

Fats are ingested mainly in the form o

SEGMENTATION CONTRACTION	PERISTALSIS
	

Active space

GI REFLEXES

1. Receptive relaxation of the stomach:

2. Gastrocolic reflex:

3. Gastroileal reflex:

4. Enterogastric reflex:

Stimuli:

- .....
- .....
- .....
- .....

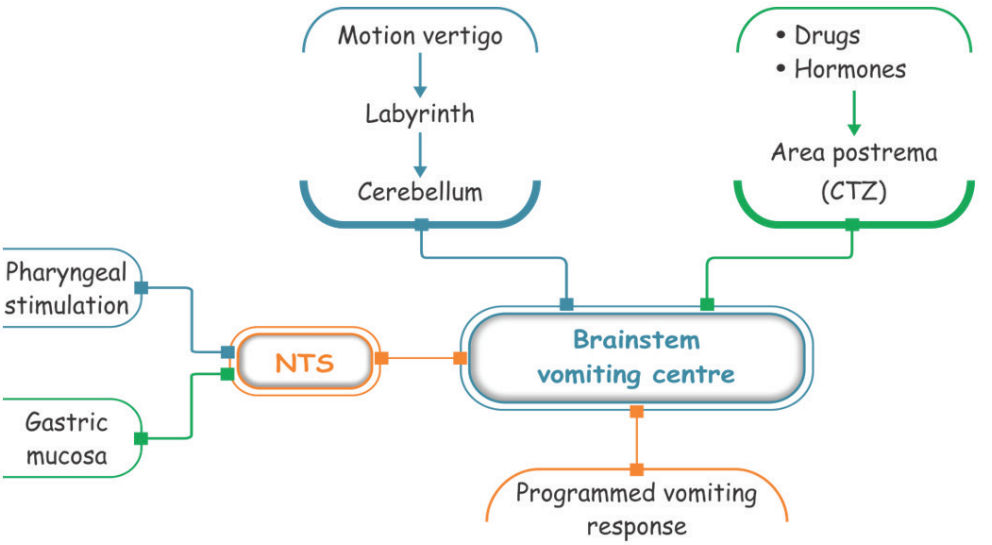
Response:

- Decrease in gastric motility
- Decrease in gastric secretion

Active space

VOMITING REFLEXES

1. Receptive relaxation o



Steps of the vomiting reflex

1. Antiperistalsis
2. Contraction of abdominal muscle & increased Abdominal Pressure
3. Relaxation of the lower oesophageal sphincter
4. Respiration stops
5. Elevation of the soft palate and closure of the upper airway by the glottis
6. Expulsion of vomitus

Active space

ENDOCRINOLOGY

INTRODUCTION

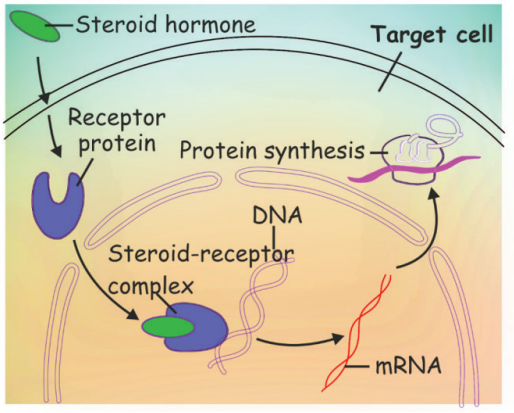
Hormones are of two types

Lipid Soluble	Water Soluble

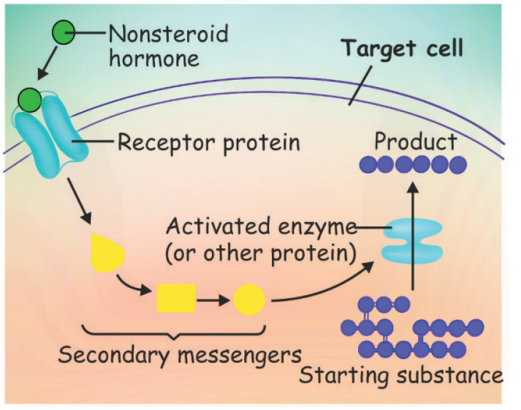
Active space

Examples of transport proteins for lipid-soluble hormones

- Testosterone and Oestrogen - Transported by .....
  .....
- Cortisol and Progesterone – Transported by .....
  .....
- T4 is mainly transported with .....
  .....
- T3 is mainly transported with .....
  .....



a. Steroid Hormone Response



b. Nonsteroid Hormone Response

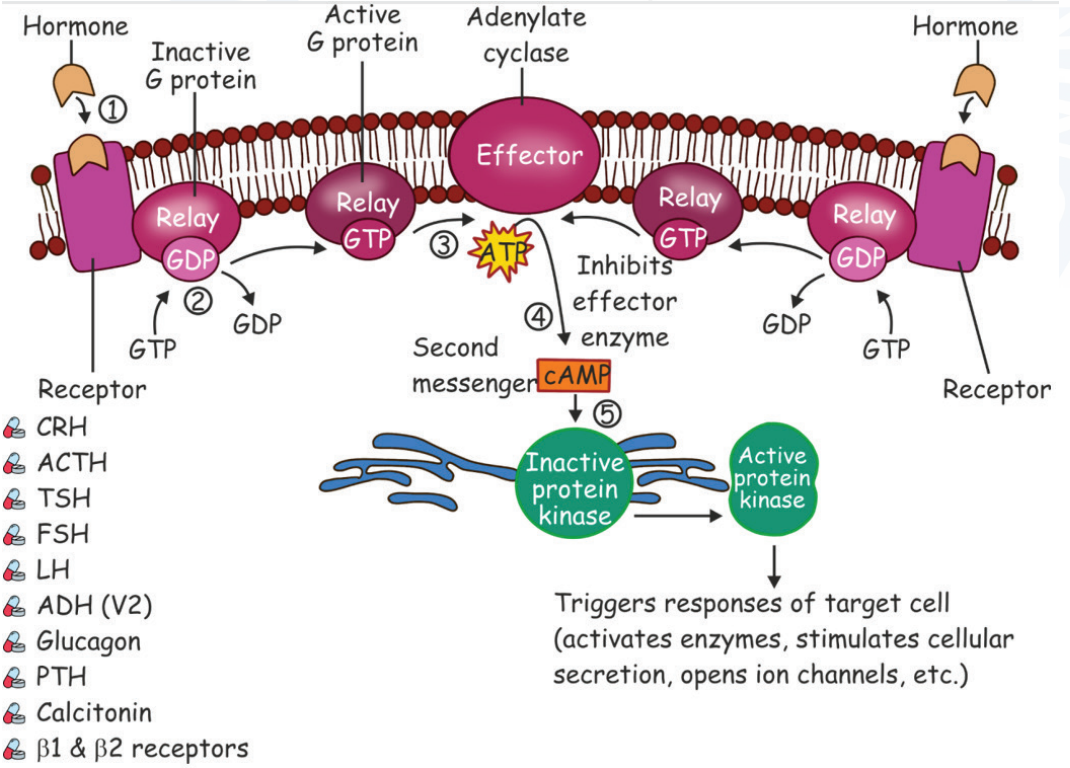
cAMP		Phospholipase C (IP3/DAG)
CRH (Corticotrophin Releasing Hormone)	Hypothalamic Hormone	GHRH GnRH TRH

Active space

MIST

ACTH MSH TSH FSH LH	Anterior Pituitary Hormones	
ADH (V2)	Posterior Pituitary Hormones	ADH (V1 & V3) Oxytocin
Glucagon PTH Calcitonin α_1 & α_2 Receptor[Q] Hcg		Angiotensin II α_1 adrenergic receptor

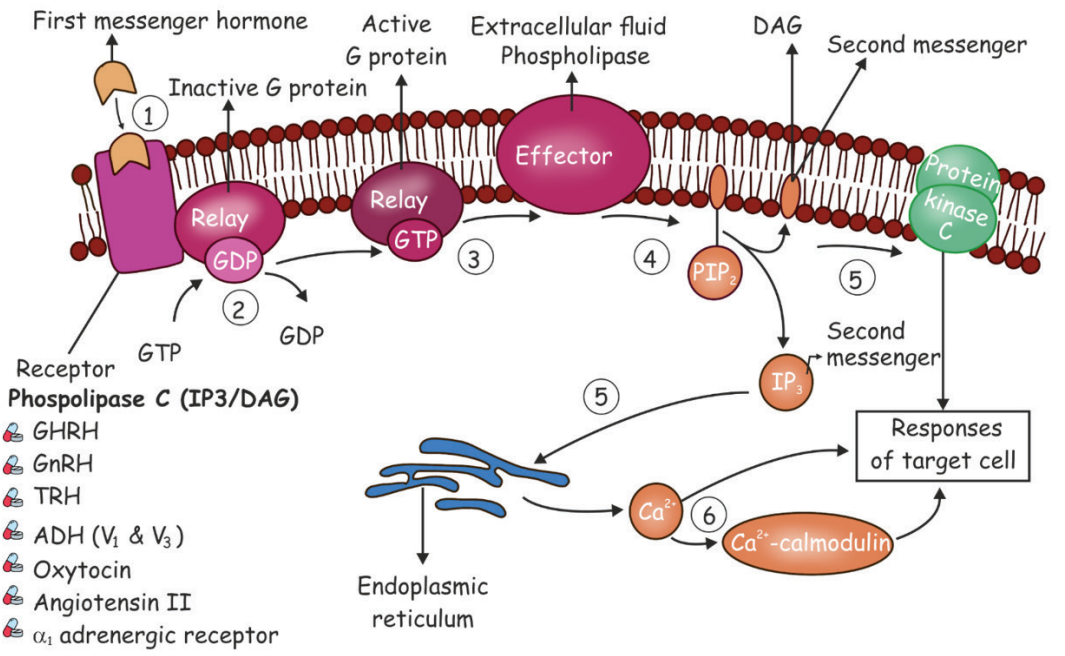
cAMP as a second messenger :



- CRH
- ACTH
- TSH
- FSH
- LH
- ADH (V2)
- Glucagon
- PTH
- Calcitonin
- β1 & β2 receptors

Active space

IP3/DAG (Phospholipase C) as second messengers



Hormones using cGMP as second messenger-

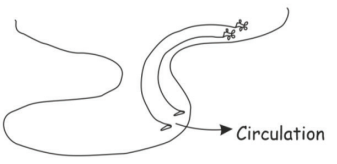
ANP, BNP

Role of Hypothalamus in Posterior Pituitary Hormone:

Neurosecretion: It involves two nuclei of hypothalamus

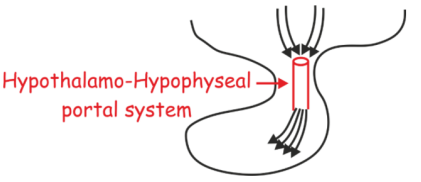
	ADH	Oxytocin
Supraoptic Nuclei		
Paraventricular Nuclei		

These hormones are transported by "axoplasmic transport" into the posterior pituitary, and from the posterior pituitary, they come into circulation.



Active space

Role of Hypothalamus in Ant Pituitary Hormone secretion



Regulatory & inhibitory hormones enter the hypothalamo-hypophyseal portal system and affect the secretion of anterior pituitary hormone.

Nuclei	Role
Paraventricular Nuclei	
Arcuate Nuclei	

Stalk Effect: If there is a thrombus in the hypothalamo-hypophyseal portal system or an accidental/traumatic section of the pituitary stalk, hypothalamic hormones cannot reach the anterior pituitary. As a result, all anterior pituita ry hormones will be reduced except Prolactin.

ANTERIOR PUTUITARY HORMONES

CELLS OF ANTERIOR PITUITARY:

1. Acidophils:
   a. Somatotropes → .....
   b. Lactotropes → .....
2. Basophils:
   a. Corticotropes → .....
   b. Gonadotropes → .....
   c. Thyrotropes → .....

Knowledge Capsule

In a patient with choriocarcinoma, hCG levels are very high. This patient can present with thyrotoxicosis because hCG and TSH have similar ∝-subunits.

- ..... are maximum in number
- ..... are minimum in number
- FSH
- LH
- TSH
- hCG

Similar α-subunits, but different β-subunits

GROWTH HORMONE

- Regulation of growth hormone secretion:
  - ✦ ..... increases growth hormone secretion
  - ✦ ..... decreases growth hormone secretion
- Growth Hormone Releasing Hormone (GHRH) is the major regulating hormone (not Somatostatin)

Stimuli which increase Growth Hormone secretion

Active space

Stress Hormones:

- Increase blood glucose levels
- Increase free fatty acid levels

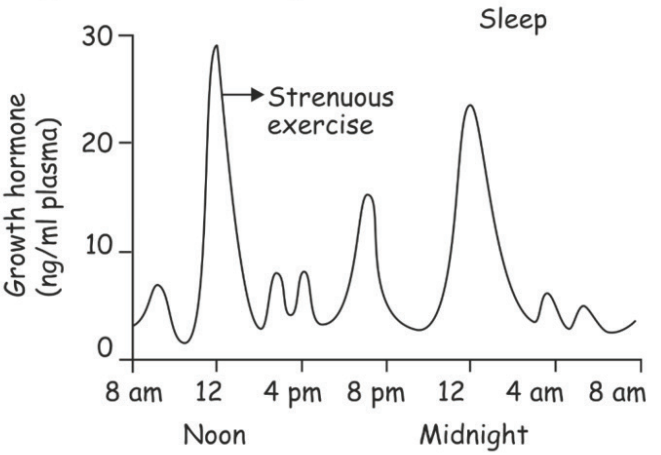
1. Catecholamines
2. Glucocorticoids
3. Growth Hormones
4. Glucagon
5. ADH
6. Thyroid (±)

ACTIONS OF GROWTH HORMONES

Direct Action	Indirect Action

Active space

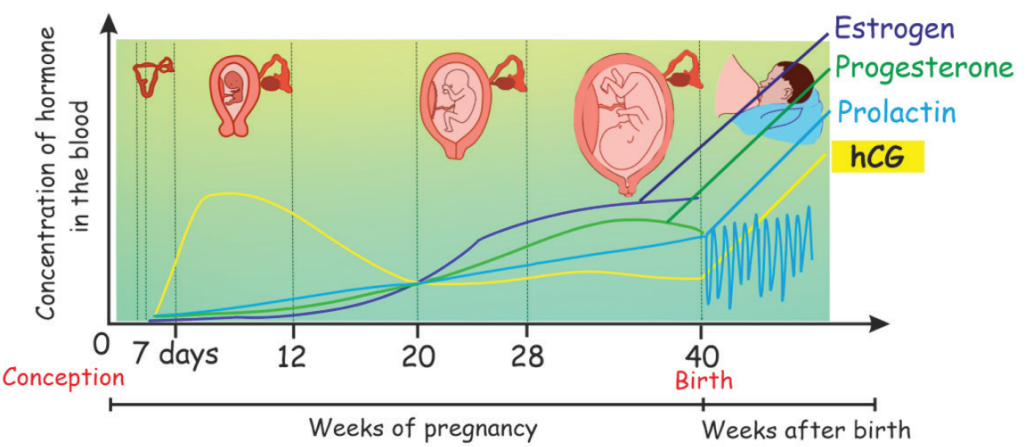
Typical variations in growth hormone secretion



5 to 20yrs	6ng/ml
20 to 40yrs	3ng/ml
40 to 70yrs	1.6ng/ml

PROLACTIN

Stimuli which increase Prolactin (PRL) secretion are:

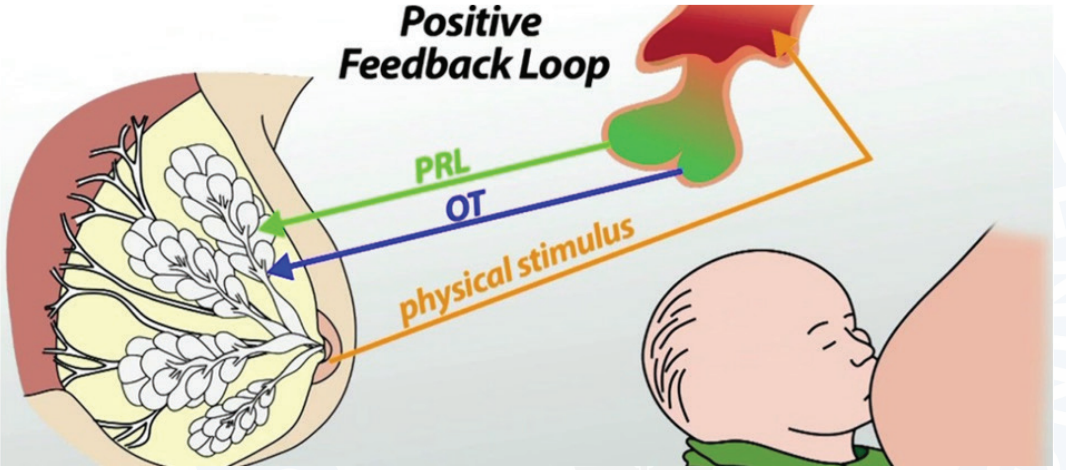


Active space

After delivery, prolactin levels rapidly drop to pre-pregnant levels, but every time the mother nurses the baby, there is a sharp increase in prolactin. This increase in prolactin occurs only at the time of nursing and does not occur in the intervals between nursing.

Actions of Prolactin (PRL):

- PRL causes .....
- PRL increases .....



1. Oxytocin: Causes contraction of Myoepithelial cells which surround the ducts causing the 'Milk Let Down' effect

POSTERIOR PITUITARY HORMONES

INTRODUCTION

ADH (Anti-Diuretic Hormone)/Vasopressin

- Mainly synthesised in the 'Supra optic nucleus' of hypothalamus
- Stimuli which stimulate/increase ADH secretion

Active space

1. Increase in plasma osmolality:

   Normal plasma osmolality: 280- 290mOsm/L

   Increase in plasma osmolality (like in sweating) by just +5 mosm/L is sufficient to increase ADH secretion.

2. Decrease in blood volume: 10-15% of blood loss → Increase in ADH secretion

3.

4.

5.

6.

7.

8.

9.

Active space

Receptors for ADH and their second messengers

- V1- IP3 & Ca2+
- V2- cAMP
- V3- IP3 & Ca2+

→ All ADH receptors are GPCR (G-protein coupled receptors)

I. V1 Receptor:

Locations	Actions
1. Vascular Smooth muscle	

II. V2 Receptor:

Locations	Actions
1. Collecting Ducts	
2. Vascular endothelium	

III. V3 Receptor

Locations	Actions
1. Anterior Pituitary	

Active space

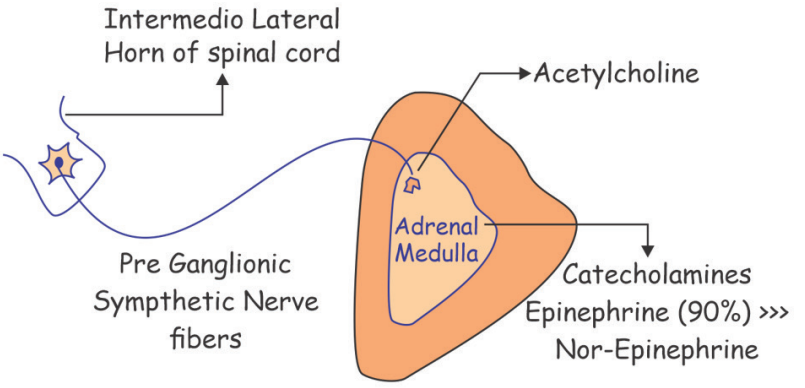
Cellular Mechanism of Action of ADH

- Adenylate cyclase converts the ATP into cAMP
- cAMP acts on protein kinase, and this kinase will cause phosphorylation of Aquaporin-2
- Once Aquaporin-2 gets phosphorylated then, they are inserted into the luminal membrane, which increases H_2O reabsorption
- Aquaporin-3 & 4 are constitutionally present on the Baso-lateral membrane. (They are in fixed numbers, whereas numbers of Aquaporin-2 can be changed.)

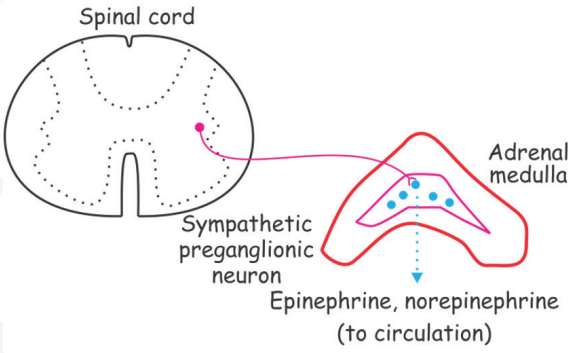
Active space

ADRENAL HORMONES

INTRODUCTION OF ADRENAL GLANDS



- Adrenal Medulla is considered as displaced sympathetic ganglion because, during embryological origin, some of the Neural Crest Cells (NCC) migrate and reach the adrenal gland and gives rise to the adrenal medulla



- Adrenal Medulla is under the control of the sympathetic nervous system
- All "preganglionic" sympathetic or parasympathetic nerve fibres are ..... so the neurotransmitter secreted by the sympathetic nerve fibres innervating the adrenal glands is .....
- 90% of the secretion from the adrenal medulla is ....

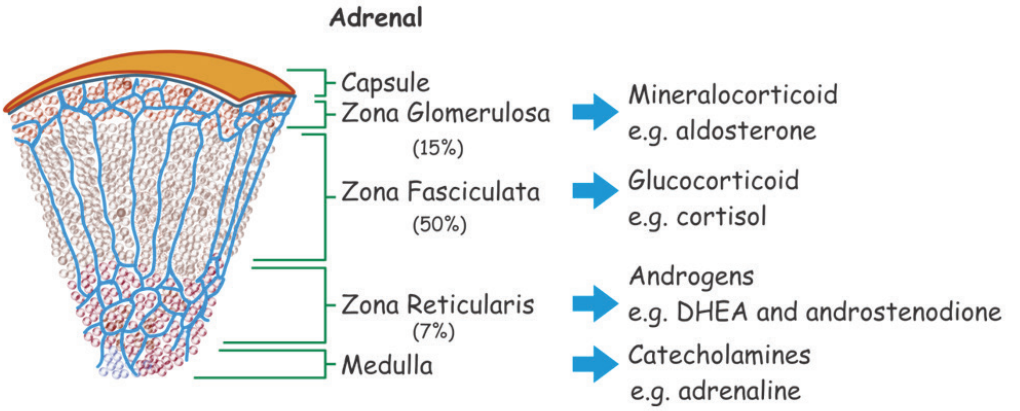
Active space

Knowledge Capsule

Main site of secretion of Norepinephrine (NE) are the postganglionic sympathetic nerve fibres

INTRODUCTION OF ADRENAL GLANDS

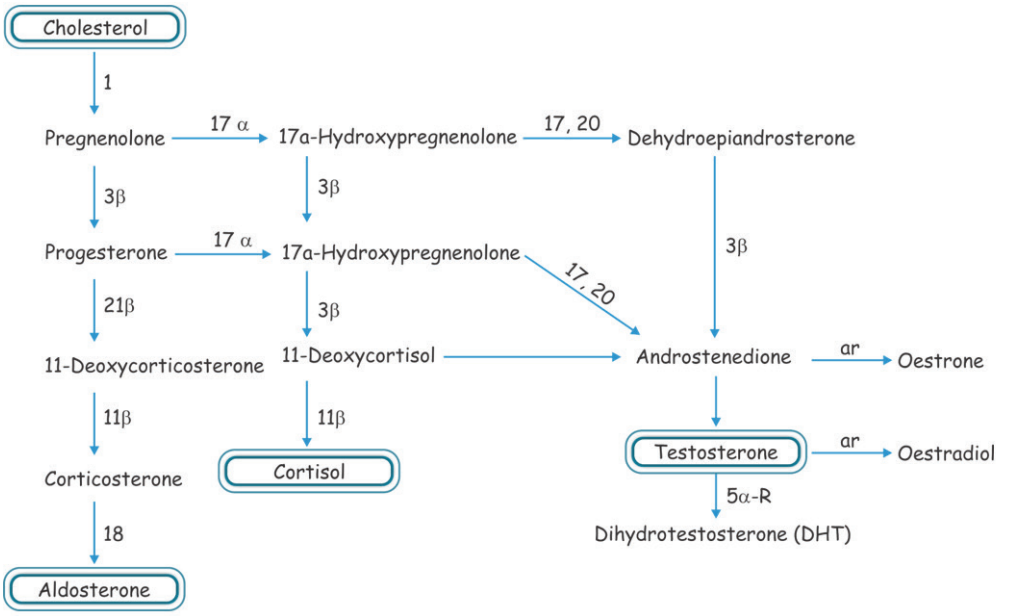
Zones	Secretes	Under control of
Zona Glomerulosa	Aldosterone	
Zona Fasciculata	Glucocorticoids + Androgens	
Zona Reticularis	Androgen + Glucocorticoids	



- ACTH is synthesised as a very large precursor molecule called Pro-opiomelanocortin (POMC) which first breaks into ACTH, lipotropins and endorphin

Active space

Steps in the synthesis of adrenal cortical hormones



Enzyme Deficiency

Deficiencies	Aldosterone	Cortisol	Androgen
21 β-hydroxylase 90% Of Congenital Adrenal Hyperplasia (CAH)			
11 β-hydroxylase 5% of CAH			
17 a-hydroxylase			
17, 20-lyase			

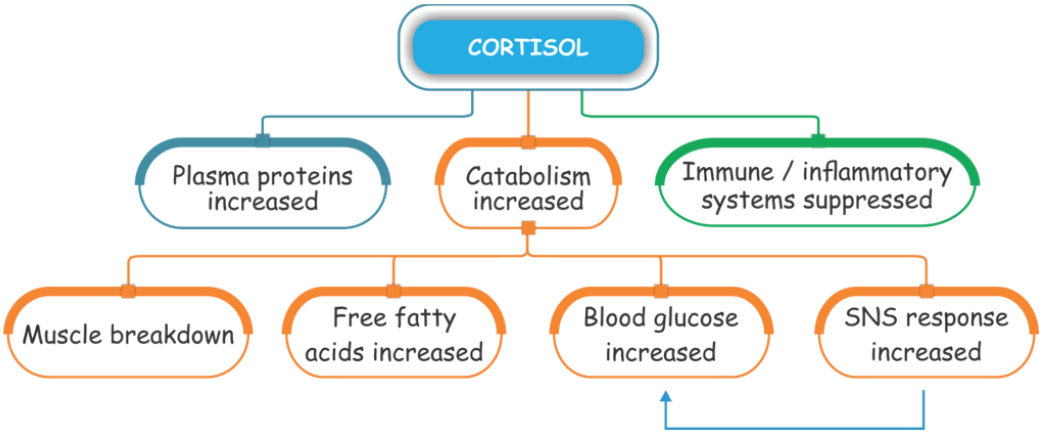
- 11-Deoxycorticosterone has mineralocorticoid effect

MIST

Active space

	21 β-hydroxylase deficiency	11 β-hydroxylase deficiency
11-Deoxycorticosterone		
Mineralocorticoid activity		
Blood Pressure		
Plasma K^+		

ACTIONS OF CORTISOL



- Cortisol Reduces: B =

  E =

  L =

- Cortisol Increases: ..... , ...

  , ... & .....

Active space

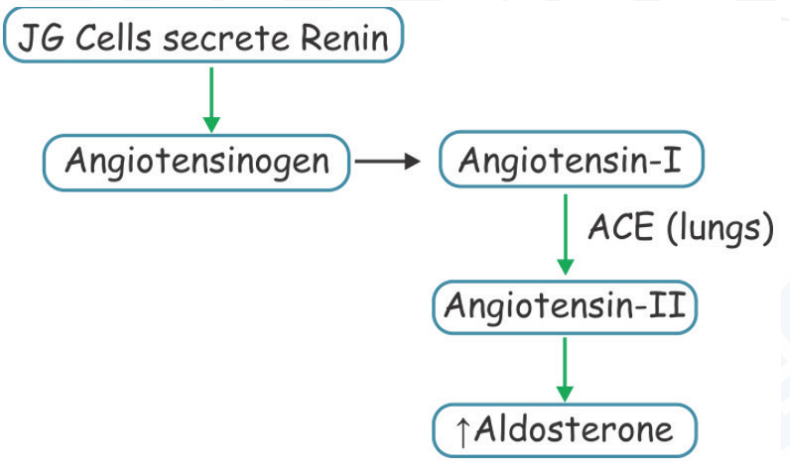
PERMISSIVE ACTION OF GLUCOCORTICOIDS

- Glucocorticoids have permissive action, i.e. enhance the action of:

1. Glucagon:

2. Catecholamines:

ALDOSTERONE



Normal Plasma Potassium = 3.5–5.0 mmol/L

Active space

Stimuli which increase Renin:

Aldosterone Receptors are present in

1.

2.

3.

4.

5.

6.

Active space

PANCREATIC HORMONES

CELLS OF PANCREAS

α – cells (20% of total cells)	1.
β – cells (60-75% of total cells)	1. Insulin 2. Amylin[Q] Ratio of insulin: amylin secretion is ....
δ – Cells	
F- Cells	
β – Cells	

Mechanism of Insulin Secretion

Glucose enters into the beta cells via Glut-2

↓

Phosphorylated by glucokinase

↓

Metabolised to pyruvate in the cytoplasm

↓

Pyruvate enters mitochondria and is metabolized via citric acid cycle

↓

MIST

Active space

Formation of ATP by oxidative phosphorylation
(high ATP/ADP ratio)
↓
ATP blocks ATP-sensitive potassium channels
↓
Closure of ATP-sensitive potassium channels
↓
Decreased potassium efflux
↓
Accumulation of positive charges within the
beta cells causes depolarisation
↓
Opening of voltage-gated calcium channels
↓
Influx of calcium
↓
Exocytosis of Insulin and c-peptide

- ..... act on ATP-sensitive K+ channel & block them, which lead to increased insulin secretion independent of blood glucose level
- Insulin: C-peptide is secreted in a ..... ratio
- Endogenous insulin secretion in IDDM patient who is receiving injectable insulin is measured by C-peptide levels via radioimmunoassay, which provides an index of β-cells function in patients receiving exogenous insulin

Active space

Regulation of Insulin Secretion

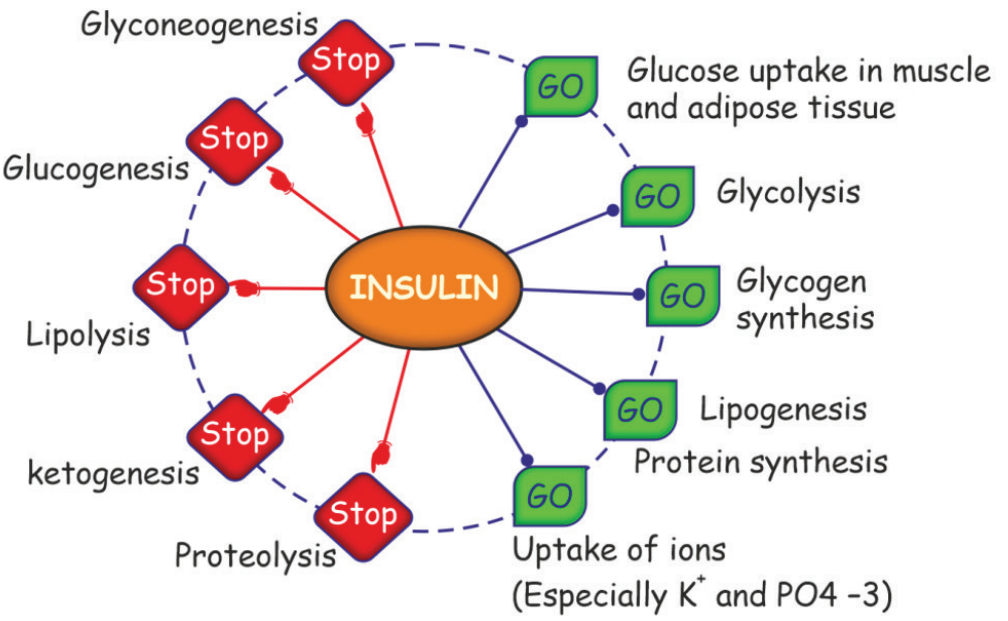
Stimulators of Insulin Secretion	Inhibitors of Insulin Secretion
1. Glucose, Mannose 2. Amino Acids – Leucine, Arginine 3. β-keto acids 4. Intestinal hormones gastrin, CCK 5. GIP, secretin, glucagon, GLP-1 6. cAMP & various cAMP-generating substances β adrenergic agonists, glucagon, theophylline 7. Sympathetic stimulation (β2 adrenergic receptors) 8. Sulfonylureas 9. Ach via M4 Receptors	1. Somatostatin 2. Sympathetic stimulation [a2 adrenergic receptors] 3. Propranolol 4. K+ channels depletion 5. Phenytoin, Alloxan, and microtubules inhibitors 6. Leptin

Knowledge Capsule

Overall effect of sympathetic stimulation: Decreased insulin secretion

Active space

Actions of Insulins



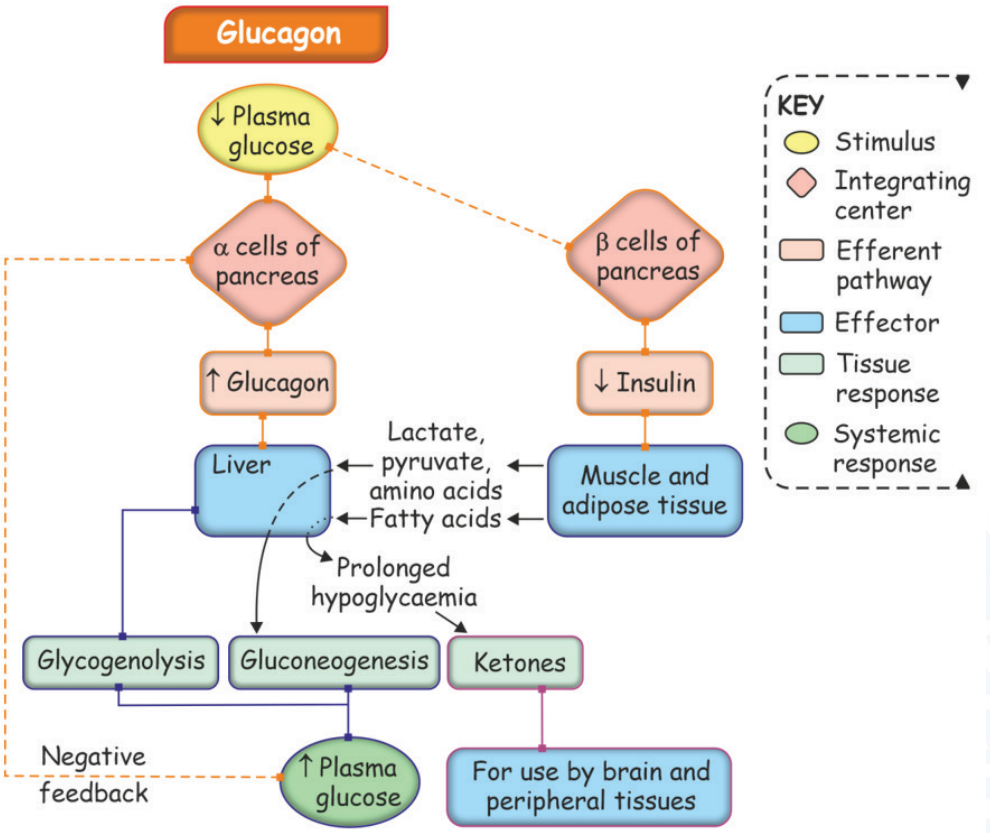
Patient with Hyperkalaemia is treated with insulin (which increases the uptake of potassium ions), along with dextrose to prevent hypoglycaemia

Target organs for Insulin are: ....., ...
.....,.....

Glucagon

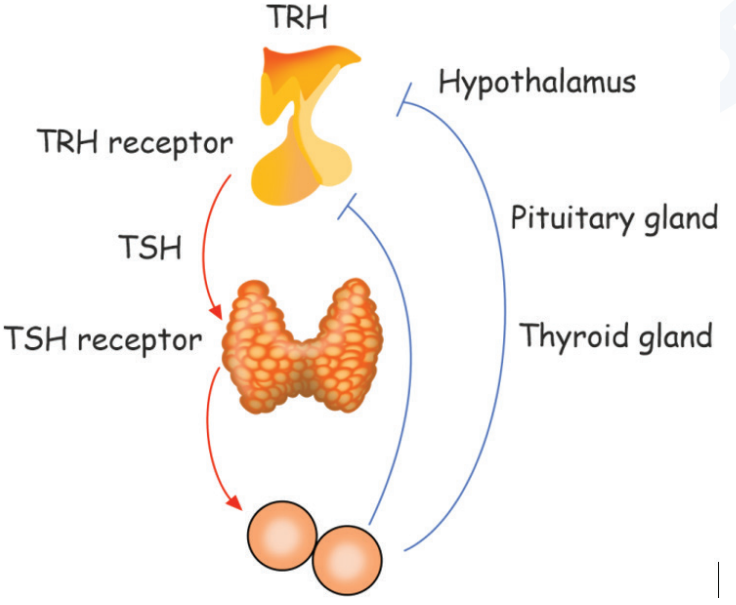
Target organs for Glucagon – .....

Active space



THYROID AND PARATHYROID HORMONES

REGULATION OF THYROID HORMONE SECRETION



Active space

Synthesis of Thyroid Hormones

Thyroid Cells:

a.

b.

- Sodium Iodide Symport (NIS) is an example of a .....
  .....
- Drugs which inhibit NIS are:
  1.
  2.
  3.
- I-/Cl- exchanger is also known as .....
  .....
- Transport of iodine from blood to thyroid cells is via .....
  .....
- Transport of iodine from the cell into colloid is via .....
  ..... Colloid

Colloid

a.

b.

c. Coupling Reaction (also require thyroid peroxidase enzyme)

DIT + DIT → T4

DIT + MIT → T3

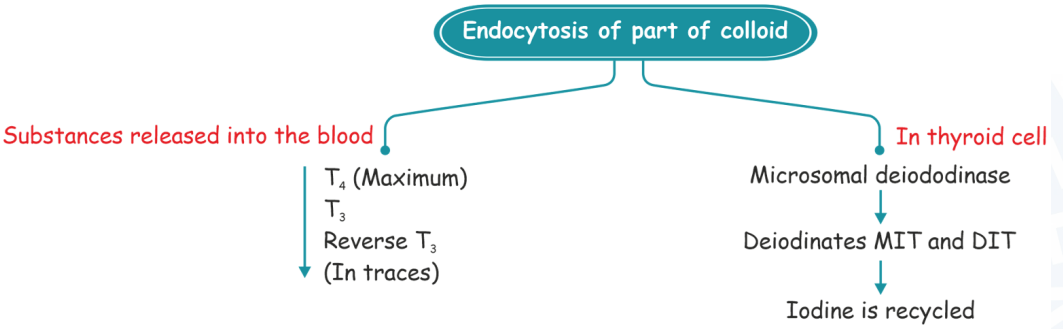
MIT + DIT → Reverse T3 (Physiologically Inactive)

Active space

Knowledge Capsule

Clinical: In Hashimoto's thyroiditis, antibodies against thyroid peroxidase enzyme cause its deficiency resulting in hypothyroidism.

- Storage
- Secretion- TSH acts on thyroid cells:

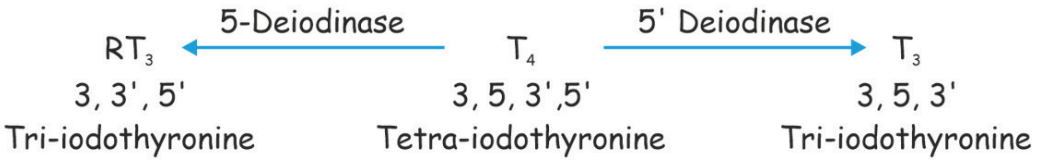


- In microsomal deiodinase deficiency, ...

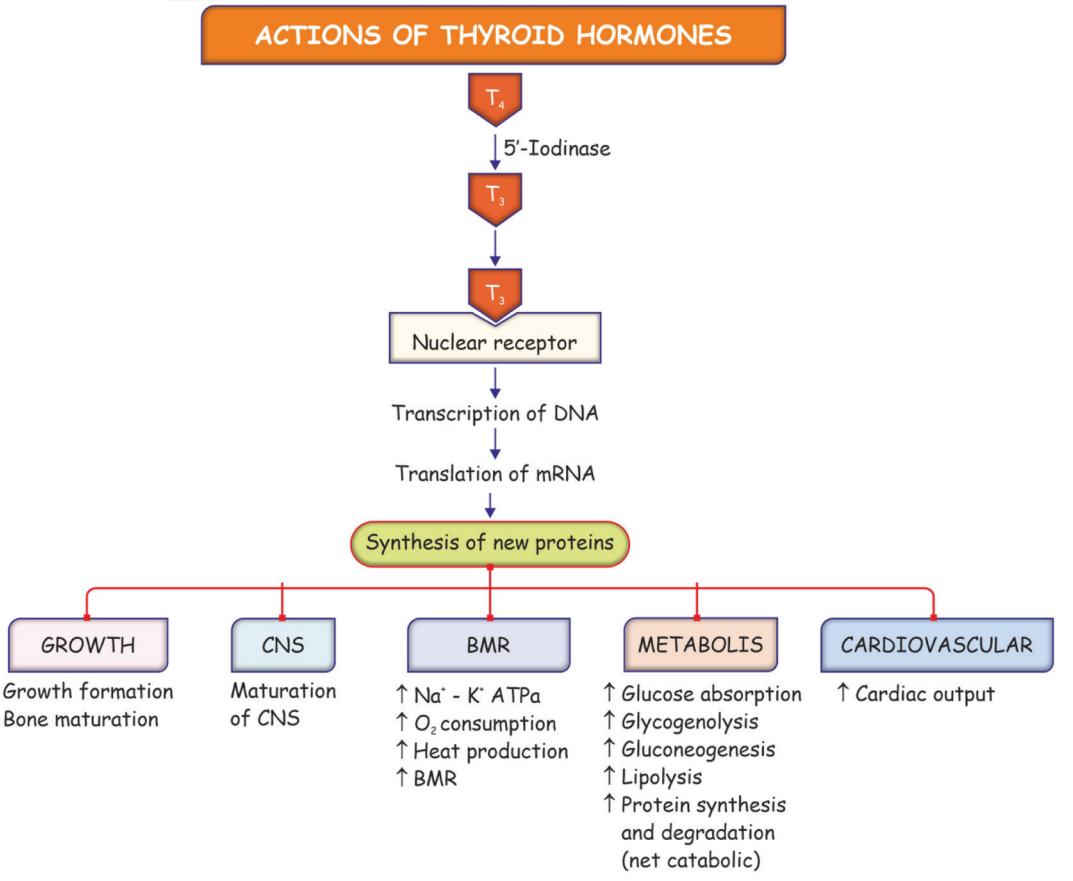
	T4	T3
Secreted by gland		
Binding with protein 1. Transthyretin 2. Albumin 3. TBG		
Free Hormone		
Binding with Receptor		

Active space

In Peripheral Tissues



Action of Thyroid Hormones



- O2 consumption increases in all tissues except .....
  .....
  .....

1. Thyroid hormones increase LDL receptors which decrease plasma cholesterol (patients with hypothyroidism have hypercholesteraemia)

Active space

2. Thyroid hormone acts on Carotene → converted into Vitamin A. Patients with hypothyroidism have carotenaemia (yellowish discolouration of the skin).

   Jaundice and carotenaemia present with yellowish discolouration of the skin, but in carotenaemia, sclera is spared- this helps to distinguish carotenaemia from jaundice.

Parathormone

Stimuli which stimulate the secretion of parathormone:

1. Decrease Plasma Ca^{2+}
2. Increase Plasma Pi (inorganic phosphorus) or Plasma phosphate

- Calcitonin comes from .....
- Parathormone receptors are present on ....
- Calcitonin receptors are present on ....

✦ Inhibits the "Osteoclastic activity"

✦ Calcitonin is used in Paget's disease. In Paget's, there is increased osteoclastic activity leading to increased bone resorption- calcitonin has an inhibitory effect on osteoclasts.

Actions of PTH on kidney

- PTH stimulates the activity of 1a-hydroxylase, which increases the conversion of 25 hydroxycholecalciferol to 1,25-dihydroxycholecalciferol (active form of vitamin D). 1,25-dihydroxycholecalciferol increases absorption of Calcium and phosphate from the GI tract and kidney.

Active space

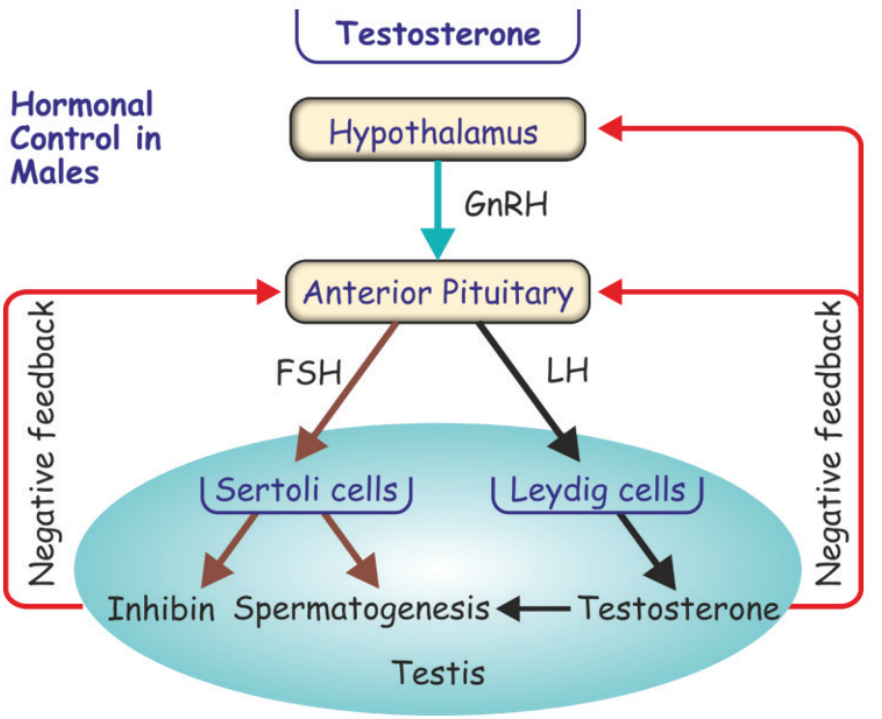
- On the kidney, parathormone has two actions on (dual effects on)
- In PCT: .....
- In DCT: .....

PTH action on Osteoblast

MALE REPRODUCTIVE HORMONE

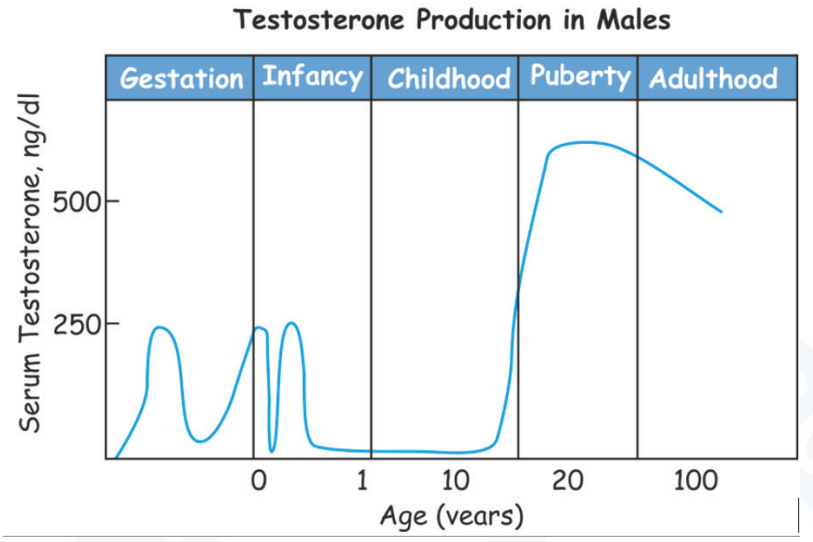
INTRODUCTION

Testosterone is involved in Gonadotropin regulation & increases Sertoli cell activity.



Active space

Testosterone Production in Males



Actions of Testosterone & Dihydrotestosterone

- Testosterone is converted into Dihydrotestosterone (DHT) by 5-α reductase

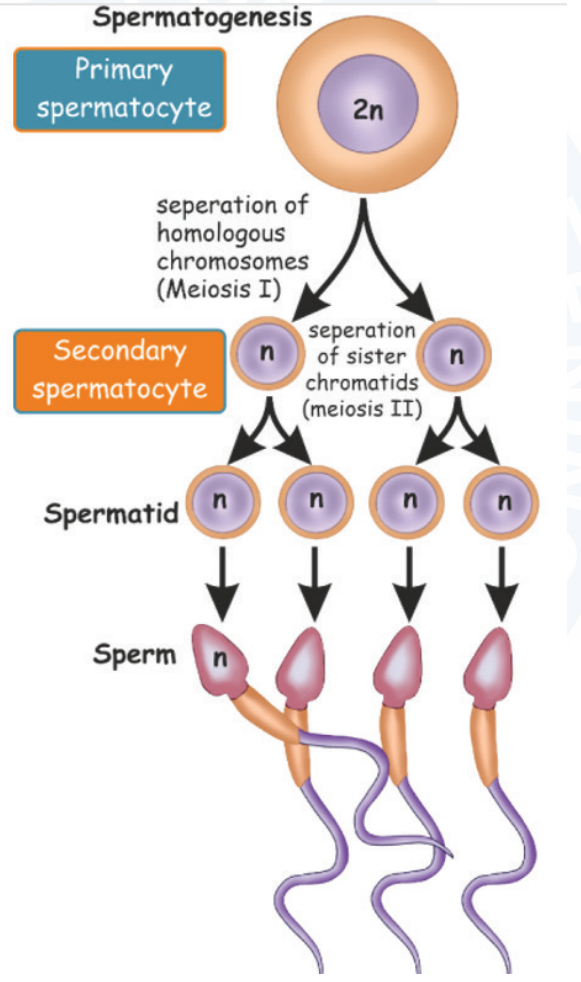
	TESTOSTERONE	DHT
In Utero		
After Puberty		

Penis-at-12 syndrome – Congenital deficiency of 5-α reductase, i.e. no DHT; hence external genitalia becomes female type, but internal genitalia is male type. The patient is born & raised as a girl until puberty when there is a sharp increase in testosterone resulting in clitoral enlargement, which begins to look like the penis (penis-at-12). MIST

Active space

SPERMATOGENESIS & OOGENESIS

- Duration of spermatogenesis = ...
- Last step of spermatogenesis, i.e. spermatid into sperm is dependent on ..... & .....
- All earlier steps in sperm development are independent of androgens



Active space

FUNCTION OF SERTOLI CELLS

1.

2.

3. Secretory Role: Secrete

   i. MIS: Mullerian Inhibiting substance (in fetal life)

   ii. ABP (Androgen Binding Proteins): Increase local concentration of androgens within the testis

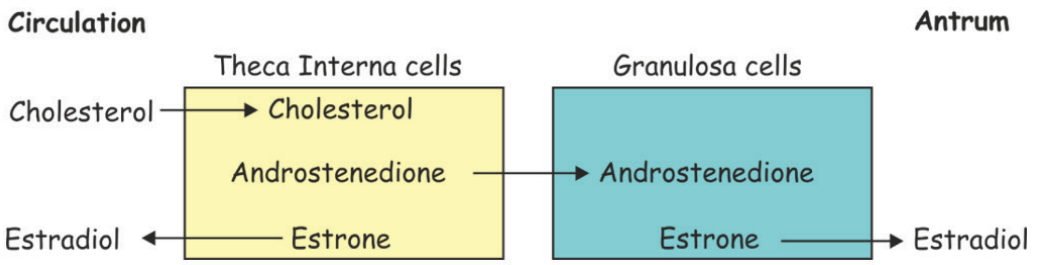
   iii. Aromatase: It acts on testosterone & converts it into oestrogen

FEMALE REPRODUCTIVE HORMONES

OVARIAN HORMONES

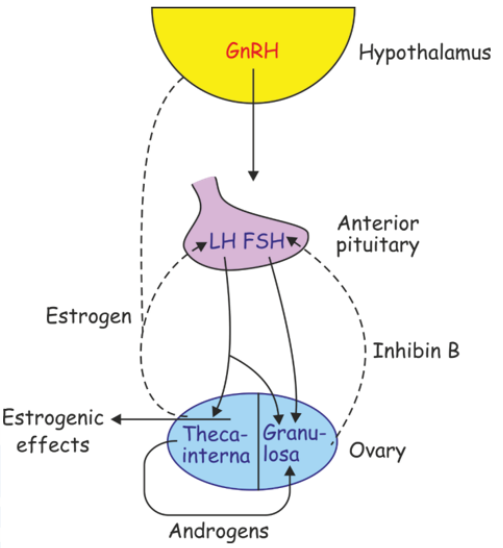
- Naturally occurring oestrogens are oestradiol, oestrone, and oestriol
- Secreted primarily by ..... of the ovarian follicles, corpus ..... & placenta
- ..... converts testosterone into oestradiol & androstenedione into oestrone
- Conversion of androstenedione to oestrone also occurs in fat, liver, muscles, and brain

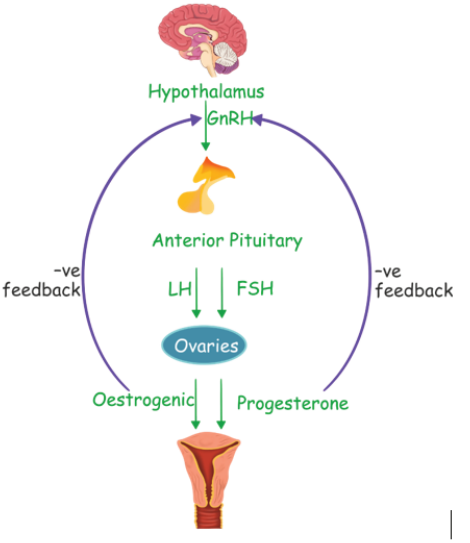
OESTROGEN PRODUCTION



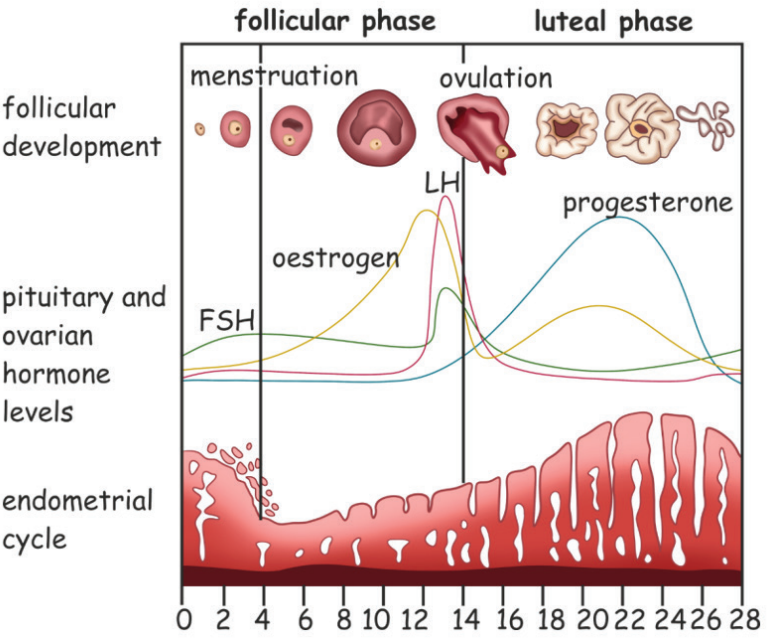
Active space

REGULATION OF FEMALE SEX HORMONE





VARIATION IN SEX HORMONES IN THE MENSTRUAL CYCLE



Active space

MIST

Changes in Gonadotropic Hormones During the Menstrual Cycle

- During each menstrual cycle, there is a cyclic increase and decrease of FSH and LH
- During the first few days of each monthly cycle, concentrations of both FSH and LH secreted by the anterior pituitary ..... slightly to moderately, with the increase in FSH slightly greater than the increase in LH and preceding it by a few days
- FSH and LH (especially FSH) cause accelerated growth of 6-12 primordial follicles each month

Follicular Changes During the Menstrual Cycle

- Early growth of the primary follicle up to the antral stage is stimulated mainly by..... alone
- Greatly accelerated growth then occurs, leading to still larger follicles called vesicular follicles- this accelerated growth is caused by

  (i) oestrogen secreted by the granulosa cells into the follicle, causes granulosa cells to form increasing numbers of FSH receptors, which causes a positive feedback effect,

  (ii) pituitary FSH and oestrogens combine to promote LH receptors on granulosa cells making them sensitive to the action of both FSH and LH,

  (iii) increasing oestrogens from follicle and LH from anterior pituitary act together to cause proliferation of the follicular thecal cells, which secrete both oestrogen and progesterone

- Ovum also enlarges in diameter another 3-4 fold

Active space

- Only one out of 5-11 follicles outgrows all others, reaching a diameter of 1-1.5 cm at the time of ovulation
- Ovulation normally occurs on the 14th day in a normal 28-day sexual cycle
- ..... surge is necessary for final follicular growth and ovulation
- LH starts increasing about 2 days before ovulation, rising 6 to 10 fold and peaking about 16 hours before ovulation
- FSH also increases about 2 to 3 fold, and FSH and LH act synergistically to cause ovulation
- During the first few hours after the expulsion of ovum, the remaining granulosa and theca cells change rapidly into lutein cells
- The corpus luteum, thus formed, secretes large amounts of progesterone and oestrogen with more amounts of progesterone
- If fertilisation does not occur, the corpus luteum involutes and becomes corpus albicans

Changes in the endometrium

- Proliferative/preovulatory/oestrogen phase of the endometrial cycle
- Endometrium proliferates and increases in thickness owing to increasing numbers of stromal cells and rapid proliferation of epithelial cells
- Progressive growth of endometrial glands and new blood vessels
- Endometrial glands, especially those of the cervical region, secrete a thin, stringy mucus
- Endometrium becomes 3-5 mm thick

Active space

Secretory/post-ovulatory/progestational phase

- Oestrogens cause a slight increase in cellular proliferation of the endometrium, whereas progesterone causes marked swelling and secretory development of the endometrium
- Glands increase in tortuosity and increase their secretions. Cytoplasm of stromal cells increases.
- Lipid and glycogen deposits increase greatly in the stromal cells
- Blood vessels to the endometrium become highly tortuous
- Endometrium increases in thickness by about 5-6 mms

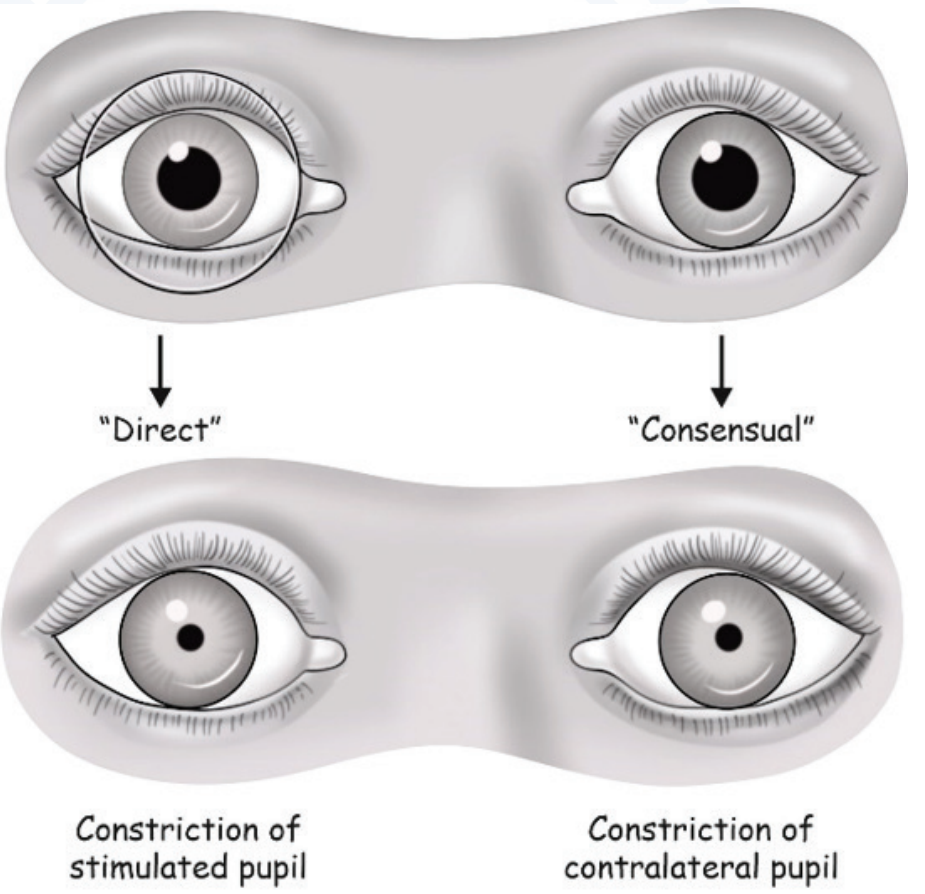
Menstruation

If the ovum is not fertilised, about 2 days before the end of the monthly cycle, the corpus luteum involutes, and oestrogen and progesterone decrease and menstruation follows.

Active space

SPECIAL SENSES

PUPILLARY LIGHT REFLEX



Active space

Pathway of pupillary light reflex

Light falls on retina

↓

Optic Nerve

↓

.....

↓

.....

↓

Preganglionic parasympathetic fibers in the oculomotor nerve

↓

Ciliary ganglion

↓

Postganglionic parasympathetic fibers

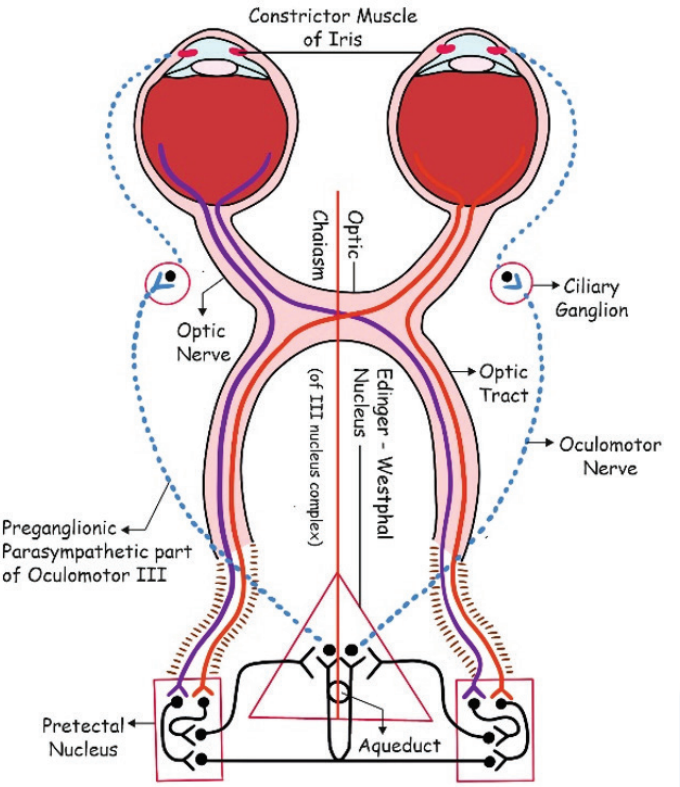
↓

..... pupillae

↓

..... of pupil

Active space



Knowledge Capsule

Dilator pupillae is under the control of sympathetic nerve fibre & constrictor pupillae is under the control of parasympathetic nerve fibres.

Accommodation Reflex

- Fixation of eyes on a near object

i. Pupillary constriction

ii. Contraction of ciliary - Relaxation of lens ligaments - Lens becomes more spherical - Increases refractive power of lens

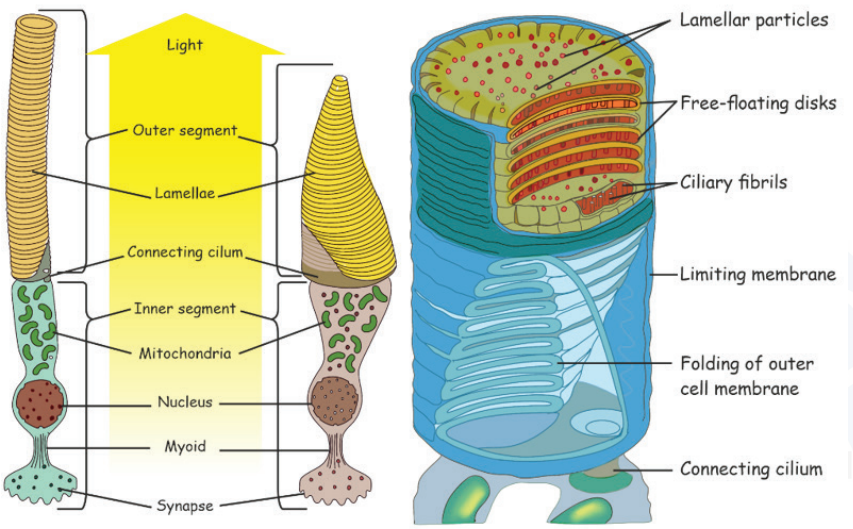
iii. Contraction of medial recti muscles - Convergence of eyeballs

- Argyll Robertson Pupil seen in Syphilis
  - ✦ Accommodation reflex is ..... but pupillary reflex is .....
  - ✦ In this condition, lesion possibly in pretectal nucleus

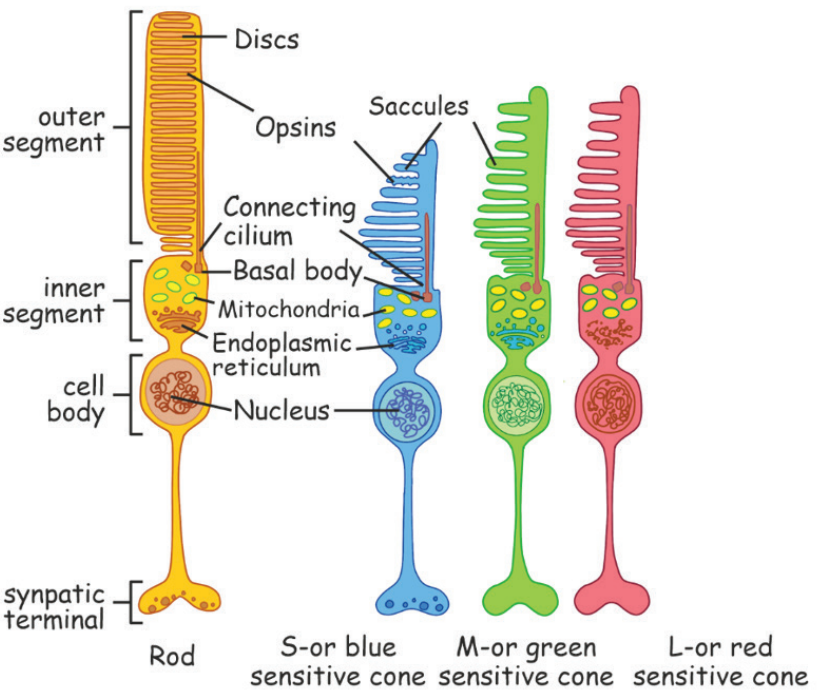
Active space

Photoreceptors

Parts of Photoreceptors



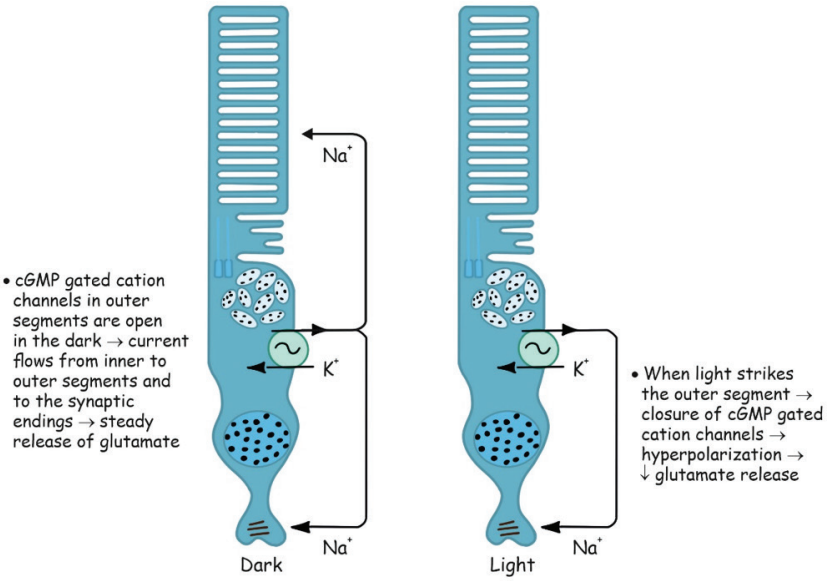
Inner Segment: Visual pigment is produced here



Active space

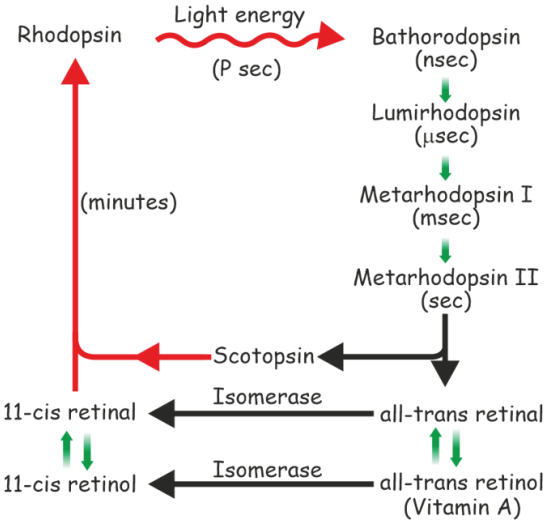
Saccules are formed by the infolding of the membrane.

Ionic Basis of Photoreceptor Potential



- 3 Na+ out of the cell & 2 K+ into the cell via Na+ K+ pump in inner segment occurs leading to resting membrane potential of photoreceptors is -40 mV
- Only photoreceptors, when excited, undergo hyperpolarisationQ whereas the rest of all receptors on excitement undergo depolarisation

Visual Cycle (00:13:00)



Active space

Phototransduction (00:16:35)

Light falls on retina

↓

Structural change in the retinal of photopigment (11-cis retinal to all-trans retinal)

↓

Conformational change of photopigment (Separation of opsin from retinal)

↓

Activation of G-Protein transducing (GDP exchanged for GTP

subunit separates)

↓

Activation of cGMP Phosphodiesterase

↓

Decrease intracellular cGMP (cGMP converted to 5′ GMP)

↓

..... of Na+ channels

↓

.....

↓

Decreased release of glutamate

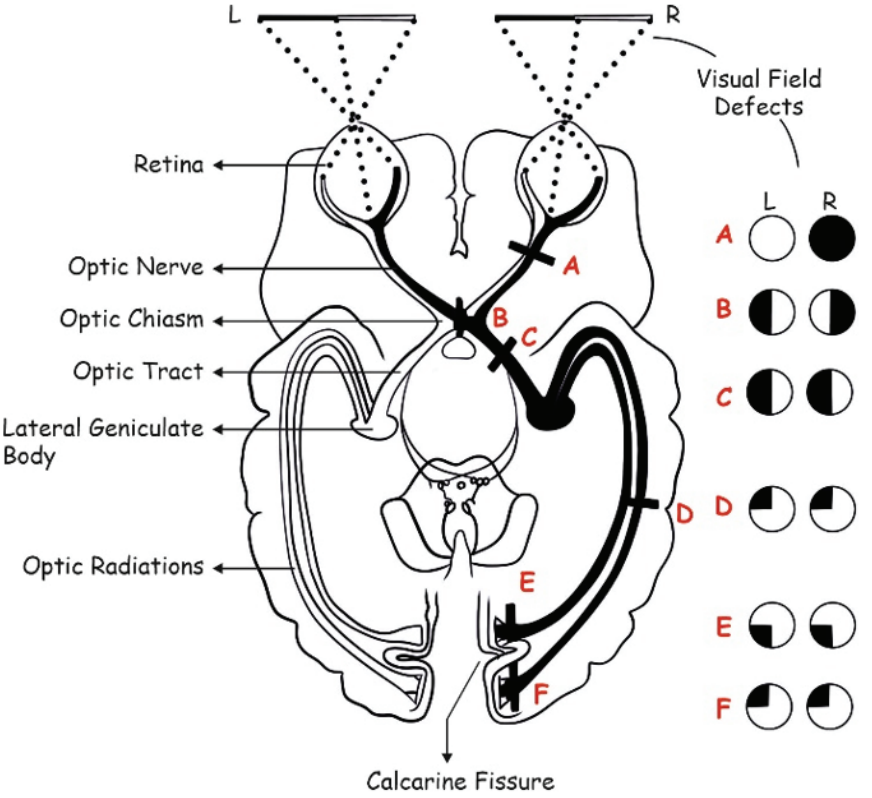
↓

Response in Bipolar Cells

Active space

Visual pathway (00:20:25)

- Light from the nasal visual field falls on temporal retina & light from the temporal visual field falls on the nasal retina
- Fibres from the nasal half & temporal half will form the optic nerve
- Optic chiasma: Fibres from the nasal half of both eyes decussate, i.e. cross over to the opposite side
- Fibres from the temporal half continue on the same side
- Both fibres are now known as the Optic tract going to the lateral geniculate ganglion
- Lateral geniculate ganglion: Visual relay nucleus in the thalamus. From here, the fibres arise & reach the visual cortex.



Active space

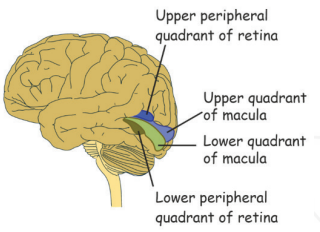
Visual field defects

A: Loss of vision in .....

B: ..... halves of both fields is affected. Also known as ..... Hemianopia.

C: Temporal half on the left side & nasal half on the right side are affected, hence left visual field of both sides is lost. Also known as ..... hemianopia.

D: Leads to .....

• Optic Nerve fibres from upper retinal quadrants subserving vision in the lower half of the visual field terminate in the medial half of LGB • Geniculocalcarine fibers from medial half of LGB terminate in superior lip of the calcarine fissure	 • Fibres from LGB that subserve the macular vision separate and end posteriorly	• Optic nerve fibres from lower retinal quadrants sub-serving vision in the upper half of visual field terminate in the lateral half of LGB • Geniculocalcarine fibres from the lateral half of LGB terminate in inferior lip of calcarine fissure

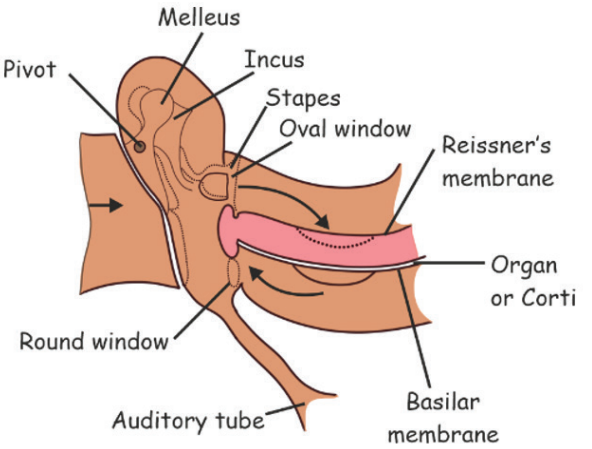
- Macular sparing- Loss of peripheral vision with intact macular vision is common with occipital lesions because macular representation is very large & separate

Active space

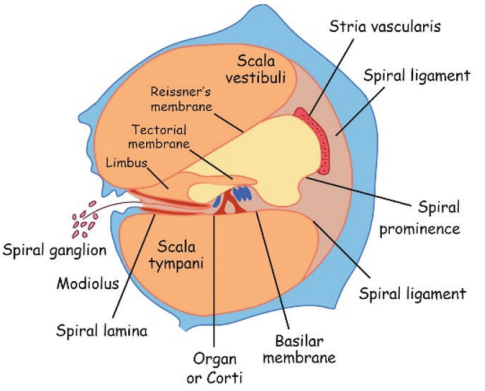
COCHLEA

Parts of Middle Ear

- 35 mm long tube
- Makes two & three-quarters turns
- Upper scala vestibuli & lower scala tympani contain perilymph and communicate with each other at the apex via the helicotrema
- Middle cochlear chamber, also called scala media, is continuous with a membranous labyrinth & contains endolymph and does not communicate with the other two scalae



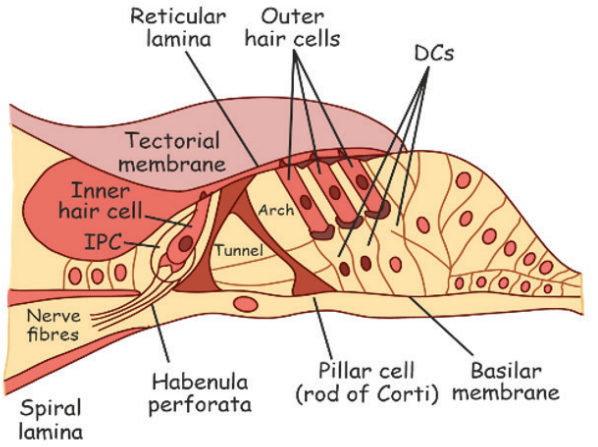
- Scala media & scala vestibule are separated by Reissner's membrane
- Between scala media & scala tympani is a basilar membrane
- Organ of corti rests on the basilar membrane in the basal turn of the cochlea
- Enolymph (rich in potassium) is produced by stria vascularis



Active space

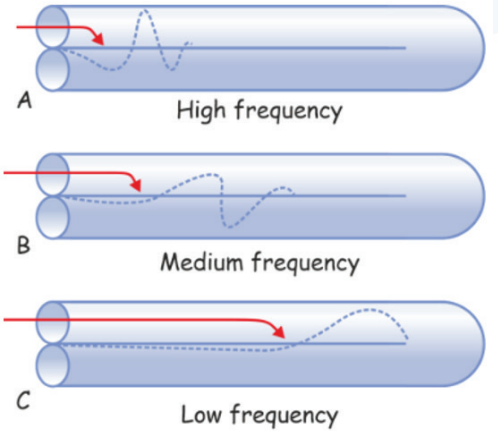
Hair cells of Ear

- Inner hair cells are primary sensory receptors
- Outer hair cells amplify sound vibrations
- IPC- Inner pharyngeal cells support inner hair cells
- DC- Deiter's cells (Outer pharyngeal cells) support outer hair cells



TRAVELLING WAVES

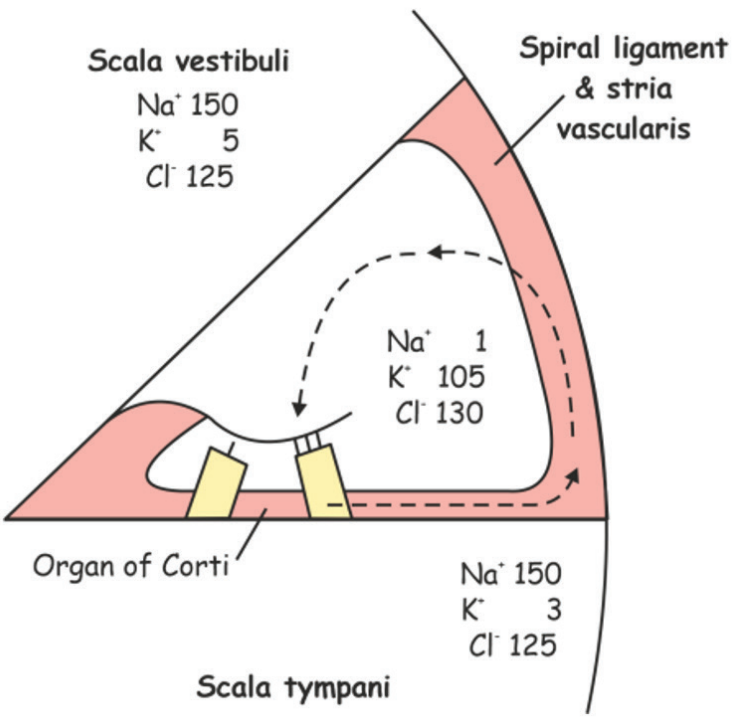
Sound Detection: Whenever sound falls on the tympanic membrane, amplification of sound is done by ossicles. The stapes rest on the oval window & generate waves in perilymph which in turn causes movement in the basilar membrane & that generates impulses in inner hair cells.



Active space

- High-pitched or high-frequency sounds reach maximum height near the base
- Low-pitched or low-frequency sounds generate waves that peak near the apex

- Perilymph: Present in scala vestibuli & scala tympani, which communicate with each other by helicotrema. It is rich in sodium & potassium.
- Endolymph: Present in scala media, which does not communicate with either of scala vestibule or scala tympani. It is extremely rich in potassium. Striae vascularis has sodium-potassium ATPase pumps, which cause movement of K+ into the lymph generating potential of +80 mV.



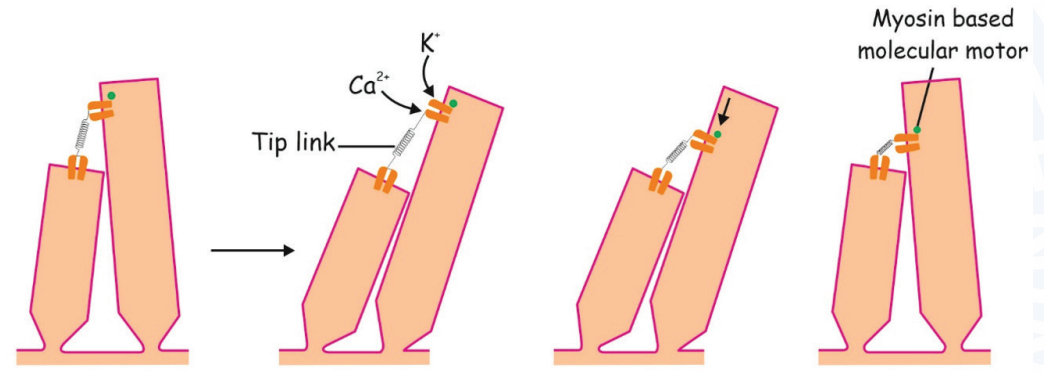
Significance of endolymphatic potential: The hair cell has a potential of -60mV. Since the difference in potential is too high, it makes hair cells very sensitive.

Active space

ELECTRICAL RESPONSES IN HAIR CELLS

- Hair cells have shorter & taller stereocilia
- Stereo cilium pushed towards taller ones → Tip link is stretched, leading to the opening of cation channels
- Only area of the body where depolarisation is by influx of potassium, whereas most skeletal muscle depolarisation is due to the influx of sodium channels

Electrical responses in hair cells



Resting position

Shorter stereocilium pushed towards taller ones → opens cation channels → depolarisation → release of glutamate

Myosin based molecular motor in taller stereocilium → moves channel towards base → releases tension in tip link → channel closes → resting state

Auditory pathway

Afferent fibres in auditory division of VIIIth nerve

↓

Dorsal and ventral cochlear nuclei

↓

Inferior colliculi (Centre for auditory reflexes)

↓

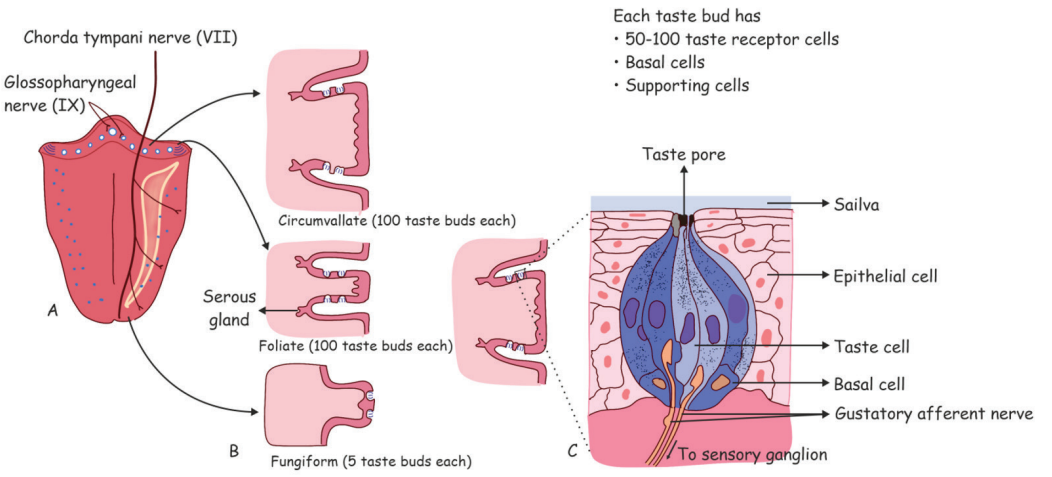
Medial Geniculate Body (Thalamus)

↓

Auditory cortex (Superior temporal gyrus of the temporal lobe)

Active space

ELECTRICAL RESPONSES IN HAIR CELLS



- Circumvalate Papillae are located in the V-aspect of tongue, i.e. posterior aspect
- Foliate papillae are present on the edge
- Fungiform are towards the anterior border along the sides of the tongue

Taste pathway

Taste Pathway

Taste fibers from anterior two-thirds of tongue → Chorda tympani branch of VIIth nerve

Taste fibres from posterior one-third of tongue → IXth nerve

Taste fibres from other areas → Xth nerve

↓

Gustatory portion of nucleus of tractus solitarius in medulla

↓

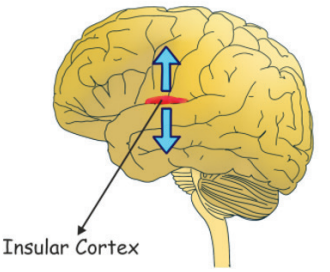
Ventral postero medial nucleus of thalamus

↓

Anterior insula and frontal operculum in ipsilateral frontal cortex

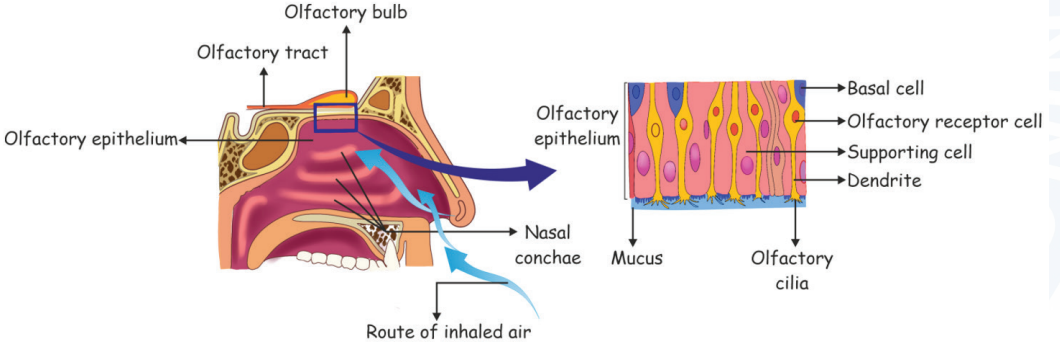
Active space

- Insular cortex: lies deep within the lateral sulcus
- Insular cortex concerned with
  - ✦ Taste
  - ✦ Visceral sensation
  - ✦ Autonomic control
  - ✦ Vestibular information
  - ✦ Equilibrium



Olfaction (00:03:48)

- Specialised part of the nasal mucosa
- Covers an area of 10sq cm in the roof of the nasal cavity
- Olfactory sensory neurons (bipolar), supporting cells, basal stem cells



Odorant signal transduction

Odorant molecule binds to its G-Positive coupled receptor

↓

G-Protein subunits (α, β, γ) dissociate

↓

α-subunit activates adenyl cyclase

↓

cAMP opens cation channels

Active space

↓

Increased membrane permeability to Na, K, Ca

↓

Inward-directed Ca++ current

↓

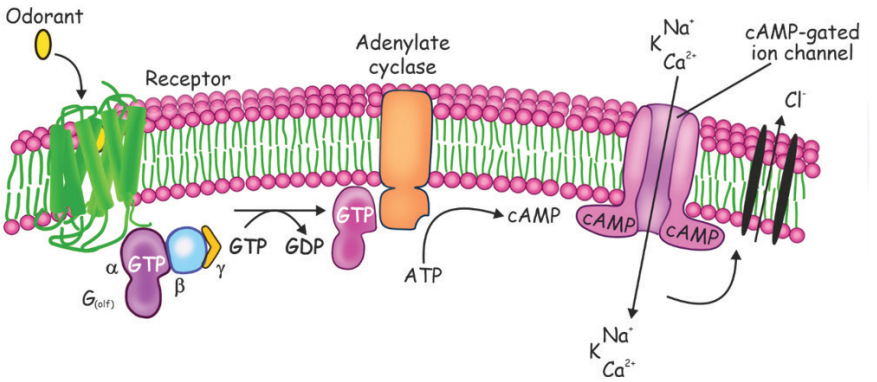
Depolarising local potential

↓

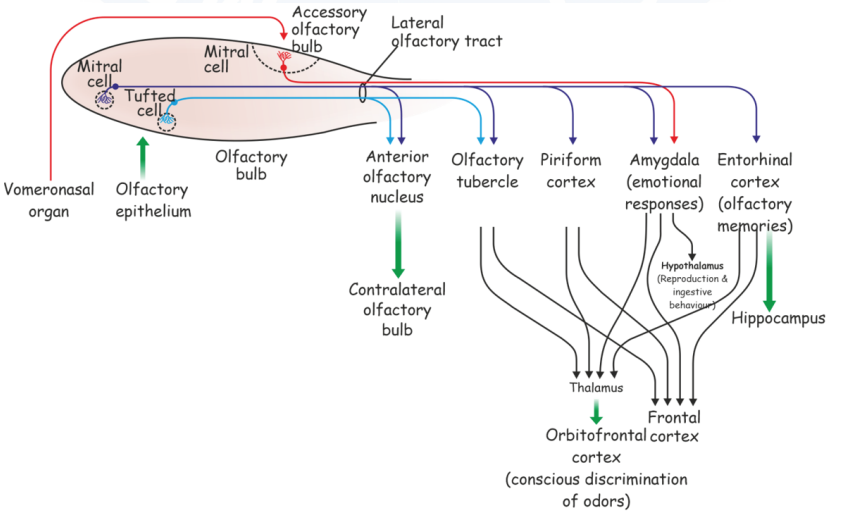
Open Ca++ activated chloride channels

↓

Further depolarisation



Olfactory pathway



Active space

MIST

Workbook Sheet

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Sheet

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